TALABAYBAYIN: A MOBILE LEARNING APPLICATION FOR GUIDING FILIPINO LEARNERS IN MASTERING AND PRESERVING BAYBAYIN

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"So whether you eat or drink or whatever you do, do it all for the glory of God." - 1 Corinthians 10:31 (NIV).

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T.L.B.Y.B.Y.N

DEDICATION

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ABSTRACT

This study, TalaBaybayin: A Mobile Learning Application for Guiding Filipino Learners In Mastering And Preserving Baybayin focuses on a mobile learning app that helps Filipino learners master and preserve the ancient pre-colonial writing system known as Baybayin. It promotes keeping this script alive and getting good at it through an interactive setup with gamified elements on a phone app. The app targets mainly Grade 7 students since they cover Baybayin in the Matatag curriculum from the Department of Education, which stresses learning and grasping the old Filipino script. Built with React Native Skia and following the Agile Software Development Life Cycle, the app mixes in multimedia lessons, handwriting recognition, quizzes, and fun features like ranks, badges, and leaderboards to boost involvement and make it easier to use.

The system's effectiveness was evaluated using the ISO/IEC 25010:2023 software quality standard through a mixed-method approach involving 100 respondents composed of students, teachers, IT professionals, and industry experts. The TalaBaybayin application received an average rating of 3.64 with a standard deviation of 0.74, indicating that users generally find the system highly acceptable in terms of functionality, performance, compatibility, and security. These findings demonstrate that TalaBaybayin provides an effective, interactive, and culturally relevant learning experience that supports both educational and cultural preservation goals.

K*eywords*: *TalaBaybayin, Baybayin, mobile learning, ISO/IEC 25010, cultural preservation.*

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter presents a comprehensive discussion of the problem and the contextual factors that prompted the development of the project. It identifies the target users and clarifies the study’s purpose, along with the methodology used in designing and developing the system. Furthermore, it specifies the project’s objectives, scope, and boundaries, and acknowledges the stakeholders who are anticipated to gain from the outcomes of this work.

Introduction

The fast evolution and advancement of technology have greatly transformed the software industry, shifting from personal computers to widely available mobile applications (Kemp, 2025). Traditionally, it is limited to cybercafes and costly personal computers. Digital access developed with the rise of smartphones, allowing users across the globe to engage with mobile apps. This transition drove the rapid advancement of the mobile app development sector, which now plays a significant role in communication, education, and business. As mobile technology continues to advance, it continues to push innovation forward, shaping various industries (Thomas & Devi, 2021).

These technological advancements have transformed mobile learning into a dynamic and strategic educational tool, offering unparalleled flexibility, accessibility, and personalized learning experiences. The proliferation of high-speed internet and smart devices has empowered students to engage with digital content anytime and anywhere, fostering self-directed learning and enhancing academic outcomes. Mobile learning applications have further revolutionized higher education by creating interactive and adaptive environments that cater to diverse student needs, reshaping traditional teaching methodologies (Lazaro & Duart, 2023).

Recent empirical research highlighted that integrating mobile devices into educational interventions was more effective than relying solely on desktop computers or traditional, technology-free learning environments. This underscores mobile learning's possibilities for enhancing student participation and educational performance through increased accessibility and interactivity (Kaur et al., 2022).

Concurrently, gamification in mobile applications has emerged as a significant strategy for enhancing user engagement, motivation, and behavioral outcomes through different domains, including healthcare, education, business, and ecological sustainability. Mobile apps can enhance user engagement and encourage sustained participation when they incorporate game-inspired features like scoring systems, achievement badges, rankings, and activity monitoring. Recent studies have shown that gamification not only helps retain users and improves their overall experience, but also tackles challenges such as declining interest and the demand for more customized forms of interaction (Oliveira et al., 2024).

Similarly, gamification in education has surfaced as a powerful approach that integrates game-based teaching strategies to enrich learning experiences, foster problem-solving skills, and boost student engagement. Educational institutions are increasingly adopting gamification to promote interactive and immersive learning environments, demonstrating its potential to make education more engaging and effective (Abdeen & Albiladi, 2021). Building on this theme of innovative educational methods, mobile technology has also played a pivotal role in preserving indigenous writing systems.

North America leads global efforts to use mobile technology for maintaining indigenous writing systems. Through its effort, Motorola (2022) designed a Cherokee language interface that embeds the Cherokee syllabary within mobile devices to deliver native language app features. The project enables smoother language learning while helping cultural protection by creating easier access to all users of traditional languages. These initiatives show how including traditional scripts with modern technology allows mobile learning to connect heritage languages with digital communication while protecting indigenous scripts in current technology usage.

Through digital initiatives, Thailand accomplished significant success in protecting its writing systems, like what Motorola does in North America. A 2022 study established an administrative management model that ordered the editing of Tai Tham palm-leaf manuscripts, including digitalization, database development, and manuscript revision activities. This model supports quality improvement practices to keep Tai Tham manuscripts accessible for scholarly research and public educational purposes. Thai digital preservation initiatives work toward protecting cultural heritage within the country's overall digital agendas (Kittipalo et al., 2022).

According to Dao (2023), the shift toward digitizing minority Tai languages such as Tai Tham is essential for safeguarding both cultural and linguistic heritage in today’s digital age. The availability of mobile apps, virtual keyboards, and online repositories has simplified access to the script, bridging the divide between traditional manuscript-based instruction and contemporary learning approaches. These efforts not only safeguard Tai Tham for future generations but also reinforce its role in maintaining the cultural identity of Northern Thailand. Through its presence on digital platforms, Tai Tham supports the engagement of younger people with their ancestral heritage, thus actively building literacy and cultural awareness.

Aksara Bali, also known as the Balinese script, started as the ancient Indian Brahmi script and developed following influences from the Sanskrit and Pallava scripts (Andika, 2021). The transmission of writing systems took place through trading networks, and with the spread of Buddhism and Hinduism, which entered the Indonesian island chain. The Balinese script evolved under local religious and cultural conditions and developed a distinct writing system that Balinese scholars used in religious manuscripts, literature, and official documentation for centuries.

The preservation and revitalization of Aksara Bali have been boosted through modern digital technology recently. Mengenal Aksara Bali serves as a notable example since it presents Balinese script education through interactive gameplay in its application platform (Andika, 2021). The mobile app enables digital preservation of traditional scribbles through interactive flashcards and translating exercises, combined with gamified learning challenges. The initiative demonstrates the power of modern digital tools to connect historical cultural heritage with modern educational practices so that Aksara Bali can live on through new generations.

The growing success of mobile learning technologies in safeguarding endangered writing systems, such as the Cherokee syllabary in North America, Aksara Bali in Indonesia, and the Tai Tham script in Thailand, highlights their potential as effective instruments for cultural and linguistic preservation. In the Philippine context, Baybayin, an indigenous pre-colonial script, exemplifies a valuable subject for such efforts. The word Baybayin comes from the Tagalog term baybay, which means “to spell.” This writing system was commonly used throughout the Philippine islands prior to the arrival of Spanish colonizers in 1565. One of the earliest printed texts in the Philippines, the *Doctrina Christiana* (1593), spans 74 pages and features content in Spanish, Tagalog written using the Latin alphabet, and Tagalog written in Baybayin. Over time, Spanish influence led to the widespread adoption of the Latin script, contributing to the displacement and eventual decline of Baybayin (Escote, 2023). While many mistakenly call Baybayin *"alibata,"* the two are not the same. The term *Alibata*, frequently linked to Baybayin, was introduced by Professor Paul Versoza, a member of the National Language Institute, who claimed that Baybayin was derived from Arabic script in 1914 during his studies at the New York Public Library (Chua, 2020).In the book published in 2009 by Christian Cabuay titled “An Introduction to Baybayin”, the script, which *alibata* is usually referred to, is originated from the Brahmic scripts of India, not from Arabic. This has been established through linguistic and archaeological evidence. Despite its historical significance, the script faces challenges in modern times. Galvez et al. (2023) state that Baybayin, a pre-colonial script in the Philippines, struggles to survive due to the dominance of the Latin script in education and communication. However, the integration of mobile learning applications presents a viable solution for revitalizing Baybayin by making it more accessible, engaging, and adaptable to modern educational frameworks.

The integration of mobile learning technologies in indigenous script preservation holds significant implications for various stakeholders, particularly Filipino grade 7 high school students and educators committed to the preservation and mastery of Baybayin, an ancient writing system with substantial cultural and historical significance. This group includes grade 7 students engaging with Baybayin, independent learners seeking to develop proficiency beyond formal education, and teaching professionals responsible for incorporating Baybayin into instructional frameworks.

For grade 7 high school students, mobile learning app tools serve as an effective complement to traditional instruction, facilitating a more interactive and engaging approach to mastering Baybayin. These digital interventions promote script literacy and cultural appreciation while enhancing learning outcomes. Independent learners benefit from mobile learning applications that provide flexible, self-paced study opportunities, allowing them to develop proficiency in Baybayin outside formal educational settings. In addition, teachers hold a significant role in effectively passing on Baybayin knowledge by incorporating mobile learning tools into their teaching practices. This integration supports the development of script literacy and strengthens students' appreciation and connection to Filipino cultural heritage.

The revival of Baybayin has gained momentum in recent years, notably through the 15th Congress's First Regular Session, held in Quezon City. Congressman Leopoldo N. Bataoil proposed House Bill 4395 in 2011, aimed at introducing Baybayin in schools to promote awareness of the Philippine linguistic heritage (Solares, 2024). Furthermore, legislative initiatives have been introduced to support the preservation and promotion of Baybayin. In 2016, Representative Leopoldo N. Bataoil filed House Bill No. 1022, or the National Writing System Act, which aims to recognize Baybayin as the Philippines' national writing system and outlines measures for its promotion, protection, and conservation. The proposal gained endorsement from several institutions, including the Department of Education (DepEd), the National Commission for Culture and the Arts (NCCA), and the advocacy organization Baybayin, Buhayin. Likewise, Senate Bill No.

1866, authored by Senator Loren B. Legarda on February 13, 2023, seeks to encourage the use of Baybayin in various public domains such as signage, product labels, and educational resources, reinforcing its role in cultural development. As of February 13, 2023, Senate Bill No. 1866 remains pending in legislative status in the Committee. However, although these legislative advancements have been made, there remains a significant gap in interactive and accessible digital tools for learning Baybayin. The educator and cultural advocate interviewed by the researchers highlighted that incorporating digital platforms into Baybayin teaching is essential for improving students' interest and skill in using the script. He highlighted the need to incorporate the ancient script into teaching practices to promote linguistic awareness and ensure its continued relevance in education.

Despite the availability of existing resources such as books, Portable Document Format (PDF), and online references, these materials primarily serve as passive learning tools that lack engagement and hands-on practice. Meanwhile, digital tools like Baybayin keyboard applications focus on transliteration rather than providing structured guidance for learners. Without proper instructional support, users may struggle to understand and apply the script effectively. Addressing these gaps through innovative digital approaches is essential to ensuring meaningful and sustainable Baybayin education.

The lack of interactive learning tools presents a significant gap in Baybayin learning, writing, and recognition. Dela Cruz et al. (2020) address the lack of interactive learning platforms for Baybayin that help users recognize and practice writing scripts. Incorporating gamified elements and interactive assessment tools in a mobile application could enhance user engagement and improve learning outcomes. While efforts have been made to promote Baybayin learning as part of Filipino heritage.

Project Context

The research team proposed that a mobile learning application integrating a Baybayin keyboard with interactive writing practice tools, assessment quizzes, and gamified features can significantly enhance the learning, preservation, and mastery of the ancient script. By incorporating elements such as achievement rank and interactive challenges, the application aims to improve user engagement and learning. This approach not only modernizes Baybayin learning but also provides an effective and immersive platform for learners to develop proficiency while contributing to the script's long-term preservation.

However, despite the growing interest in Baybayin, existing applications have primarily focused on text conversion, character recognition, and transliteration, lacking interactive features that facilitate active script writing and real-time assessment. These resources offer only passive learning experiences, limiting learners' ability to develop accuracy and fluency in writing Baybayin. Recognizing these setbacks, the researchers identified the need for a more comprehensive mobile learning application that caters to both foundational learning and skill enhancement. The gaps resulting from the absence of an interactive and engaging platform highlight this intervention's necessity. To address these challenges, the research team is adopting a proactive approach in developing and enhancing the mobile application, which has the potential to significantly benefit Filipino learners while also preserving and promoting the nation's cultural heritage. In response to the absence of an engaging and culturally relevant digital-learning platform, the researchers are actively developing and refining a mobile application aligned with the MATATAG Curriculum. This curriculum, introduced by the Philippine Department of Education

(DepEd), seeks to elevate instructional quality and overall learning outcomes. Its gradual implementation will begin in School Year 2024–2025, initially covering Kindergarten and Grades 1, 4, and 7. With a focus on holistic learner development, the curriculum encompasses several core developmental areas, including literacy, language and communication, cognitive growth, socio-emotional development, creative and aesthetic expression, physical and motor abilities, and values formation. The phased implementation approach is intended to facilitate a smooth and effective transition to the updated educational standards.

The MATATAG Curriculum, which aims to develop foundational competencies while nurturing national identity and personal growth. More specifically, the application is anchored on the Grade 7 learning module, which guides students in examining written texts such as literature from the Pre-Colonial period up to the Spanish Colonial eraas well as informational, academic, and visual materials. The module aims to help learners produce meaningful multimodal outputs by applying linguistic elements they have learned, ultimately fostering personal identity and strengthening Filipino values. As part of this instructional material, within this context, the learning of Baybayin, the indigenous script of the Philippines, is embedded as a key cultural and linguistic element. enabling students to explore its historical significance and practical application. This inclusion supports culturally responsive teaching by integrating Filipino linguistic heritage into modern instruction. Through engagement with these materials, students are encouraged to critically analyze the cultural and historical foundations of Filipino identity, thereby fostering personal development and a deeper appreciation of Filipino values.

To support this educational framework and further empower learners, the researchers align their initiative with global efforts to create inclusive and forward-thinking educational solutions. The 17 Sustainable Development Goals (SDGs) established by the United Nations provide a worldwide framework for fostering a more sustainable and equitable future. These goals, endorsed by all UN Member States in 2015 under the 2030 Agenda for Sustainable Development, aim to tackle critical global issues such as poverty, inequality, climate change, environmental harm, and the promotion of peace and justice. Each goal is interconnected, designed to ensure inclusivity and prevent anyone from being overlooked. Achieving these objectives requires immediate and collaborative efforts from governments, businesses, civil society, and individuals.

In line with this global vision, the present project emphasizes Goal 4: Quality Education and Goal 9: Industry, Innovation, and Infrastructure. By utilizing digital tools to develop an innovative mobile learning platform, the researchers aim to enhance both the accessibility and quality of education, while also integrating cultural heritage into contemporary learning experiences. The application encourages inclusive, lifelong learning and fosters digital innovation in the educational sector, particularly by revitalizing indigenous knowledge systems like Baybayin through interactive, tech-driven methods. This initiative not only promotes educational equity but also strengthens the digital infrastructure for culturally grounded instruction.

Given these circumstances, the researchers recognized the necessity of developing a mobile learning application designed to support Filipino grade 7 high school students and individuals seeking to retain and enhance their knowledge of Baybayin. This application integrated interactive writing exercises, structured assessments, and gamified features to establish an compelling and effective learning engagement. By providing a comprehensive and accessible platform, the system addressed the limitations of existing learning resources, which primarily focus on passive instruction and translation. To ensure alignment with current educational standards, the researchers utilized the *Modyul* in the MATATAG Curriculum that specifically addresses the teaching of Baybayin. This served as the foundational framework for content development, ensuring that lessons are culturally relevant, educationally sound, and supportive of DepEd's goals for holistic education. Through interactive practice and adaptive learning mechanisms, users developed proficiency in reading and writing Baybayin and contributed to the script's preservation and cultural revival in modern education. Furthermore, the application incorporated culturally contextualized content within its assessments and quizzes, focusing on lessons that teach users how to read and write Baybayin accurately. These evaluations reinforced script recognition, proper stroke order, and syllable construction. Time progression and progress tracking personalized the learning experience, encouraging continued engagement and skill development.

Purpose and Description

This study aimed to help students, teachers, and others understand and master Baybayin, the traditional indigenous script of the Philippines, by developing an interactive application to enhance the learning experience. With the goal of preserving culture and making Baybayin more accessible to modern learners, the study promotes a deeper appreciation of this ancient script, which represents syllables through 17 symbols, including three vowels and 14 consonants. According to Camba (2020), digital tools can improve language learning participation, making them valuable for cultural preservation.

Nowadays, there are attempts to revive and promote Baybayin through education, online platforms, and artistic works. Specifically, the system benefited the following:

Grade 7 High School Students. This application incorporates gamified elements to enhance learner motivation and improve knowledge retention through interactive challenges. Additionally, students can also track their progress and find areas for growth on the site, allows students to explore Baybayin through lessons, exercises, and quizzes, making learning fun and effective.

Teaching Professionals. This program gives educators a guided medium to incorporate Baybayin into their curriculum, giving them resources like quizzes, writing activity pages, and discussion boards to increase learning engagement.

Baybayin Enthusiasts. This application can provide self-directed learning for those who want to learn Baybayin independently, providing structured modules, tests, and self-learning materials. It seeks to foster a sense of community among learners by encouraging the sharing of insights and progress through integrated forums and discussion spaces.

Researchers. This research can serve as a reference to future scholars, especially those in Information Technology, wanting to develop similar systems, portals, applications, or games. It will provide insights into the design, implementation, and challenges of integrating gamified learning experiences with cultural preservation efforts. By studying the methodologies and technologies used in this system, IT researchers can further innovate and improve digital education platforms for indigenous languages and scripts. Additionally, the findings could inspire future collaborations between technology and cultural experts to create more engaging and effective educational tools for preserving endangered languages and traditions. This research underscores the potential of technology to link the disparity between modern education and cultural heritage, highlighting the importance of innovative approaches in the preservation of linguistic diversity.

Project Objectives

This section discusses the following general and specific objectives that the researchers aim to achieve in this paper. Objectives are essential in shaping the direction and scope of the study. They also help in maintaining a clear focus during the research process. These objectives serve as a foundation for assessing how effectively the study addresses the research problem, ensuring that the study remains aligned and serves its intended purpose.

General Objective

This study aims to promote preservation and mastery in the Baybayin script through the development of TalaBaybayin, a mobile learning application intended to give Filipino learners an engaging and interactive method of learning Baybayin. Through the use of contemporary technology, the application aims to make learning both engaging and easy to access, while encouraging a stronger appreciation of Filipino cultural heritage. TalaBaybayin aims to revive interest in this traditional writing system by integrating it into modern educational tools and practices.

Specific Objectives

Specifically, the researchers aim to achieve the following goals:

1. To determine the current state of studying Baybayin and its preservation among Filipino learners, particularly in terms of accessibility, interest, and the effectiveness of available materials, through surveys and interviews.

2. To create and build an all-inclusive platform for learning Baybayin, incorporating features that enhance learner engagement, facilitate progress tracking, and improve knowledge retention, such as:
    2.1. Login/Sign-up;
    2.2. Interactive Lessons;
    2.3. Leaderboard;
    2.4. Time-Based Scoring;
    2.5. Quizzes;
    2.6. Progress tracking;
    2.7. Gamification features;
    2.8. Mini-games;
    2.9. Multimedia sounds ;
    2.10. Common Phrases and
    2.11. Completion Certificate.
3. To include a specialized feature that improves the user experience and learning process, including:
    3.1. Baybayin Keyboard;
    3.2. Handwriting Feature;
    3.3. Text to Baybayin Translation; and
    3.4. Text-to-Speech.
4. To assess the acceptability of the TalaBaybayin app based on the ISO/IEC 25010:2023 software quality standard by conducting User Acceptance Testing (UAT) using a 4-

point Likert scale questionnaire aligned with the ISO characteristics, focusing on the evaluation of key quality attributes:

4.1. Functional Suitability;

4.2. Performance Efficiency;

4.3. Compatibility;

4.4. Interaction Capability;

4.5. Reliability;

4.6. Security;

4.7. Maintainability;

4.8. Flexibility; and

4.9. Safety.

Scope and Limitations

The primary focus of this study is to help Filipino students, teachers, and language learners master and preserve Baybayin. The application provided several features developed to improve the learning process, such as lessons on the background and usage of Baybayin, and materials like quizzes for mastering the writing of the script. Users are able to access the learning and engagement features of Baybayin. The functionalities include Login and Sign-up, where users can either log in to an existing account or create a new one. Interactive Lessons, wherein users have access to detailed lessons on Baybayin, including its history, writing system, and usage, which incorporate quizzes, to reinforce learning. Leaderboard features a dynamic leaderboard that ranks users based on their accumulated experience points (XP) and quiz scores. Time-based scoring tracks the time it takes for users to answer and take quizzes. Faster completion with correct answers rewards the user with bonus points or achievements. Handwriting Feature allows users to select a Baybayin character to practice writing on a canvas, then save their outputs, allowing them to revisit and continue. Users can manually trace the character, erase parts of their strokes, clear the entire canvas if needed, and save their current progress. Quizzes, where users can take quizzes to measure their knowledge and comprehension of the subject matter, are provided after each lesson. Progress Tracking, wherein users are able to see their learning progress and achievements in a dashboard display.

Gamification Features, where users can engage in learning Baybayin through game-like elements to make the experience more fun and effective. Learners can unlock lessons and quizzes, earn experience points (XP), achieve badges, and rank up based on their progress. To further engage learners, mini games are included to provide fun practice opportunities, while multimedia sounds enhance the experience for auditory learners. Additionally, common phrases in Baybayin are incorporated, allowing learners to practice useful expressions for real-world application. In addition, the Baybayin Keyboard is a special feature where users have the ability to type in Baybayin characters, enhancing their practical application of the script. Text to Baybayin translation, where users can input a word, and it converts to Baybayin, and vice versa. This comprehensive method guarantees that users acquire both theoretical understanding and practical abilities, fostering continuous involvement with the Baybayin script and Text-to-Speech functionalities to enhance accessibility and engagement in learning Baybayin. For text-to-speech, Baybayin characters are transliterated into words, which are then converted into audible speech using a standard voice engine. This feature supports auditory learners, enables hands-free interactions, and promotes a multi-sensory approach to mastering Baybayin. In addition to these core features, the application incorporates mini games that promote active engagement while reinforcing learning. These include a memory match game, a word scramble game, a character quiz game, a pattern recognition game, and a drag-and-drop game. Unlike the main gamified elements of the system, these mini games are not tied to the leaderboard, XP, or rank progression, but are intended to provide learners with enjoyable practice opportunities. The inclusion of multimedia sound resources, such as audio materials, further enhances motivation and provides an enriched learning environment. The mobile learning application incorporates common Tagalog phrases, including greetings, polite expressions, and basic conversational lines, enabling users to learn the script through practical applications in real-life conversations. In addition to the already mentioned features, a Completion Certificate will be given to the users, serving as a significant motivational tool in the mobile learning app. Upon finishing all lectures and quizzes, users receive a certificate that validates their effort and achievements, boosting their sense of accomplishment. This certificate, provided in PDF format, offers a tangible recognition of the learner's commitment and progress. As an e-certificate, it can easily be shared or printed, providing a clear acknowledgment of the user's success and dedication to learning.

While TalaBaybayin is designed with extensive functionality to support the educational purposes of Baybayin mastery learning, certain limitations need to be considered. The app is designed for students, teachers, and Baybayin enthusiasts. This focus may limit its appeal to a broader audience, such as non-Filipino learners or those interested in other indigenous scripts. In addition, the app is for students, teachers, and language learners at basic to intermediate levels of Baybayin proficiency and does not include the advanced linguistic study or scholarly research functionality. The app's reliance on technology means that users without access to smartphones or tablets may be excluded from the learning experience. This could limit the app's reach, particularly in rural areas where technological access is limited.

Users with older devices or limited internet connectivity may experience difficulties accessing multimedia materials or interactive features. Additionally, the application does not have a web-based version available and is only accessible through supported Android mobile platforms. The effectiveness of gamification features and discussion forums relies heavily on user engagement. Computerized testing gives instant feedback but might be less insightful than individualized assessment by seasoned teachers. Users may benefit from additional context that goes beyond the technical aspects of writing and reading the script. Users may desire more tailored learning paths based on their individual interests and proficiency levels. The interactive functionalities of the app are educational and do not include functionalities of formal certificate generation or professional credentials for Baybayin mastery.

CHAPTER II

REVIEW OF RELATED LITERATURE/SYSTEMS

This chapter explores relevant literature and studies that support the feasibility of the research. It provides insights from previous works related to the study and presents the conceptual framework, which guides the research.

Related Literature

This study presented works of literature that provided the foundation for the development of TalaBaybayin: A mobile learning application for guiding Filipino learners in mastering and preserving Baybayin. It provides a strong theoretical and empirical foundation that supports its development, effectiveness, and significance in promoting Baybayin literacy among Filipino learners.

Baybayin and Its Underlying Factors in Preservation, Accessibility, and Efficiency of Learning Materials

Over the past several years, the Philippines has experienced renewed interest in studying other languages due to globalization and the increasing need for multilingual competent professionals. The ability to adapt is crucial, and while adapting to these modern changes, the ancient writing system of Baybayin has continued to fade into obscurity, largely forgotten by many Filipinos.

According to Prado (2024), the need to revive Baybayin is emphasized as a cultural and educational tool. Despite being an indigenous script, Baybayin has largely been excluded from modern educational systems. Prado's study examined Junior High School students' awareness and competence in reading and writing Baybayin, revealing significant improvement, as shown by the notable difference between pretest and posttest scores. These findings highlight the effectiveness of structured educational interventions in promoting Baybayin literacy.

According to Solares (2024), the proposal of the Philippine Congress to declare Baybayin as the national writing system inspired more studies on improving Baybayin literacy. One such study conducted at Adamson University was titled "Development and Acceptance of a Proposed Module for Enhancing the Reading and Writing Skills in Baybayin: The Traditional Writing System in the Philippines." evaluated a curriculum designed to help students become more proficient readers and writers in Baybayin. The results showed substantial improvement in literacy, reinforcing that incorporating Baybayin into educational modules not only enhances students' skills but also fosters greater cultural appreciation and contributes to the revival of the script (Solares, 2024).

In alignment with these efforts, the Department of Education has included Baybayin instruction within the Grade 7 Filipino curriculum through the official *Modelong Banghay Aralin sa Filipino, Baitang 7, Kuwarter 1: Aralin 1* (para sa Unang Linggo), SY 2024–2025. Developed under the MATATAG K to 10 Curriculum pilot implementation, this module was authored by Rachel C. Payapaya and Ma. Teresa P. Barcelo and reviewed by Dr. Joel C. Malabanan under the Philippine Normal University and its affiliated research centers. It supports instruction on *mga tekstong nasusulat tulad ng panitikan sa Panahon ng Katutubo hanggang sa Panahon ng Pananakop ng Espanya*, including indigenous scripts like Baybayin. By guiding learners to create makabuluhang tekstong multimodal using elementong panlingguwistika, the module positions Baybayin as both a literacy component and a cultural anchor, thereby reinforcing Filipino identity and values.

In addition, a qualitative case study by Galvez et al. (2023) surveyed Junior High School instructors in Bulacan regarding their orientation toward the Baybayin script. The study found that instructors value Baybayin for its cultural significance and potential as a communication tool. However, it also revealed challenges such as inconsistent teaching practices and Baybayin’s struggle to adapt to contemporary linguistic and pedagogical paradigms. Galvez et al. (2023) research advocates for Baybayin literacy through curriculum integration, online activism, and cross-disciplinary efforts.

Similarly, Eslit (2024) found that digitizing Baybayin is a crucial step in preserving Filipino heritage, particularly in the face of post-pandemic challenges. Eslit’s study explores the importance of integrating Baybayin into higher education, emphasizing its transformative effects on cultural literacy and identity. His research proposes enhancing digital literacy, promoting participation, and fostering interdisciplinary collaboration to improve the appreciation of Baybayin within educational contexts.

Building on this idea of promoting digital literacy and cultural awareness, this paper focuses on creating a mobile application that would enhance learning and preservation of the old Philippine writing system known as Baybayin(Lacorte et al.,2024). The study highlights the importance of creating engaging and interactive learning materials to spark interest and enhance Baybayin writing skills. With its inclusion of a spiral model for software development and a comprehensive set of features such as interactive lessons and quizzes, the app aims to achieve similar goals of cultural awareness and historical understanding.

In line with the goal of promoting cultural awareness through innovative tools, the mobile app not only aids in the preservation of Baybayin but also connects to the broader theme of Baybayin's role in strengthening Filipino identity, particularly for those affected by colonial mentality. Baybayin acts as a cultural anchor, helping to restore a sense of pride and resilience by reconnecting Filipinos with their indigenous heritage (Camba, 2021). By making the learning process interactive and accessible, the mobile app offers a modern way to engage with Baybayin, potentially countering the inferiority complex rooted in colonial history. The app's contribution to preserving Baybayin not only aids in personal and collective cultural empowerment but also supports ongoing efforts to ensure its relevance and use, particularly among groups like the Mangyan tribes in Mindoro who continue to use the script.

Collectively, the studies stress the urgency of preserving Baybayin as a key aspect of Filipino culture, identity, and emotional well-being. While significant strides have been made in education, digital tools, and cultural advocacy, challenges remain, particularly in terms of accessibility, resources, and teacher training. Continued research and development are essential to ensure Baybayin's revival and integration into modern Filipino life, helping to reconnect Filipinos with their roots while also enhancing global understanding of their rich cultural heritage.

Components and Development of a Mobile Learning Application for Learning and Retaining Baybayin

The researchers discussed the components of the application for mobile learning and how they help create a fun and gamified Baybayin learning environment platform. The discussion highlights the essential modules that promote effective learning and user engagement, supported by insights from related literature and best practices in mobile-assisted language learning.

To develop a functional and user-centered mobile learning application, a developer must define components that align with established standards, frameworks, and design principles suited to the users' needs. For a mobile-based system, this process begins with selecting an appropriate technology stack, including the programming languages, libraries, and tools required for development.

Mobile application development offered flexibility but also came with complexities, requiring careful selection of technologies that ensure performance, usability, and scalability. A mobile-based system consisted of several core components, including application structure, user interface design, data processing, and storage management. Each of these components plays a vital role in ensuring the system's seamless performance, efficiency, and overall user satisfaction.

The framework required for this project is React Native. According to Liu et al. (2024), React Native is an open-source framework that has become highly popular since its launch in 2015 because it bridges traditional mobile development with the flexibility of Node.js. Its core principle is to provide cross-platform JavaScript APIs that can access platform-specific features by invoking Objective-C/Swift for iOS or Java/Kotlin for Android, ensuring smooth integration with native components. Similarly, Ramachandrappa (2024) explains that React Native, developed by Facebook, is a widely adopted framework for building cross-platform mobile applications using JavaScript and React. Furthermore, this project also makes use of Expo. As noted by the open-source learning platform Softworth Solutions Pvt. Ltd. (2023), Expo serves as both a framework and a platform designed for developing React Native applications. Its primary goal is to simplify the development process by handling much of the setup and configuration, thereby reducing the complexities typically involved in building a React Native project.

In addition, this project is the integration of gamification to enhance user engagement. According to Akther et al. (2024), the secret is gamification; elements like leaderboards, prizes, and badges foster an engaging and inspiring environment. The main focus is on tracking footsteps, with gamified features like personalized consultations to enhance the experience. It distinguishes itself technically by utilizing state-of-the-art technologies, particularly React Native, to guarantee a smooth and adaptable platform.

On the other hand, the library for design is CSS NativeWind. According to (Shadi F, 2025) NativeWind emerges as an innovative solution that brings the popular utility-first approach of Tailwind CSS to React Native, unifying mobile and web styling under a single, expressive syntax. In this deep dive, we explore what NativeWind is, why it’s important, its cross-platform advantages (including Expo support), and how its advanced features, such as platform modifiers and explicit style declarations, elevate the developer experience.

In recent years, advancements in cloud computing have led to the emergence of modern commercial cloud services, including Mobile Backend as a Service (MBaaS). According to Dahunsi et al. (2021), MBaaS plays a vital role in providing reliable backend data storage and management for mobile applications. Furthermore, Nandyal and Rafi (2020) explained that MBaaS delivers a cloud-based server infrastructure equipped with Application Programming Interfaces (APIs) and Software Development Kits (SDKs) that facilitate development across mobile, web, and desktop platforms. This significantly reduces both development costs and time by simplifying backend processes, allowing developers to concentrate on enhancing functionality, enhancing the application’s visibility and optimizing the overall user experience.

According to Rajappa et al. (2020), Firebase is Google-powered toolkit that offers a wide range of services, including analytics, databases, and authentication. It is recognized as a dependable and reputable platform used by major companies such as Alibaba, The New York Times, Duolingo, and Shazam. Shukla et al. (2021) described Firebase as a NoSQL database that enables users to store and retrieve data efficiently through stable connections. The platform provides multiple services, with Firebase Authentication being particularly beneficial for both developers and users, as it simplifies the process of managing and developing login systems through an easy-to-use API. Additionally, Firebase includes a real-time database feature that supports data backup. Since storing media files like videos, text, and images can be challenging and costly for new developers, Firebase offers a cloud-hosted platform for convenient and affordable storage. As noted by Pujari and Shetty (2023), Firebase integrates essential features such as user authentication, cloud services, real-time database functionality, and application crash reporting.

Login/Sign-up**.** The first feature of the proposed system is the login functionality, where users authenticate their accounts. According to Ma’roef et al. (2024), the login page of a mobile learning platform generally prompts users to input their registered username and password, ensuring a simple and secure authentication process.

The researchers proposed a mobile application that utilized Firebase Authentication, which provides secure sign-ins through various methods, including email/password and social login options. Social login enhances user convenience by allowing authentication via existing social media accounts, reducing the friction associated with traditional registration (Kim et al., 2024). Additionally, Firebase Realtime Database is integrated to enable dynamic and synchronized data updates, ensuring a seamless user experience (Das et al., 2024).

Recent studies showed that login systems in learning applications do more than secure access, they also help personalize the learning experience. Escober and Monzon (2025) developed an adaptive multimedia learning system that adjusted lessons based on each user's preferences and learning style. Their research found that having individual accounts allowed the system to track user behavior and improve engagement and retention. In TalaBaybayin, the login feature can likewise support personalized learning and progress monitoring while ensuring data privacy.

As a Backend-as-a-Service (BaaS), Firebase efficiently manages authentication and real-time data storage, streamlining the backend processes of the application (Saraf, 2022). The integration of Firebase Authentication within the proposed mobile application enhances security and user accessibility by offering multiple authentication mechanisms, including email/password and social login options. The implementation of social login reduces registration barriers, thereby improving user adoption and engagement. Furthermore, Firebase Realtime Database facilitates dynamic and synchronized data updates, ensuring a seamless and responsive user experience while optimizing backend processes and minimizing development complexity.

Interactive Lessons**.** One of the key features of the proposed system is the integration of interactive lessons designed to enhance Filipino learners' comprehension and retention of the ancient script. This approach aligns with the findings of Barnett-Itzhaki et al. (2023), who found that interactive learning strategies positively influence students' engagement and performance evaluations.

Further supporting this perspective, Verawati et al. (2024) examined the effectiveness of interactive learning media in increasing students' interest in physical education. Their study revealed that conventional teaching methods often fail to provide sufficient engagement, whereas interactive tools significantly enhance motivation and participation. Through a structured analysis of pre- and post-treatment data, the researchers found that incorporating interactive learning media led to a measurable increase in student interest. Notably, they emphasized that the effectiveness of interactive learning depends not only on its frequency but also on the diversity of strategies employed. This underscores the importance of using varied and dynamic approaches to foster comprehension and engagement, reinforcing the need for diverse interactive techniques in educational frameworks.

Similarly, a study by the faculty of the Human-Computer Interaction Institute at Carnegie Mellon University (Aupperlee, 2021), revealed that students achieve better learning outcomes when participating in active learning approaches. Methods such as interactive exercises, discussions, personalized feedback, and AI-powered tools have been proven to enhance academic performance more effectively than traditional lectures or passive reading. Furthermore, Tarigan et al. (2023) reinforced the effectiveness of interactive learning through research conducted with 180 eleventh-grade students, showing that those who utilized interactive digital learning modules demonstrated significantly higher retention and academic performance compared to peers who relied on non-interactive or text-based materials.

Collectively, these studies underscore the significant impact of interactive learning in enhancing student engagement, motivation, and overall educational achievement. By integrating a variety of dynamic and participatory teaching techniques, educators can create more dynamic and engaging learning environments that cater to the diverse and ever-changing needs of modern learners.

Practice Exercises. Another key feature proposed for the mobile learning application is the inclusion of interactive practice exercises that allow users to actively engage with Baybayin and track their learning progress. These exercises are designed to strengthen understanding through retrieval practice—a research-backed approach known to enhance memory and retention. Huang (2024) reported that retrieval practice significantly improves long-term vocabulary retention among English as a among learners of English as a Second Language (ESL), with those engaging in active recall outperforming those using traditional study techniques. The study further affirmed that retrieval-focused strategies are more effective than passive review methods. Likewise, Agarwal et al. (2021) demonstrated that retrieval practice consistently boosts learning outcomes across various age groups and academic disciplines, underscoring its value as a fundamental educational technique.

The integration of practice exercises in the mobile learning application leverages retrieval practice to enhance Baybayin learning and retention. Research by Huang (2024) and Agarwal et al. (2021) confirms that active recall significantly improves long-term memory across various subjects and age groups. This evidence supports the effectiveness of retrieval-based learning in reinforcing Baybayin mastery.

Leaderboard. In addition to the above-mentioned features, the app integrated into a leaderboard where players can view and compare their progress with others in real time. This gamified element is designed to increase user engagement by encouraging a sense of motivation and accomplishment among learners. Leaderboards serve as an effective mechanism to promote healthy competition, facilitate social comparison, enhance involvement, and offer goal and performance-oriented feedback to users (Ninaus et al., 2020). The inclusion of a leaderboard in the TalaBaybayin app aligns with established gamification principles that emphasize motivation through competitive dynamics. By allowing users to see how their performance compares to others, leaderboards help increase motivation. As a result, this feature keeps users more involved and helps them build regular ways of learning, making it easier and more enjoyable to learn Baybayin.

Another feature integrated into the mobile learning application is the Time-Based Scoring system. As noted by Wallin et al. (2024), time constraints can influence learners' response behavior, affecting both performance and engagement. In TalaBaybayin, this feature is utilized in quizzes to reward users who answer correctly within a shorter period. The faster a learner responds accurately, the higher the score earned. This mechanism encourages mastery through quick recall and reinforces learning efficiency. Moreover, the element of time introduces a sense of challenge and competitiveness.

Handwriting Feature. Also the proposed application integrated handwriting exercises to enable practical writing of the Baybayin script. These activities aim to engage users with the traditional Filipino writing system while enhancing their cognitive involvement in the learning process. The findings indicated that handwriting, in contrast to typing, enhances cerebral connectivity in ways that facilitate learning and memory retention. Handwriting specifically enhanced cerebral activity in regions associated with concentration and memory retention, underscoring its significance in fortifying the brain's learning and recall processes (Malesu, 2024). Incorporating handwriting activities inside the application correlates with cognitive research showing the neurological benefits of hands-on writing. This method not only preserves culture but also improves retention of information, making the learning of Baybayin more efficient and everlasting. Handwriting activates brain regions associated with attention and memory, converting passive recognition into active mastery, thereby enhancing instructional effectiveness and user satisfaction.

Quizzes. The TalaBaybayin mobile learning application includes a quiz feature inspired by the Prolog-based quiz system developed by Ram et al. (2025). Their system dynamically generates questions, provides real-time feedback, and supports level-based progression, offering a structured approach to measuring learning outcomes. By applying K-Means clustering analysis to data from 140 students, Ram et al. identified three learner categories: foundational, intermediate, and advanced. This approach demonstrated the potential of quiz-based systems and clustering techniques in supporting personalized learning. Likewise, integrating quizzes, including the common phrases, into TalaBaybayin can enhance its capability to evaluate learner proficiency and adjust instructional content accordingly, leading to improved engagement and retention in digital learning environments.

Progress Tracking. The project adds progress tracking in the features, which helps users to track their Baybayin learning progress, like the Learning Path Dashboard by Silva et al. (2024). Their system is designed to speed up skill acquisition by offering progress tracking and carefully chosen educational resources. By implementing a progress tracking system, TalaBaybayin can provide users with personalized feedback and insights into their learning journey. This approach mirrors the findings of Ram et al. (2025), who highlighted the importance of adaptive learning environments in enhancing student engagement and retention of knowledge. Incorporating progress tracking supports a more personalized learning experience by enabling users to recognize their strengths and pinpoint areas that require further improvement or practice.

Gamification features. Like Duolingo, TalaBaybayin integrated gamification elements to increase user engagement (Steiner, 2024). Users who utilize Duolingo will be motivated by all the game features, such as collecting points, badges, achievements, levels, and rankings. Examining how digital platforms such as Duolingo can affect users' motivation for mastery is crucial because user motivation is a key component of successful skill acquisition. By offering instant feedback and rewarding positive behavior, both of which are essential for learning, gamification techniques can maximize progress.Users are likely to experience a sense of accomplishment as they advance through stages and accrue points, which can inspire them to keep studying. This supports research showing that by making learning more fun and interesting, gamified learning environments can enhance academic results. By adding these gamification elements to TalaBaybayin, the platform not only complies with contemporary educational technology trends but also meets the demand for creative methods of language acquisition. The app can produce a dynamic and efficient platform for learning and maintaining Baybayin by utilizing the motivating elements of gamification.

Mini-games. Phadke et al. (2024) emphasized that mini-games, when integrated into larger game environments, can serve as powerful tools for both engagement and knowledge reinforcement. Their research demonstrated that mini-games help simplify difficult concepts and make use of human intuition and pattern recognition to support problem-solving. In the context of TalaBaybayin, incorporating mini-games such as drag-and-drop activities or character quizzes can provide students with interactive practice opportunities. These activities not only sustain learner motivation but also strengthen their mastery of Baybayin characters in an enjoyable and meaningful way.

Multimedia Sounds. Multimedia refers to the integration of different forms of media such as text, symbols, images, audio, video, and animations leveraged through technology to improve comprehension and retention. When properly implemented, these elements create interactive learning experiences that enable learners to absorb and remember information more effectively (Abdulrahaman et al., 2020). In this context, Jian-hua and Hong (2012, as cited in Abdulrahaman et al., 2020) highlighted the advantages of employing multimedia tools in teaching Physics, a subject that many students find challenging. Their research demonstrated that incorporating graphs, text, sound, and images through multimedia technology improves the presentation of information, making lessons more engaging and enabling students to learn actively through multiple sensory channels.

In line with these findings, the use of background music in TalaBaybayin is intended to create an immersive learning environment that maintains user focus and interest throughout the learning process.

Common Phrases. To translate whole phrases or groups of words, but not individual characters, Alcantara et al. (2024) constructed A machine learning model designed for transliterating groups of Baybayin words into their Tagalog equivalents. This innovation demonstrates that phrase-based learning could be incorporated into digital tools and help to master the script more efficiently. With the ability to transliterate at the phrase level, students are able to practice more frequent Tagalog phrases in Baybayin, including greetings, polite phrases, and simple conversational lines. This allows the users to learn the script within meaningful language contexts, which makes the learning more practical and interesting. By including these typical expressions in the TalaBaybayin application, it is possible to improve reading fluency, writing, and cultural understanding by manipulating the transition between the study of the traditional script and its practical application in real life.

Completion Certificate. The integration of certificates plays an essential role in encouraging learners to actively participate in the content and finish their courses. Shaikh, Rehman, and Rehman (2020) conducted a study that examined learners' perceptions in Massive Open Online Courses, focusing specifically on the motivational effects of obtaining certificates. The findings indicated that participants uniformly acknowledged that MOOCs facilitated lifelong learning and research opportunities, while also serving as a motivator through certificate issuance. The research demonstrated a significant correlation between learners' intentions to obtain a certificate and their actual achievement of that certificate, indicating that receiving a certificate is a crucial motivator for learners to complete MOOCs. The study identified goal orientation and grit as significant predictors of course completion, with grit serving as a robust independent predictor that influences course completion similarly to the intention to complete. The study demonstrates a significant correlation between the intention to obtain a certificate and the successful acquisition of one, indicating that certificates function as essential motivational instruments in the educational process. These features correspond with the design of Talabaybayin, which motivates students to remain dedicated and strive to achieve their educational objectives. The provision of a certificate upon completion serves as a powerful incentive to continue learning and a tangible acknowledgment of their achievement, ensuring that users not only complete the lessons but also experience a sense of accomplishment in their journey of learning.

Integration of Specialized Features in Mobile Learning Applications

In this section, the researchers discussed the integration of specialized features that enhance the user experience and learning process in mobile applications. These features are designed to support interactive and engaging learning experiences by incorporating essential tools that facilitate ease of use, accessibility, and retention. The discussion highlights core functionalities such as custom input methods, interactive exercises, and real-time feedback mechanisms, informed by insights from related literature and best practices in mobile-assisted learning.

The implementation of the proposed Writing Feature utilizes React Native Skia, a cross-platform 2D graphics library that enhances rendering capabilities within the React Native ecosystem, ensuring smooth performance across iOS, Android, macOS, Windows, Linux, and web browsers (Friyia, 2023). Over the past two years, Shopify Engineering has actively contributed to its development, ensuring smooth integration into React Native and expanding its capabilities for developers seeking highly interactive and visually rich applications. A key strength of React Native Skia is its optimization for dynamic user interactions, offering greater flexibility than the standard React Native View component, which is limited to rendering basic geometric shapes. Unlike traditional rendering methods, Skia allows developers to manipulate complex graphical elements, ensuring smooth animations while maintaining optimal performance and a stable frame rate through its seamless integration with react-native-reanimated (Friyia, 2023).

In addition to its advanced rendering capabilities, React Native Skia is built on a strong foundation of quality assurance. Candillon (2022) highlights that the library, developed using TypeScript and C++, undergoes extensive static code analysis and end-to-end testing to ensure consistent rendering across different platforms. By comparing drawings and Skia object outputs, the test framework identifies platform-specific variations, such as font rendering differences, contributing to the library's reliability. Initially tested within Node.js, the framework was progressively expanded to include mobile platforms, significantly enhancing React Native Skia's stability and performance. These advancements position React Skia as a powerful and efficient tool for applications requiring complex custom drawings and high-performance graphics, making it particularly well-suited for the proposed Writing Feature in this study.

In addition to this feature, the proposed system also introduces a custom Baybayin keyboard, designed to enhance the accessibility and usability of the Baybayin script in digital communication. The integration of custom keyboards in mobile applications has played a crucial role in improving user input efficiency, particularly in the preservation and promotion of indigenous writing systems. As digital communication continues to evolve, the need for specialized keyboards that support non-Latin scripts has become increasingly evident.

In this context, Weerakoon (2023) examined the development of a custom mathematical keyboard within a React Native environment, showcasing the effectiveness of dynamic rendering techniques and state management in facilitating a seamless and intuitive user experience. By implementing a system that allows smooth transitions between standard and domain-specific keyboard interfaces, the study demonstrated the potential of custom keyboards in improving accessibility and usability for specialized input methods. Building on these principles, the proposed TalaBaybayin application seeks to modernize and facilitate the use of the Baybayin script through a custom keyboard developed in React Native. By leveraging React Native's capabilities, the study focuses on optimizing state management and implementing a user-friendly interface to ensure a seamless and efficient input experience for Baybayin characters and letters. This project is part of efforts to add native writing systems to modern digital devices. It makes these scripts easier to use and helps keep cultural traditions alive.

Baybayin Keyboard. The integration of a Baybayin keyboard within the TalaBaybayin app significantly enhances user experience by enabling seamless typing in the Baybayin script. This feature allows users to practice writing and composing text directly in the app, fostering a more immersive learning environment. With an intuitive keyboard layout that mirrors the structure of the Baybayin script, users can easily access the necessary characters, promoting fluency and familiarity. Incorporating a Baybayin keyboard aligns with the increasing use of technology in language learning, providing learners with a practical way to apply their skills in real-world contexts.

Handwriting Feature. The inclusion of a writing feature in the TalaBaybayin app serves as a vital improvement designed to help users master the Baybayin script while deepening their appreciation of its cultural importance. This feature included interactive writing exercises, such as tracing activities to learn proper stroke order. By implementing this element, the TalaBaybayin app does not only promotes writing proficiency but also enriches users' appreciation for the Baybayin script, This creates a more engaging and immersive learning experience that aligns with the renewed interest in the Baybayin script, encouraging users to reconnect with their cultural heritage while enhancing their fundamental writing skills.

Text to Baybayin Translation. Since the proposed mobile application is designed to translate words into Baybayin, it aims to provide Filipino learners with an accessible tool to engage with and better understand this ancient script. Effective translation requires more than just word-for-word conversion; it must consider broader discourse, cultural context, and linguistic nuances. Naveen and Trojovský (2024) emphasize that machine translation must account for factors such as polysemy, idiomatic expressions, and ambiguous references to ensure accuracy and naturalness. Without contextual awareness, translations may fail to preserve the intended meaning, making it essential to integrate a system that processes linguistic structures carefully when converting Text into Baybayin. The creation of a mobile application for translating text into Baybayin aims to provide Filipino learners with an accessible tool for engaging with and understanding this ancient script. However, effective translation extends beyond direct lexical conversion and requires a comprehensive approach that considers cultural context, discourse structures, and linguistic nuances. As emphasized by Naveen and Trojovský (2024), machine translation must account for complexities such as polysemy, idiomatic expressions, and ambiguous references to ensure both accuracy and naturalness in the output. Without a contextualized processing system, translations may fail to convey the intended meaning, underscoring the necessity of integrating linguistic analysis and structural processing to enhance the quality and reliability of Text-to-Baybayin translation.

Text-to-Speech. In addition to the above-mentioned features, the proposed TalaBaybayin application integrated a Text-to-Speech (TTS) function to significantly improve both accessibility and user engagement. These features allow users to input text and hear its pronunciation. This process not only facilitates accurate linguistic translation but also supports multisensory learning, bridging oral and written communication.

TTS technology, also known as speech synthesis, enables computer systems to convert written text into spoken language, allowing users to receive audio feedback via output devices like speakers or phones. This is particularly beneficial for language learners and visually impaired users, as it promotes better pronunciation and understanding (Khanam et al., 2022). By integrating TTS, TalaBaybayin creates a comprehensive language tool that helps users learn, practice, and preserve the old Baybayin script. It offers interactive, voice-driven experiences that connect present Filipino speech to ancient literature, making the process both simple and culturally significant.

Utilizing a Software Quality Standard Model in Evaluating Mobile Applications

The ISO/IEC 25010:2023 provides a complete quality model that functions as the fundamental measurement in the Systems and Software Quality Requirements and Evaluation (SQuaRE) series. ISO/IEC 25010:2023 delineates eight primary quality characteristics, namely Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility, and Safety. Each main quality characteristic includes specific sub-characteristics that measure product assessment; for instance, Functional Suitability contains Functional Completeness, Functional Correctness, and Functional Appropriateness, while Performance Efficiency includes Time Behavior, Resource Utilization, and Capacity assessment.

According to Jadaone et al. (2023), Aralin Baybayin is an Intelligent Tutoring System (ITS) designed to help individuals such as historians, foreigners, and Filipinos learn the Baybayin script. The study, which included 130 participants, evaluated the system's quality and user experience using the ISO 25010 standard. Results indicated that the system provided a motivating and satisfying learning experience through well-structured, interactive lessons. It also demonstrated high efficiency in optical character recognition, achieving a training accuracy of 99.16% and a testing accuracy of 98.60%, successfully transliterating 412 out of 450 words for an overall accuracy of 91.55%. The researchers concluded that *Aralin Baybayin* is an effective platform for promoting efficient and interactive Baybayin learning.

Building on this foundation, ISO/IEC 25010:2023 provides an essential evaluation method that modern software development requires for complete assessments. The standard directs professionals through various tasks from requirements collection to the formulation of design objectives, testing, quality tracking, and configuration of acceptance criteria. By enabling organizations to detect weaknesses, the quality model facilitates informed decision-making aimed at enhancing software product quality. In today's dynamic ICT atmosphere, where system performance, security, and usability are top priorities, this standard's detailed categories and sub-characteristics allow for a thorough evaluation of multiple operational aspects, ensuring that both functional delivery and maintainability are rigorously measured.

This evaluation approach was directly incorporated into the TalaBaybayin project. TalaBaybayin developers adopted the requirements from ISO/IEC 25010:2023 in establishing the fourth evaluation objective of their mobile learning app. Here, the assessment of acceptability requires strict measurement based on the quality characteristics identified in the standard. For example, under Functional Suitability, the evaluation checks whether TalaBaybayin contains complete educational functionality, including interactive components, quizzes, and gamification features that promote user involvement. Furthermore, the examination of Performance Efficiency analyzes the app's speed and resource conservation across different mobile platforms, while Compatibility and Interaction Capability assess how well the app functions across various environments and provides an intuitive interface. This connection between the standard and its implementation ensures that the app not only fulfills educational objectives but also upholds stringent quality benchmarks.

Moreover, enterprises must assess additional dimensions such as Reliability, Security, Maintainability, Flexibility, and Safety according to the ISO/IEC 25010:2023 criteria. In this context, the performance of TalaBaybayin under varying operational conditions is closely examined to ensure it is protected against potential threats and can be maintained and adapted to evolving educational requirements. This systematic approach allows all stakeholders, including developers, educators, quality assurance personnel, and independent evaluators, to objectively measure the application's ability to fulfill both its functional and nonfunctional requirements. As a result, the app experiences sustained improvement, creating an environment that ensures better delivery of services to its targeted users.

In conclusion, the evaluation of TalaBaybayin in accordance with the requirements established by ISO/IEC 25010:2023 is directly in line with the purpose of developing an interactive mobile learning system that satisfies exceptional quality standards.

Related Systems

In this section, the researchers reviewed the related studies and research related to the proposed mobile learning app in mastering and preserving Baybayin.

eBaybayMo: An E-Learning Mobile Application Tool for Transliterating Baybayin Characters to Latin Letters using k-NN Algorithm

The development of eBaybayMo by Perin et al. (2024) marked a notable advancement in character recognition, especially for transliterating Baybayin script into Latin letters. This mobile application utilizes the k-Nearest Neighbors (k-NN) algorithm to improve the accuracy and efficiency of identifying Baybayin characters from images captured on mobile devices. By integrating machine learning, eBaybayMo improved both the precision and accessibility of Baybayin translation, making it a valuable resource for individuals seeking to learn and preserve this indigenous writing system. Moreover, the application contributes to ongoing efforts to revitalize traditional scripts, ensuring their continued relevance in modern digital platforms.

To evaluate its effectiveness, Perin et al. (2024) conducted a series of tests prior to the public release of eBaybayMo. They used a 2x2 confusion matrix to assess the model's capability in recognizing primary Baybayin characters, achieving a notable accuracy rate of 95.47%. These results underscore the considerable potential of machine learning–based character recognition in supporting language preservation and education. Based on their findings, the researchers recommended integrating eBaybayMo into educational institutions that teach Baybayin, as the application could serve as an innovative learning tool to enhance student engagement and comprehension.

The recommendation to implement eBaybayMo in academic settings aligns with broader efforts to incorporate cultural heritage into modern education. As future research explores its impact on students' learning efficiency and academic performance, eBaybayMo could pave the way for further technological advancements in the preservation of endangered writing systems. By connecting traditional scripts with modern digital technologies, this application holds the potential to strengthen cultural identity while promoting innovative methods of language learning.

Similarly, the proposed study on TalaBaybayin, an educational mobile application for learning and preserving the Baybayin script, shares the same objectives as eBaybayMo. Both applications seek to leverage technology to enhance the recognition, transliteration, and accessibility of Baybayin, ensuring its survival in the digital age. Through these advancements, they contribute to the broader mission of revitalizing and preserving indigenous writing systems while making them more accessible to modern learners.

bAI-bAI: A Context-Aware Transliteration System for Baybayin Scripts

The study by Bernardo and Estuar (2025) focuses on creating a context-aware transliteration system for Baybayin scripts, addressing the challenge of word ambiguity through the use of word embeddings and part-of-speech tagging. Utilizing the bAI-bAI system, the researchers enhanced Baybayin transliteration by integrating a disambiguation technique that identifies the most contextually accurate Latin equivalent. The study compared two approaches: a conventional method employing word embeddings with part-of-speech filtering, and an advanced approach utilizing a large language model, the Generative Pre-trained Transformer (GPT-4o mini). Results showed that while GPT-based disambiguation achieved higher accuracy, the traditional method was significantly faster. The study of Bernardo and Estuar provides the researchers with a deeper understanding of how an AI-powered transliteration system can significantly improve Baybayin recognition and processing. It proves that a well-developed system not only aids in preserving the Baybayin script but also enhances accessibility and usability through the application of modern techniques in natural language processing.

This study aims to improve the recognition and processing of ancient scripts by developing a mobile learning application that enables more efficient and precise transliteration.

Enriching Students' Vocabulary through Quizlet in Reading Class

The study by Pitriani et al. (2024) examines the effectiveness of Quizlet as a gamified platform for vocabulary improvement in reading classes, emphasizing its ability to increase student engagement through interactive and dynamic learning methods. The platform incorporates flashcard-based learning with progress tracking, allowing students to focus on mastering challenging words through a spaced repetition algorithm that reinforces retention. Additionally, Match Mode and Live Mode introduce speed-based challenges and collaborative gameplay, promoting active recall, competition, and teamwork to further support vocabulary acquisition. Although the Gravity game was discontinued, it previously strengthened rapid word association through time-constrained exercises.

While Quizlet does not include an official badge system, it has a reward-based learning structure, enabling students to track progress, improve quiz times, enhance accuracy, encourage self-motivation, and sustain learning engagement.

These gamified elements collectively transform vocabulary acquisition into an interactive, engaging, and dynamic process, fostering higher motivation and improved retention rates among students. The study by Pitriani et al. closely aligns with the researcher's proposed mobile learning application for Baybayin, as both emphasize the integration of gamification and digital tools to enhance language acquisition, student engagement, and interactive learning experiences. By incorporating similar game-based learning strategies, the proposed Baybayin application aims to facilitate an engaging and effective approach to learning the ancient script, mirroring the positive impact observed in Quizlet-based vocabulary instruction.

Learning Management Systems (LMS) Towards Helping Teachers and Students in the Pursuit of Their E-Learning Methodologies

This project seeks to enhance the educational experience for both teachers and students by providing a well-organized platform that facilitates effective e-learning practices. While the primary objective of LMS is to support a wide range of online learning environments, its principles can be relevant to systems like Aralin Baybayin and gamified language learning systems.

Learning Management System (LMS) platforms provide essential tools for creating, managing, and delivering educational content, making them indispensable in the digital era. These features encompass course management, assessment tools, and communication channels that foster meaningful interaction between educators and learners. This functionality supports a collaborative learning environment, which is especially advantageous for language acquisition systems that depend on active participation and engagement. Furthermore, incorporating gamification elements within an LMS can significantly enhance the overall learning experience. By integrating game mechanics inspired by frameworks such as Octalysis, LMS platforms can encourage users to engage more deeply with the curriculum. This approach caters to diverse learning styles and preferences, helping learners better understand and retain complex language concepts.

Overall, the learning management system (LMS) approach advances the objectives of language education by providing a flexible and adaptive framework for both teachers and students. By utilizing technology to enhance e-learning practices, LMS can play a pivotal role in the success of language programs such as Aralin Baybayin and other innovative educational tools (Lee, 2024). These systems enable personalized learning experiences, allowing educators to tailor content to the diverse needs of learners. The integration of data analytics within LMS supports continuous assessment and feedback, which is crucial for monitoring student progress and refining instructional strategies. The adaptability of LMS promotes a more engaging and effective learning environment, empowering students to take an active role in their language learning journey.

Technology-Enhanced Learning Environments: Improving Engagement and Learning

This project emphasizes the role of technology in creating engaging and effective educational experiences. This concept is highly relevant to systems like Aralin Baybayin and gamified language learning frameworks, as it explores how technology can enhance learner engagement and facilitate better learning outcomes.

Technology-enhanced learning environments utilize various digital tools and resources to support instructional strategies, making learning more interactive and immersive. Features such as multimedia content, interactive simulations, and collaborative platforms help to capture learners' attention and encourage active participation. These elements are particularly beneficial for language learning systems, where engagement is crucial for mastering new concepts and skills.

Furthermore, the integration of gamification techniques within these environments can significantly boost motivation and retention. By applying game mechanics to educational content, such as rewards, challenges, and social interactions, technology-enhanced learning environments can create a dynamic atmosphere that fosters enthusiasm for learning. This approach closely aligns with frameworks such as Octalysis, which seek to enhance user engagement through the strategic application of gamification principles.

Overall, the principles of technology-enhanced learning environments provide important guidance for creating effective language learning systems. By leveraging technology to increase engagement and improve learning outcomes, educational tools can better meet the diverse needs of learners and support successful language acquisition (Duterte, 2024). Incorporating interactive features and real-time feedback further enriches the learning experience, making it more personalized and adaptable to individual learner needs

Transformation of Student Motivation Through the Use of Quizizz in Language Learning

The project titled "Transformation of Student Motivation Through the Use of Quizizz in Language Learning" highlights the important role of interactive technology in boosting student motivation and engagement in language education. This is especially relevant to platforms like Quizizz, which use gamified elements to create a more dynamic and engaging learning environment.

Quizizz incorporates features like real-time quizzes, instant feedback, and gamification strategies to create an engaging and enjoyable educational experience. By enabling students to participate in activities that are both competitive and collaborative, the platform effectively captures learners' attention and encourages active participation in the learning process. This approach is especially valuable in language learning contexts, where maintaining motivation is essential for developing proficiency in vocabulary and grammar.

Moreover, incorporating gamification elements in Quizizz such as points, leaderboards, and customizable avatars substantially enhances student motivation and knowledge retention. These game mechanics create an enjoyable and supportive learning environment that encourages enthusiasm, aligning with educational theories emphasizing the role of engagement and enjoyment in effective learning.

Overall, the principles underlying the use of Quizizz in language learning offer meaningful insights for the creation of effective educational tools. By harnessing technology to enhance motivation and engagement, platforms like Quizizz can better accommodate diverse learner needs and contribute to more successful language acquisition (Munirah & Sahriani, 2024).

A Game-Based Learning System Based on the Octalysis Gamification Framework to Promote Employees' Japanese Learning

The project utilizes the Octalysis gamification framework to improve Japanese language learning among employees. This system aims to increase engagement and motivation by incorporating game mechanics that provide a fun and interactive learning experience. With features like rewards, challenges, and social interaction opportunities, the system effectively encourages active participation and maintains learners' interest throughout the language learning process.

A key feature of this learning system is its gamification strategy, which employs the Octalysis framework to target both intrinsic and extrinsic motivation. By integrating game design principles with educational goals, the system offers an immersive experience that sustains learner motivation and focus. This approach not only enhances language skill retention but also cultivates a sense of community and collaboration among learners.

Additionally, the game-based learning system provides immediate feedback and personalized learning paths, allowing employees to progress at their own pace according to their individual needs. This customized approach is crucial for addressing diverse learning styles and proficiency levels, ensuring that all users gain maximum benefit from the training.

Overall, this system illustrates the effectiveness of combining gamification and language learning, showcasing how engaging methodologies can facilitate the acquisition of new skills in a corporate setting. By integrating game mechanics with structured language instruction, it serves as a valuable reference for systems like Aralin Baybayin and other educational tools aimed at enhancing language proficiency (Chen et al. 2023).

Duolingo: An Enchanting Application to Learn English for College Students

This project examines the effectiveness of Duolingo as a language learning platform specifically designed for college students. The authors emphasize how the application utilizes technology to provide an interactive and engaging learning experience, which plays a crucial role in maintaining student motivation. By integrating gamified elements such as rewards, levels, and instant feedback, Duolingo transforms the often-challenging process of learning a new language into an enjoyable experience. This approach not only increases user engagement but also encourages consistent practice, an essential factor in mastering and retaining new language skills.

The study highlights the crucial role of technology in contemporary education, especially in language learning, where traditional methods often fail to sustain student engagement. Duolingo is thoughtfully designed to help students read, write, listen, and speak through a structured yet flexible approach. This adaptability is particularly valuable for university students with diverse learning styles and schedules. By delivering content in manageable, bite-sized lessons that students can complete at their own pace, Duolingo effectively addresses multiple student needs, making language learning more accessible and convenient than ever.

In addition, the authors explain how Duolingo's focus on functional use of language, through practice in translation and contextual learning, closes the gap between theory and practice. This not only makes vocabulary learning easier but also enhances grammatical ability, allowing learners to apply their skills in practical situations. The instant feedback provided by the app also greatly aids the learning process, as learners can identify and correct errors quickly, feeling a sense of achievement and momentum.

The findings of the research conducted by Permatasari and Aryani (2023) on Duolingo shed light on the growing role of technology in the process of educational language acquisition. In addition to boosting student motivation, Duolingo also increases engagement, thereby transforming the process of language learning into an experience that is both enjoyable and productive. This is accomplished through the strategic combination of gamification and interactive components. This approach is representative of contemporary education practices that place an emphasis on student-centered experiences. It highlights the ways in which websites such as Duolingo may cater to the diverse requirements of college students and encourage efficient language learning in the digital world of today.

Quizlet as a Learning Tool for Enhancing L2 Learners' Lexical Retention: Should It be Used in Class or at Home?

According to Nguyen and Van Le (2022), the rapid technological advancements of the twenty-first century have significantly transformed teaching and learning practices, particularly in second language acquisition. Numerous applications and software have been created to support and enhance learners' acquisition of L2 vocabulary. The study aimed to assess the effectiveness of Quizlet in improving lexical retention among EFL students at a private institution in Vietnam under two learning conditions: at home and in the classroom. A total of eighty-nine university students participated and were divided into three groups.

In the digital age, technology has become a fundamental component of Second Language Acquisition (SLA) through methods like Computer-Assisted Language Learning (CALL) and Mobile-Assisted Language Learning (MALL), both of which effectively aid in vocabulary retention. Tools like Quizlet have emerged as valuable resources for helping learners expand their L2 vocabulary. As a free platform accessible via both web and mobile applications, Quizlet has demonstrated its effectiveness in motivating learners and enhancing their lexical development.

Despite its promise as a learning tool, previous studies have largely concentrated on learners' perceptions of Quizlet usage, with limited empirical research examining its actual effectiveness. Moreover, most existing studies have relied on quasi-experimental designs—some even lacking control groups—and only a few have adopted mixed-methods approaches. Notably, none of these studies seem to have utilized contemporary statistical methods, such as the Generalized Linear Mixed Model (GLMM), for data analysis to generate more reliable and valid results.

Furthermore, there has been no comparative research exploring learners' attitudes toward using Quizlet in two different settings: in-class under teacher supervision and at home through independent learning. This highlights a notable research gap that calls for further investigation into Quizlet's role in L2 vocabulary acquisition. Addressing this gap requires a comprehensive study examining how these two learning modes affect learners' retention, engagement, and overall vocabulary development. This research gap emphasizes the need for a detailed comparison of the effects of these learning modes on vocabulary retention and learner engagement.

Kahoot! as a Gamified Learning Platform: Enhancing the Exam Experience for Students

According to Niza and Ardian (2022), the integration of educational technology has greatly transformed traditional learning methods, particularly using gamified platforms like Kahoot! Their study explores the effectiveness of Kahoot! in improving the examination experience by introducing fun and interactive elements into the learning process. The research includes a review of existing literature on web-based learning tools, with a particular emphasis on the application of Kahoot! in various educational contexts.Kahoot!

and other such applications provide instructors with novel approaches to student engagement as digital technologies continue to advance. Kahoot! is a gamified platform that is extensively used and provides educators with the ability to administer quizzes and examinations in a manner that is like that of a game to students. This strategy tries to improve student engagement, retention, and motivation by utilizing certain strategies. Kahoot! stands out as a contemporary tool that is especially useful for facilitating learning and assessment because of its visually appealing and dynamic design.

While Kahoot! offers a dynamic and engaging alternative to conventional testing methods, its effectiveness largely depends on factors such as reliable internet connectivity, access to compatible devices, and the users' level of technological proficiency. These technological requirements can limit their effectiveness in certain environments, particularly in underprivileged regions where resources may be scarce. Despite these constraints, Kahoot! has proven to be a valuable tool within the broader ecosystem of gamified learning platforms, offering educators and students a new way to experience exams and quizzes through interactive and engaging methods. Integrating elements of competition, immediate feedback, and collaboration, it transforms assessment into a more stimulating and participatory activity.Its accessibility across different devices further enhances its usability, making it convenient for both classroom and remote learning environments. Additionally, the platform's customizable quizzes enable educators to tailor assessments to meet specific learning objectives, fostering a more adaptive and student-centered approach to education.

To gain a deeper understanding of how Kahoot! compares to other platforms in enhancing learning outcomes and promoting student engagement across different In various educational settings, including both traditional classrooms and remote learning environments, further research is necessary. When educators gain a clearer understanding of the contexts in which Kahoot! is most effective, they can make more informed decisions about adopting the software. Stakeholders can maximize the benefits of this innovative tool by first identifying best practices, addressing potential challenges, and providing educators with the appropriate training.

Conceptual Framework

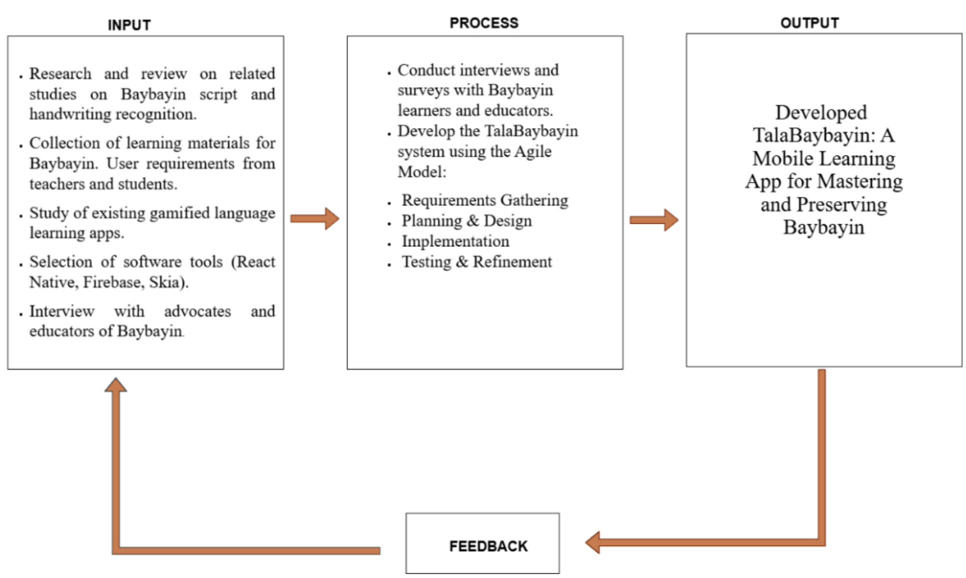
Conceptual Framework of the TalaBaybayin Application. Figure 1 illustrates the conceptual framework employed to depict and explain the overall flow of the study. The input phase serves as the foundation, beginning with data collection and requirement analysis. During this stage, the research team identified the existing challenges in teaching and learning Baybayin through observations, interviews, and surveys conducted with educators. The results of these activities validated the difficulties faced in Baybayin education and informed them of requirements required for the development of the TalaBaybayin mobile learning application.

Additionally, the researchers reviewed related literature and studies to examine existing digital learning tools, gamification strategies, and best practices in language preservation. To ensure the system's proper and efficient implementation, the study also evaluated various software development technologies, including React Native, Firebase, and Skia. Furthermore, different handwriting recognition techniques and custom keyboard integrations were explored to ensure the system could effectively support learners in accurately mastering the Baybayin script.

The process phase centered on the development of TalaBaybayin using the Agile methodology, which enabled iterative improvements based on continuous user feedback. The process began with identifying essential features such as stroke-based handwriting recognition and a custom Baybayin keyboard. The application was built using React Native and React Native Skia to support handwriting functionalities, while Firebase handled data management and user authentication. Testing was carried out with both students and teachers, guided by the ISO/IEC 25010 quality model, which informed the refinement process to improve the app's usability, functionality, and overall performance. The output phase presents TalaBaybayin, a mobile learning app designed to help Filipino learners master and preserve Baybayin. It offers an interactive way to learn through handwritten exercises, quizzes, and guided lessons. The app also includes gamified features for engagement, a custom Baybayin keyboard, and real-time handwriting feedback to improve accuracy.

Figure 1

Conceptual Framework of Talabaybayin application



CHAPTER III

TECHNICAL BACKGROUND

This chapter describes the methodological approaches, study population, research instrument, data processing, testing, and project design implemented to accomplish the objectives of the research.

Research Methodology

The research employed a mixed method to address the challenges faced by Filipino grade 7 high school students in learning and preserving the Baybayin script. Introduced in the late 1970s, mixed methods research integrates both quantitative and qualitative data collection in one study (Lall 2021). Creswell (2014) described mixed methods research as a strategy of inquiry that combines or correlates both qualitative and quantitative forms. Additionally, Battista & Torre (2023) explain that mixed methods incorporate both qualitative and quantitative approaches to achieve an in-depth understanding of complex phenomena. In uniting the strong points of both methodologies, researchers can triangulate data, enhance the validity of findings, and generate deeper insights into research inquiries. This framework facilitated the integration of varied viewpoints, enabling the collection of contextually rich qualitative data alongside robust statistical analysis, thereby strengthening the study's overall reliability.

As noted by Berman (2017), the exploratory sequential mixed-methods design consisted of an initial qualitative phase of data collection and analysis, followed by a quantitative phase, and a final phase of integrating data from the two strands. As outlined in the well-known typology of Creswell et al. (2011), mixed methods research includes three fundamental designs: the convergent, explanatory sequential, and exploratory sequential approaches. According to Curry and Nunez-Smith (2015), the exploratory sequential design starts with qualitative data collection before moving into the quantitative phase. The exploratory sequential design is employed to inform the development and administration of a new survey instrument and to interpret the two sets of results in conjunction. Furthermore, the exploratory sequential design supports the creation of a new survey instrument and the joint interpretation of qualitative and quantitative data. It enables the quantitative results to validate or generalize the initial qualitative findings (Plano et al., 2015). In this context, integration conducted at the methods level is known as building (Onwuegbuzie et al., 2010). Building takes when the findings from one data collection procedure guide the approach of the other method (Fetters et al., 2013).

The exploratory sequential design begins with a strong emphasis on qualitative research principles, followed by the integration of quantitative methods to provide a comprehensive analysis. Qualitative research investigates human experiences and social phenomena using non-numerical data, such as observations, interviews, and textual analysis (Dahal et al., 2024). Integrating qualitative methodology into the development of the TalaBaybayin mobile learning application facilitated a thorough understanding of user experiences and cultural contexts. This approach is essential for promoting the learning and utilization of the Baybayin script.   Employing these methods enables the creation of an application that is not only user-centered but also culturally attuned, thereby enhancing user engagement and educational efficacy. Additionally, these qualitative techniques aid in identifying recurring patterns and themes within user interactions and feedback, which can inform the application's design and functionality to better meet user needs. The insights that the researchers gained from using the qualitative method guided the subsequent development of the application itself, which utilized an agile methodology within the Software Development Life Cycle (SDLC) framework to iteratively build and refine features based on user needs. This study employs purposive sampling to select 70 Grade 7 students from Caniogan National High School who have directly used the instructional module in class. Although the sample comes from a single school, it is assumed that the learning contexts and module implementation are sufficiently similar across the four schools involved, making these students a meaningful representation of the broader Grade 7 population. To ensure diverse perspectives, participants are selected from different sections within the school.

A four-point Likert scale survey is used to assess the degree of difficulty students face with various aspects of Baybayin, including character recognition, writing fluency, and cultural comprehension. The insights gathered help identify specific learning challenges and inform the design and instructional strategy of the TalaBaybayin app. Participants are chosen for their direct experience, and their feedback is used to develop insights rather than for statistical generalization.

explain that the SDLC provides a structured methodology for systematically progressing through the phases of planning, designing, developing, testing, releasing, and maintaining software solutions. The implementation of this framework in the development of TalaBaybayin ensured that each phase aligns with user requirements, beginning with the creation of an intuitive interface, followed by iterative development and testing informed by continuous user feedback. This structured approach ensured the application's effective development, incorporating feedback to meet the needs of Filipino grade 7 high school students and serve as a sustainable, high-quality learning tool for Baybayin instruction.

Building on the insights gained during the qualitative phase, the following stage of this exploratory sequential design used quantitative research tools to enhance, validate, and strengthen the initial findings. Quantitative approach allows for the systematic collection and analysis of numerical data to find patterns and trends in the development process (Dahal et al., 2024). Surveys and assessments measured crucial characteristics such as user satisfaction, learning efficacy, and grade 7 high school students' opinions of Baybayin learning. The researchers used descriptive analysis, specifically Measures of Dispersion (Variability), to synthesize and organize the data they collected. Cooksey (2020) defines descriptive statistics as the application of statistical tools to arrange and present data. This process organizes data into visual or numerical summaries to identify trends and irregularities. He also outlines the dispersion (variability) metrics used to calculate the range, variance, and standard deviation. This is used to evaluate the TalaBaybayin app, with an emphasis on user engagement, learning outcomes, and error rates. Prioritize upgrades to maintain a consistent learning experience while making data-driven decisions to improve app usability and performance. Metrics such as the interquartile range for feedback and performance variance might assist in uncovering critical issues. As these statistics are purely descriptive, they summarize and characterize the dataset without making inferences about the broader population. The quantitative analysis gives empirical information to help improve the application's user interface and instructional design, ensuring that aspects that contribute to increased engagement and learning outcomes are emphasized. Furthermore, statistical analysis of grade 7 high school student performance data provided objective insights into the application's efficacy, directing evidence-based changes. To create a well-structured and proficient learning tool, the TalaBaybayin application was developed using both qualitative and quantitative methodologies that include user behavior and educational impact.

To guarantee the quality of the developed mobile application, the researchers gathered feedback from respondents to assess its acceptability. A total of 10 teachers and 70 Grade 7 high school students from Caniogan High School participated in the evaluation. The respondents are selected to represent a range of teaching and learning experiences related to the ancient script. This selection ensures that Grade 7 students, teaching professionals, and Baybayin enthusiasts within the school who have direct experience with learning and teaching the script can provide significant observations on the system's effectiveness and acceptability in delivering interactive learning through its essential features.

In addition to the end-users of the system, the researchers employed expert sampling, a type of purposive sampling, to select the remaining respondents for the study. Expert sampling is deemed appropriate as it ensures the participation of individuals with specialized knowledge and expertise in system development, enabling an in-depth analysis evaluation of the developed application.

Through this sampling method, the researchers selected 10 IT professionals from Bulacan State University (BulSU). These professionals included instructors from the College of Information and Communication Technology (CICT) who specialize in software development-related disciplines, such as computer programming, web development, and database management. Their expertise in software development makes them highly suitable for assessing the acceptability and technical quality of the developed system.

Furthermore, the researchers engaged 10 Information Technology (IT) experts from the industry to evaluate the system. These experts included software and web developers, software engineers, and software quality assurance testers. The developers and engineers assessed the system based on programming standards and software development best practices, providing valuable insights and recommendations to enhance its overall quality. Additionally, the software quality assurance testers evaluate the system's functionality and usability, identifying potential logical errors and glitches to confirm that the application satisfies the industry standards for software performance and reliability. Their collective expertise contributed to a more robust assessment, addressing both backend efficiency and user-centric design considerations. This multifaceted feedback is instrumental in refining the system to align with current industry benchmarks and user expectations.

Table 1 shows the respondents and their corresponding frequency for the study.

Table 1

Respondents of the Study

	Respondent	Frequency (N)	Percentage (%)
Targeted User	Grade 7 Students	70	57.13
	High School Teachers	10	14.29
Technical Experts	IT Experts from the Industry	10	14.29
	IT Professionals of BulSU	10	14.29
Total		100	100

The mobile learning application was evaluated by a total of 100 respondents. Among them, 70 Grade 7 high school students from Caniogan High School served as the primary end-users and had the highest frequency of use in assessing the application's effectiveness in delivering interactive learning. In addition, 10 Filipino teachers from the same school also evaluated the system based on their teaching experience, though they have a lower frequency of interaction. Furthermore, 10 IT professionals from Bulacan State

University (BulSU), specifically faculty members from the College of Information and Communications Technology (CICT), assessed the application’s compliance with software standards. Lastly, 10 IT experts from the industry, including developers, engineers, and quality assurance testers, evaluated the application's functionality, efficiency, and usability.

A survey was conducted among teachers, Grade 7 students, university staff, and IT experts to assess the acceptability of the Baybayin mobile learning application. Using the ISO/IEC 25010:2023 model and a four-point Likert scale, the data were analyzed quantitatively to evaluate the application’s effectiveness, usability, and overall acceptance.

A four-point Likert scale was employed in the questionnaire to evaluate the acceptability of the developed system, using the ISO/IEC 25010 Software Quality and Acceptance Model. The scale ranges from 1 to 4, where 1 indicates Not Acceptable and 4 indicates Highly Acceptable. This method provides a more detailed understanding of user perceptions, allowing researchers to assess the system’s effectiveness and quality based on user feedback. Using this method, the study aims to identify areas for improvement and ensure that the system meets user expectations and needs. Additionally, the insights gained from these responses informed future developments, promoting continuous enhancements that better align with user preferences and enhance overall satisfaction. This same four-point Likert scale format was applied in the preliminary survey conducted with 70 students from Caniogan High School to assess the level of difficulty they experience in learning the Baybayin script. Using the scale, 1 represents Not Difficult and 4 represents Highly Difficult, allowing the researchers to quantify and interpret student struggles in a structured manner. The absence of a neutral midpoint encourages respondents to make a definitive judgment, thereby reducing ambiguity in responses. This design choice enhances the clarity and actionability of the collected data, particularly in identifying which aspects of Baybayin learning require targeted instructional support.

Moreover, the questionnaire items used to evaluate the system's acceptability were based on its identified attributes and organized according to the major and sub-characteristics specified in the ISO/IEC 25010 Software Quality and Acceptance Model. This structured approach ensures that the evaluation captures a thorough view of the system's performance across various dimensions, such as functionality, usability, and reliability.

The respondents answered the survey by accepting or not accepting the context of the items using a scale of 1 to 4, where 1 indicates Not Acceptable and 4 indicates Highly Acceptable. Each item from the questionnaire interprets the degree of agreement in terms of the acceptance of the respondents based on the computed mean range, with higher mean values indicating greater agreement and acceptance among them. After computing the weighted mean, the mean is a measure of central tendency that indicates the average of a data set. It is calculated by summing up all the values in the data set and dividing by the total number of values. In the case of grouped data, each data value is multiplied by its corresponding frequency, and the sum of these products is divided by the total number of (Mendenhall, Beaver & Beaver, 2012).

To interpret the table, the researchers used both the mean and standard deviation. The standard deviation measures the amount of variation in a dataset and is calculated as the square root of the variance, which represents the average squared distance from the mean (Triola, 2018).

Both the ISO/IEC 25010 Software Quality and Acceptance Model acceptance survey and the preliminary survey for assessing difficulty used these measures to evaluate respondents' answers. In the ISO acceptance survey, the mean and standard deviation assessed the level of acceptance for each item. These measures showed how users accepted the software. A higher mean meant more satisfaction. A larger standard deviation showed varied opinions. These measures helped identify aspects requiring improvement. They highlighted strengths in the system. They pointed out weak areas. They guided further enhancements. They provided clear feedback for decision-making. Each item was described as follows:

Table 2 shows the scale and corresponding descriptive interpretation used to determine the respondents perceived level of difficulty in learning the Baybayin script.

Table 2

Scale and Descriptive Interpretation level of difficulty in learning the Baybayin script

Scale	Range	Descriptive Interpretation
4	3.00-4.00	Highly Difficult
3	2.00-2.99	Moderately Difficult
2	1.50-1.99	Slightly Difficult
1	1.00-1.49	Not Difficult

Table 3 shows the scale and descriptive interpretation of the system's acceptability.

Table 3

Scale and Descriptive Interpretation of the acceptability of the system

Scale	Range	Descriptive Interpretation
4	3.00-4.00	Highly Acceptable
3	2.00-2.99	Acceptable
2	1.50-1.99	Fairly Acceptable
1	1.00-1.49	Not Acceptable

Software Development Methodology

The researchers developed the system using the processes stated in the System Development Life Cycle (SDLC) while adopting the Agile software development method to produce the developed system.

Agile software development follows an iterative and incremental approach. This results in the researchers breaking down the development process into several iterations. Each iteration should be completed in a period based on the available time limit given to the research group. Agile methodology was chosen for TalaBaybayin due to its flexibility, iterative design, and user-driven improvements. Its incremental cycles enable continuous feedback and prompt changes in usability, content, and performance to ensure an effective learning process.

Figure 2 illustrates the process of software development, highlighting the evolution from traditional methodologies to more adaptive approaches. In 2001, 17 software developers, later known as the Agile Alliance, wrote the Agile Manifesto at Snowbird, Utah, to replace traditional Waterfall software development methods that relied on heavy documentation. At its core, the Agile Manifesto emphasizes valuing individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and adaptability over rigid planning. The software development principles support adaptive progress along with continuous customer participation (Agile Alliance, 2001). These principles emphasize flexibility, promoting iterative development cycles where feedback from customers and team members is integral.

Requirements. The researchers collected essential data through interviews, survey questionnaires, and observations. The main purpose of the interviews is to identify the challenges and needs Filipino learners may encounter in learning and mastering the ancient script.

Plan. After gathering all project specifications, the researchers create an explicit project outline to develop the TalaBaybayin application during the Plan phase. An initial step entails writing down features that are included in the application: interactive learning materials, practice activities, quizzes, tracking progress mechanisms, game elements, Baybayin typing abilities, writing practice sections, and Tagalog-to-Baybayin translation capabilities. The team arranges the required features into priority order according to their essentiality for initial development. The researchers examine all necessary resources, including technology tools, member competencies, and time, and identify potential risks before implementing risk mitigation strategies. The planning phase establishes fundamental elements for developing an application that offers educational solutions for Filipino grade 7 high school students and individual learners to conserve and restore the Baybayin script.

Design. The researchers began the design phase by developing the Data Flow Diagram (DFD) for the system, which facilitated the identification of data sources and their associated processes. After the researchers established the DFD of the system, the scholars created the Entity Relationship Diagram (ERD) for the system. The ERD helps the researchers to produce a concrete database structure for the system. Additionally, a Flow Chart diagram was utilized to visualize the required algorithm. Furthermore, the research team employed a Visual Table of Contents (VTOC) diagram to systematically map and organize the mobile application's content and navigational structure.

Develop. After the design phase, the team developed the TalaBaybayin mobile app using selected tools (see Chapter 2), implementing all core features and modules (see Chapter 1). Bugs were resolved through debugging, and user acceptance testing ensured usability and effectiveness.

User and stakeholder suggestions were implemented to improve functionality and user experience. The team used React Native Expo for app development, Firebase for authentication and data management, and Expo Router for navigation. Canvas components enhanced the interface, especially for Baybayin writing, with React Native Skia powering the handwriting module.

Firebase Firestore and Cloud Storage handled real-time data and scalability. By the end of development, all major features including authentication, interactive lessons, exercises, quizzes, progress tracking, a Baybayin keyboard, writing, and Tagalog-to-Baybayin translation were fully implemented and tested.

Release, Track, and Monitor. During these phases, the system operates as intended. Maintenance follows deployment and monitoring. These phases are excluded from the study, which documents only the testing and evaluation of the mobile learning application.

The research team actively incorporated feedback from users and stakeholders, integrating valuable suggestions to strengthen the app's usability and educational impact. This collaborative process encouraged continual learning and aligned technical innovation with real user needs.

Figure 2

SDLC using Agile methodology (Manifesto for Agile Software Development, 2001)

Requirements Analysis and Documentation

The researchers analyzed and documented the users' needs for the capstone project. The requirements gathered from the interviews and questionnaires with the grade 7 high school students are carefully examined and used to identify the system's features. The analysis assisted in identifying essential features, functionalities, and limitations that must be addressed throughout system development. By prioritizing these criteria, the researchers can guarantee that the final solution satisfactorily fulfills user expectations. This systematic method promoted a more user-focused design, improving usability and overall user contentment.

User Requirements

The user requirements for TalaBaybayin were determined through an analysis of the current challenges encountered by Filipino learners, educators, and Baybayin enthusiasts in accessing interactive and engaging learning tools. Guided by the literature review and preliminary surveys, the following outlines the identified problems, recommended features, and the system's software specification requirements. This comprehensive approach ensures that the application addresses the specific needs of its users, ultimately enhancing their learning experience. By incorporating feedback from the target audience, TalaBaybayin aims to provide a more effective and user-friendly platform for mastering the Baybayin script.

The development process prioritized user-centered design to guarantee the platform's intuitiveness and accessibility for learners at all levels. Ongoing testing and incremental enhancements informed by user input further enhance TalaBaybayin, making it a dynamic and adaptive educational resource. This iterative process ensures the app continuously evolves to meet the changing needs of its users, cultivating an interactive and efficient learning setting.v

Table 4 shows the problems encountered by Baybayin learners.

Table 4

Problems Encountered by Baybayin Learners

Problem	Description
Limited Access to Interactive Tools	Traditional materials (e.g., books and static PDFs) do not offer the dynamic, interactive elements necessary for modern learning.
Insufficient Guided Practice	Learners lack structured, hands-on exercises that provide immediate corrective feedback when practicing Baybayin writing.
Lack of Real-Time Assessment	There is a significant gap in tools that allow learners to monitor their progress and receive instant evaluations.

The identified problems led to system features that create an engaging and effective learning environment. With gamification elements and progress tracking, the app boosts motivation and helps grade 7 students learn more effectively. Using challenges and rewards, it encourages students to stay engaged while offering a sense of achievement. Tracking progress helps users to recognize their strengths and areas for improvement, making the learning experience more personalized. These features aim to enhance learning outcomes while preserving the cultural importance of the Baybayin script.

The interactive features make learning fun and cater to different learning styles. With a visually appealing design, the system keeps users engaged and makes learning more enjoyable. Real-time feedback encourages active participation and keeps students motivated. Its adaptable nature allows each student to progress at their own pace, promoting success for all. By encouraging ongoing learning and self-assessment, the system helps develop lifelong learning habits. The use of technology bridges the gap between traditional learning methods and modern educational approach, making cultural preservation more relevant to today's digital learners.

In practice, students responded positively to features that transformed lessons into interactive experiences rather than passive activities. Many learners expressed greater enthusiasm for mastering Baybayin because the app transformed what could have been routine drills into dynamic, game-like activities. The instant feedback made each accomplishment feel meaningful, and the app's progression system built confidence incrementally. As a result, the platform not only elevated academic performance but also helped foster pride in Filipino heritage, making language and script learning both modern and meaningful for a new generation.

Table 5 shows the recommended features for the system.

Table 5

Recommended Features

Feature	Description
Login/Sign-up	Users may create a new account or sign in with existing credentials, with support for Google and Facebook authentication.
Lessons	Interactive multimedia lessons about Baybayin follow a sequence of historical information, structural understanding, and usage techniques, integrating written material components validated by three Filipino teachers.
Leaderboard	The system features a dynamic leaderboard that ranks users based on their accumulated experience points (XP), quiz scores. Users can view their standing relative to others, which motivates them to complete more lessons, practice exercises, and improve their skills.
Time-Based Scoring	Users earn points based on both the speed and accuracy of their responses to questions in quizzes and assessments. Learners who answer correctly in a shorter amount of time are given additional bonus points, which encourages not only accuracy but also efficiency and active engagement in the learning process.
Quizzes	The platform offers carefully designed quizzes that measure user performance at different points while providing instant feedback and tracking learning development.
Progress tracking	A dashboard displays the user's acquired rank, earned badges, and completed lessons, providing an overview of their growth and achievements.
Gamification features	The integration of game elements introduces motivational challenges that encourage learners to engage in consistent practice. Competitive and enjoyable activities allow users to earn badges and advance in rank, fostering both skill development and sustained interest in the learning process.
Mini-games	The application incorporates mini-games designed to promote active engagement and reinforce learning. These games include a memory match game, a word scrambling game, a character quiz game, a pattern recognition game, and a drag-and-drop game.
Multimedia Sounds	The inclusion of background music and other audio materials enhances learner motivation, creates a richer learning environment, and makes the experience more enjoyable and fun.
Common Phrases	It is a feature that gives a list of commonly used Tagalog expressions, translated into Baybayin script. It allows users to rehearse reading and writing phrases rather than individual characters, and the learning experience itself is more practical, conversational, and culturally relevant.
Completion Certificate	This certificate serves as formal recognition of their dedication and achievement, reinforcing their commitment to continuous learning and academic progress.
Baybayin Keyboard	The system offers integrated typing tools that allow users to type in Baybayin script as well as in the standard Tagalog keyboard.
Handwriting Features	The system allows users to select a Baybayin character to practice writing on an interactive canvas. Users can manually trace the character, erase parts of their strokes, clear the entire canvas if needed, and save their current progress. The handwriting sessions saved can be reopened for further practice or refinement.
Tagalog to Baybayin Translation	The system includes a translation feature that allows users to input Tagalog words and automatically convert them into Baybayin script, and vice versa.
Text-to-Speech	The system integrates text-to-speech functionalities to enhance accessibility and engagement in learning Baybayin. For text-to-speech, Baybayin characters are transliterated into Tagalog words, which are then converted into audible speech using a standard voice engine.

The proposed interactive features transform content delivery into an immersive educational experience by incorporating quizzes, activities, real-time feedback, and an adaptive question set that modify according to the student's progress. Gamification mechanics like badges, leaderboards, and reward systems boost motivation, while progress tracking allows students to assess their development over time. Personalized feedback and achievement milestones further enhance engagement, acknowledging effort and guiding progress toward objectives. Active components and multimedia elements such as audio, visuals, and interactive exercises cater to diverse learning styles, creating a more enriched experience. The application also adjusts its difficulty based on user performance, ensuring an efficient and personalized learning journey.

System Requirements

The system requires specific technical specifications to fulfill its aim of delivering an optimal user experience. The specified requirements enable the application to perform optimally across different devices using contemporary technologies.

Table 6 presents the Software Specification Requirements for the TalaBaybayin Mobile Learning Application

Table 6

Software Specification Requirements for the TalaBaybayin Mobile Learning Application

Software Resources	Minimum Requirements	Recommended Requirements
Operating System	Android 8.1 – Oreo	Android 16
Device	Smartphones with at least (4GB RAM)	Recent devices with faster processors and enhanced graphics for smoother performance.
Internet Connection	Stable mobile data	High-speed Wi-fi broadband
Application Environment	Compatibility with current mobile operating systems and basic accessibility features.	Compatibility with the latest OS versions and enhanced accessibility options

The minimum software requirements for the mobile application are Android 8.1 for Android devices, ensuring compatibility with current and stable mobile operating systems. This ensures that a wide range of users, including those with older devices, can access the application without issues. The recommended software requirements are Android 16 for Android devices, providing access to the latest features and enhanced performance. With Android 16, users can benefit from improved security, optimized system performance, and a smoother user experience. Additionally, ensuring compatibility with newer versions allows the app to leverage advanced functionalities such as gesture navigation, dark mode, and enhanced privacy controls, contributing to a more effective and user-friendly learning system.

Design of Software, Systems, Product, and/or Processes

The researchers' mobile learning application design structure followed excellent modeling techniques and structured design methods, combined with principles that prioritize users. During initial development, the researchers produced interactive wireframes as well as mock-ups to demonstrate an intuitive understanding of user requirements and activities. Data Flow Diagrams (DFDs) provided a visual representation of data movement between system procedures, data repositories, and external interactions for each functional element (Ibrahim & Yen, 2010), including interactive events, quizzes, gamified tasks, and Baybayin text entry. The implementation details of processes are defined through flowcharts, which ensure that execution steps enable smooth operation and work alongside other visual frameworks. Entity Relationship Diagrams (ERDs) illustrated the relational structure of the application database by connecting users, content modules, language elements, and quiz results. Combined, these diagrams created a visual framework with technical specifications for developed the application.

The application quality soared thanks to a user-centered approach, as highlighted by Giampedraglia (2023). By considering user behaviors and expectations in every decision, the system became both functional and easy to use. The team worked together to create structures and diagrams based on real-world usage patterns and mental workflows, ensuring the system was intuitive and learner-friendly. This approach allowed for improved functionality, including content updates, new educational features, and support for additional local scripts. The mobile platform, designed with user-focused thinking and diagrams, aligns the system's architecture with user experience goals. It also offers a scalable design that can easily integrate future modules and multilingual support without losing usability.

Data Flow Diagram (DFD)

Figure 3 illustrates the Context Diagram (level 0) of the system's Data Flow Diagram (DFD), which portrays the data flow within the Web-based Management Information System. A DFD, as noted by Ordonez et al. (2020), helps model software by showing how data moves and defining the system's boundaries. This diagram facilitates a high-level view of interactions between external entities and the system, making it easy for stakeholders to understand the data exchange process. It also helps developers identify inefficiencies and optimize data flow before implementation, setting the stage for more detailed DFDs. Early visualization improves communication between technical teams and non-technical parties, ensuring aligned expectations.

The Level 0 Data Flow Diagram, or Context Diagram, of the TalaBaybayin Mobile Learning Application is shown in Figure 3. This diagram will give an overall perspective of the whole system within a single process and point out the main interactions of the system with the user. The user at this level is the only outside party that interacts with the application. The user enters different kinds of input data, including authentication data (registration and login credentials), learning activity data (lesson and quiz requests), Baybayin character recognition data (handwriting inputs), text recognition data (translation), gamification features, and progress tracking feature data (activity requests).

These inputs are received and processed in the system. It is responding with outputs that have authenticated access feedback, learning content and lesson materials, quiz questions and results, Baybayin translations and text-to-speech translation outputs, gamification feedback with scores and leaderboard position, and progress reports and certificates generated upon completion of an activity. Level 0 DFD focuses on the data flow between the user and the application in general and does not present what is inside of the application or the specific processes that process these functions. This level provides a clear overview of how data moves throughout the system. It also serves as the foundation for creating more detailed data flow diagrams in the next levels. It helps in understanding the combined interaction between the user and the system's core functions. It ensures that all major system processes are properly defined and interconnected. Additionally, it provides a clear overview of how information moves within the system, highlighting the key data sources and destinations.

Figure 3

Context Diagram of the TalaBaybayin: A Mobile Learning App for Guiding Filipino Learners in Mastering and Preserving Baybayin

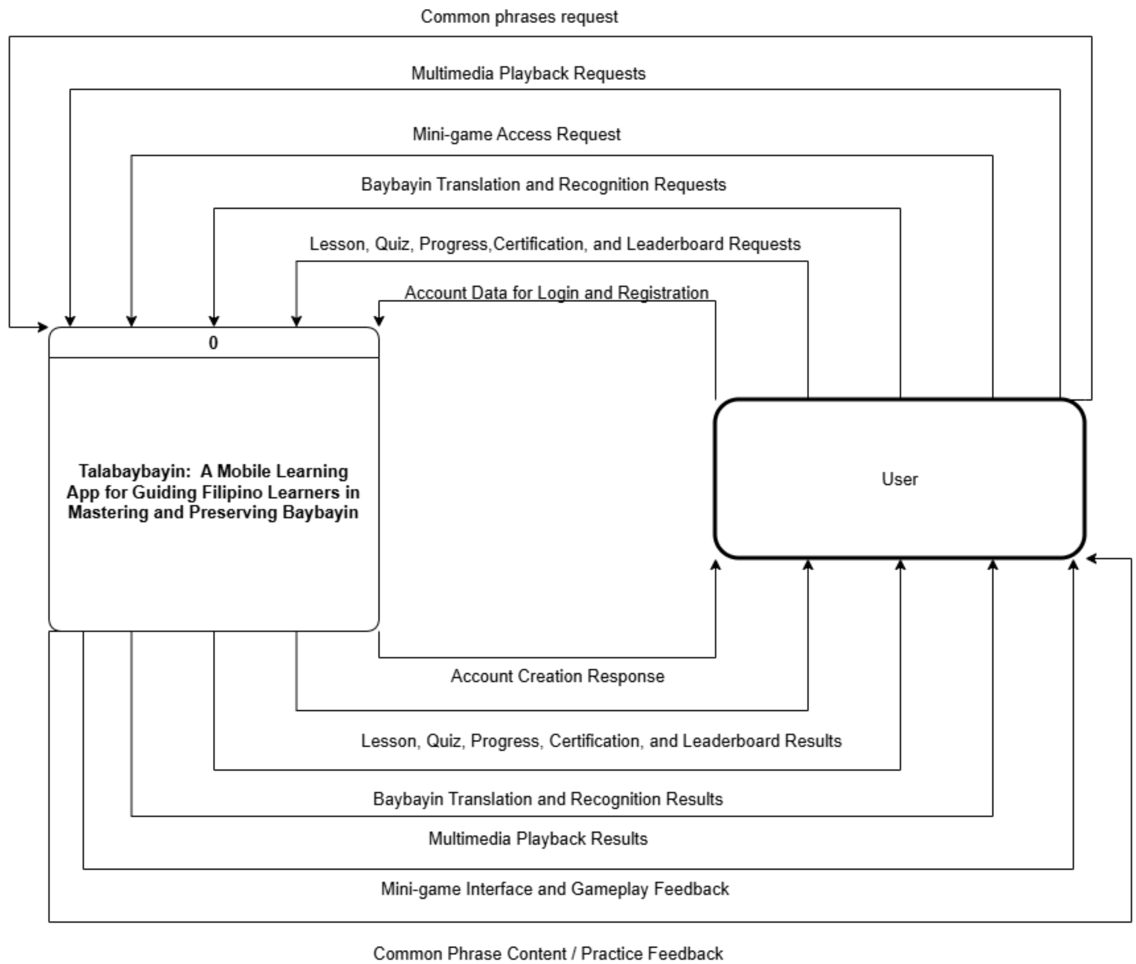


Figure 4 shows Level 1 of the Data Flow Diagram (DFD) for the Learning Application, highlighting user interactions with features like the Baybayin Keyboard, Writing Tool, Translator, and Progress Tracker. It illustrates the flow of data between the user and the system, helping developers and stakeholders understand the application's functions. To improve the app, adding a handwritten script database and Tagalog support is essential. The handwritten database stores images of Baybayin characters, while the

Tagalog database maps Tagalog words to Baybayin. These additions preserve cultural heritage and ensure the app's functionality in a digital learning environment. Incorporating these features, the app will become more versatile and engaging for users, allowing them to experience Baybayin in both modern and traditional contexts.

The level one DFD of the developed TalaBaybayin system shows that the user entity begins by either registering or logging into the system. User interaction begins with authentication via user credentials, leading to an authentication status. The core learning experience involves interactive lessons, where lesson interaction data is captured and lesson content is delivered. Quizzes gauge user understanding, with answers collected and results generated, contributing to overall progress tracking.

The key processes are Account Management, Lesson and Quiz Module, Baybayin Conversion and Recognition, Mini-Game Module, Leaderboard and Gamification, Progress Tracking and Certification, and Common Phrases Module. The processes are used to perform certain functions, including user authentication, delivery of lessons and quizzes, text translation and handwriting recognition, management of gamified learning processes, user progress, and the provision of crucial Baybayin phrases to practice. Each process contributes to creating an engaging and efficient learning experience for users. The Account Management process secures user access, while the Lesson and Quiz Module delivers learning content and assessments. The Baybayin Conversion and Recognition feature enables translation and handwriting practice for better language comprehension. Gamification elements like the Mini-Game Module and Leaderboard enhance motivation through rewards and rankings

Figure 4

Data Flow Diagram of the TalaBaybayin: A mobile Learning App for Guiding Filipino Learners in Mastering and Preserving Baybayin

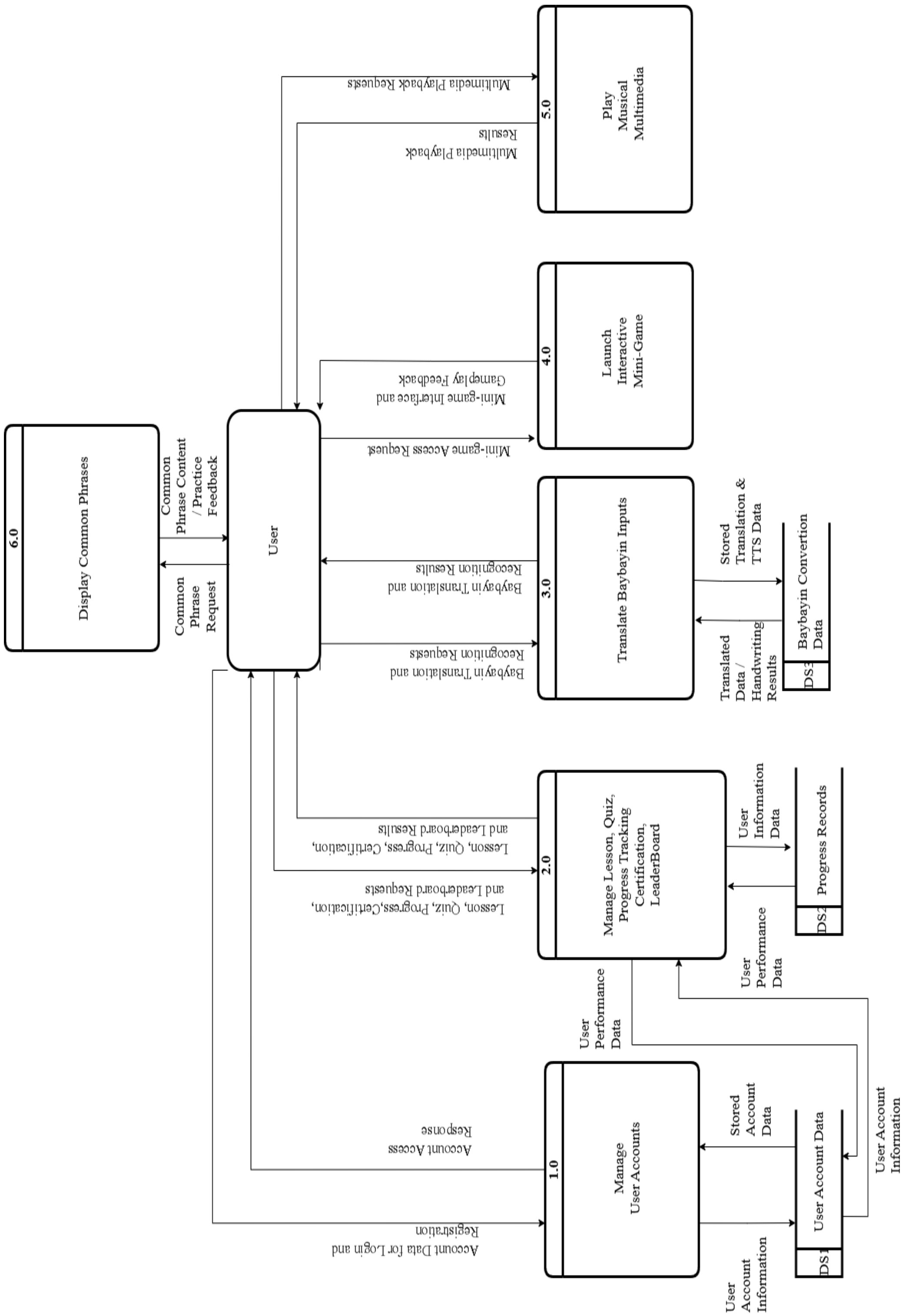


Figure 5 shows the entity-relationship diagram (ERD) that was used to design and model the database of the developed Baybayin learning system. The ERD displays entities representing each database table and the relationships between tables using Crow's Foot notation. The diagram also highlights the relationships between entities such as one-to-many and many-to-many, ensuring efficient data organization and access. The ERD shows the major table of the database, which is the User table, containing the basic information about a user, such as the username, email, and password. The User table is connected to several other tables, which manage the different activities of the system, such as lessons, quizzes, writing activities, keyboard inputs, translations, and user progress. This structure ensures that all user-related data is efficiently organized and easily retrievable, supporting smooth operation and accurate tracking of user interactions within the system.

The Lesson table contains information about each Baybayin lesson, including the title, content, and duration. The UserLesson table functions as an associative entity that connects users to the lessons they have accessed or completed, allowing the system to monitor individual learning progress and history. Similarly, related tables like Quiz and UserQuiz manage the quiz content and record user scores, while the WritingActivity table tracks handwriting exercises to assess the learner's writing accuracy and proficiency. The Translation table manages data related to text conversions between Baybayin and the Latin alphabet, supporting both learning and practice activities. The KeyboardInput table records user interactions with the virtual Baybayin keyboard, allowing the system to analyze typing accuracy and frequency. The Progress table consolidates user achievements and learning milestones, providing a basis for generating reports and certificates. Relationships among these tables ensure data consistency and integrity across all system modules. Overall, the

ERD establishes a well-structured database design that supports seamless functionality and scalability of the Baybayin learning system. By organizing these interconnected tables, the system can easily adapt to new features and support growing numbers of users without compromising performance.

Figure 5

Entity Relationship Diagram of the TalaBaybayin: A mobile Learning App for Guiding Filipino Learners in Mastering and Preserving Baybayin

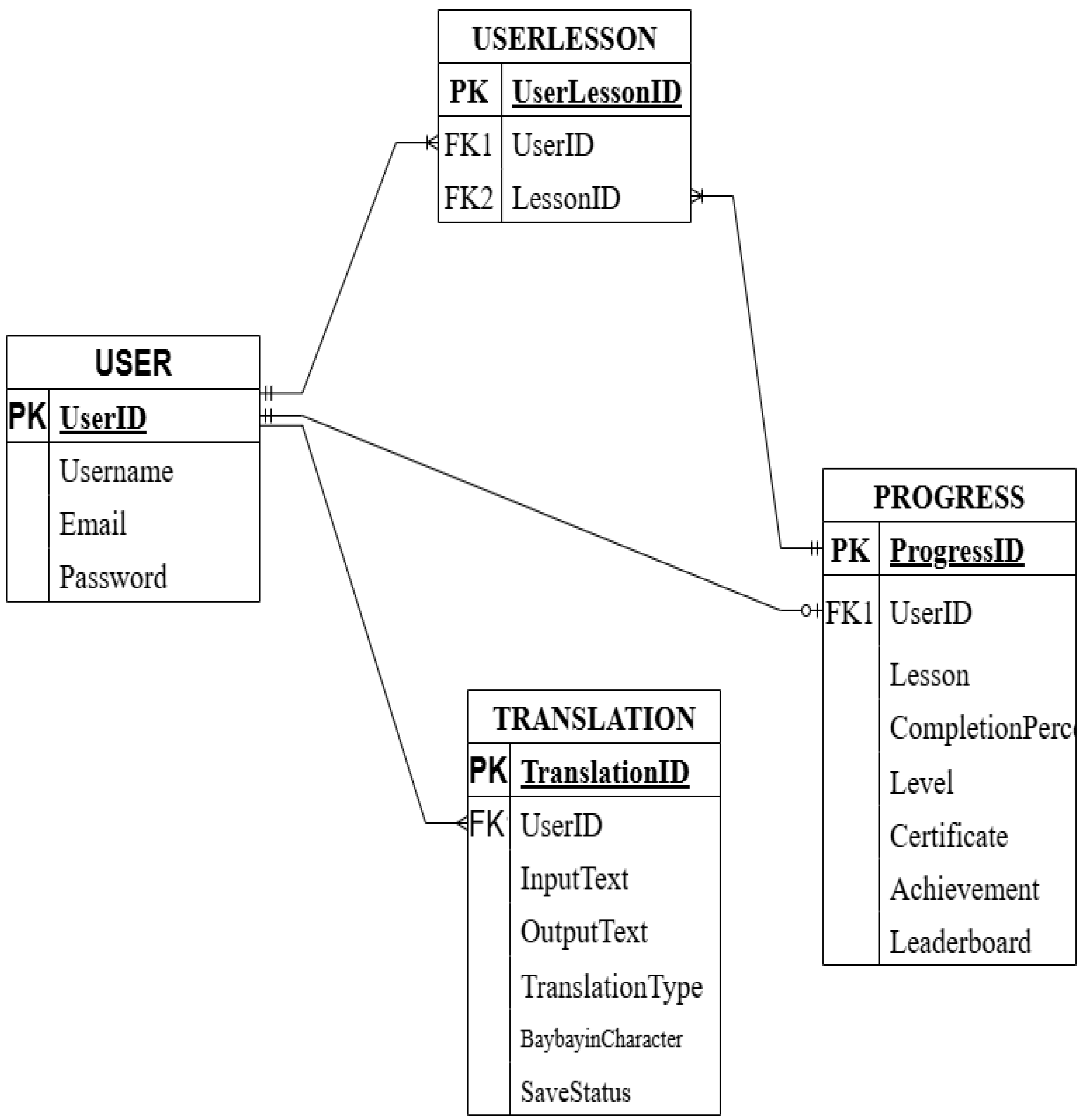
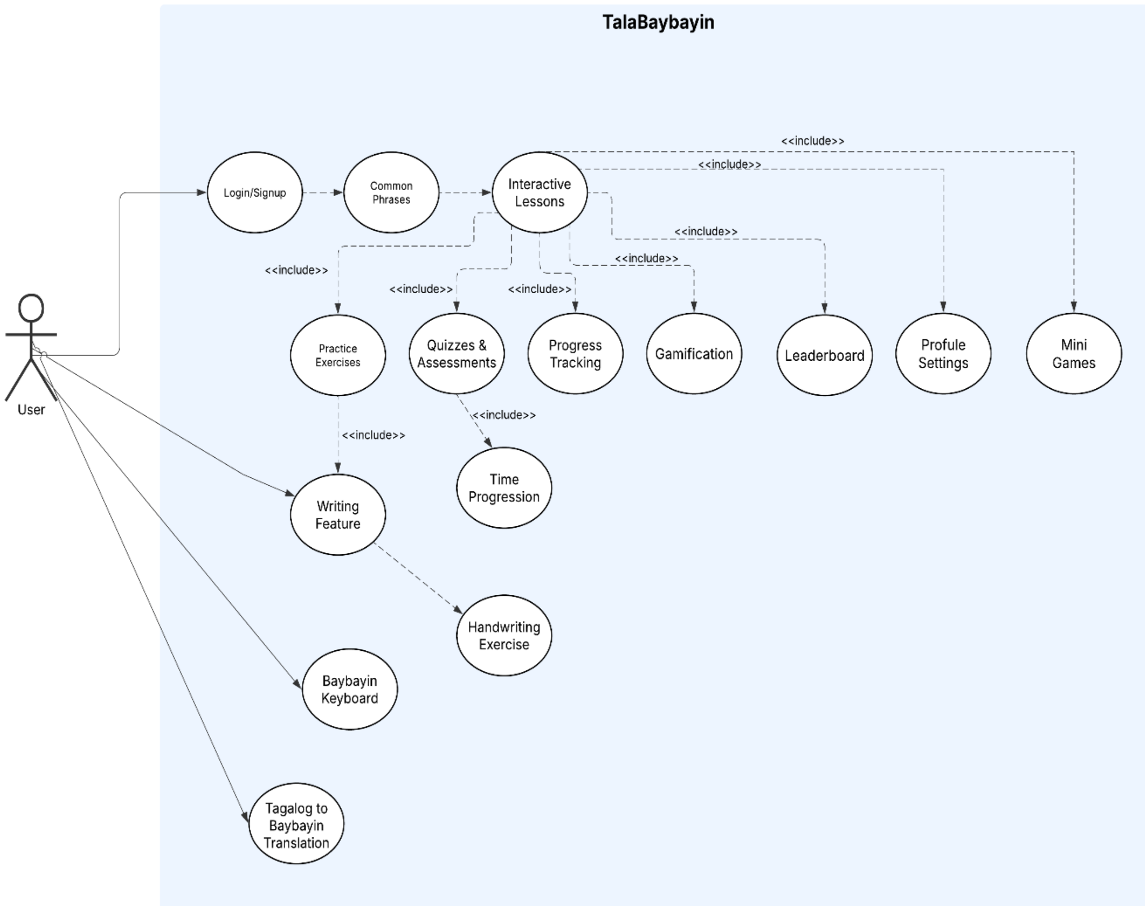


Figure 6

Use Case Diagram of TalaBaybayin: A mobile Learning App for Guiding Filipino Learners in Mastering and Preserving Baybayin



In the use case diagram for TalaBaybayin, the system boundary encompasses all core features: Login/Sign-up, Interactive Lessons, Practice Exercises, Quizzes & Assessments, Progress Tracking, Baybayin Keyboard, Writing Feature, Handwriting Features, Leaderboard, Profile Settings, Mini Games, Time Progression, and Tagalog to Baybayin Translation. This integrated structure ensures a comprehensive and engaging learning experience for users. Each feature in the diagram represents a specific interaction between the user and the system that contributes to overall learning efficiency. As users interact with different features, the diagram highlights the essential pathways that guide their progress and engagement.

Figure 7

Flowchart of the Login process of TalaBaybayin app

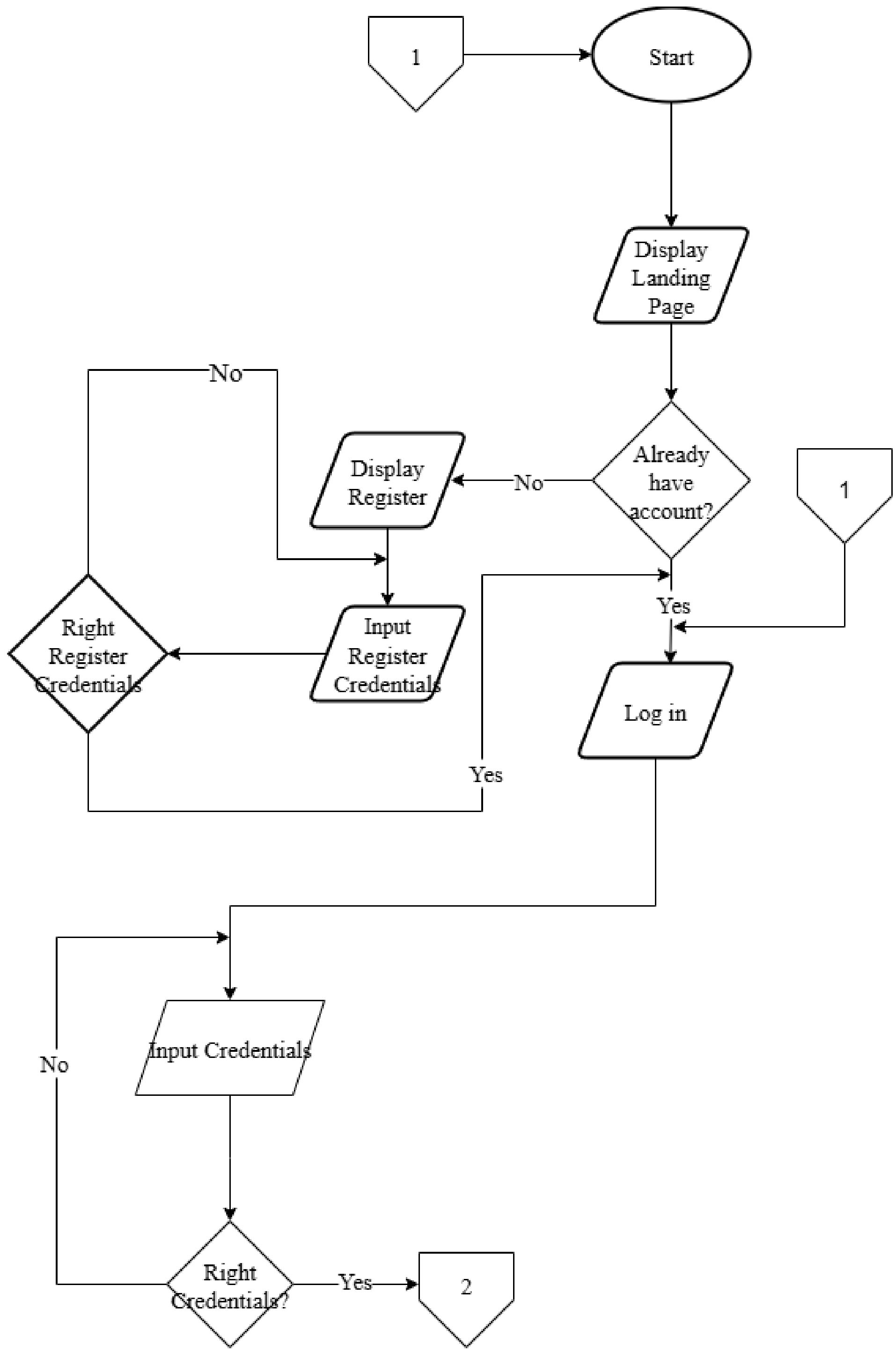


Figure 8

Flowchart of the Dashboard of TalaBaybayin app

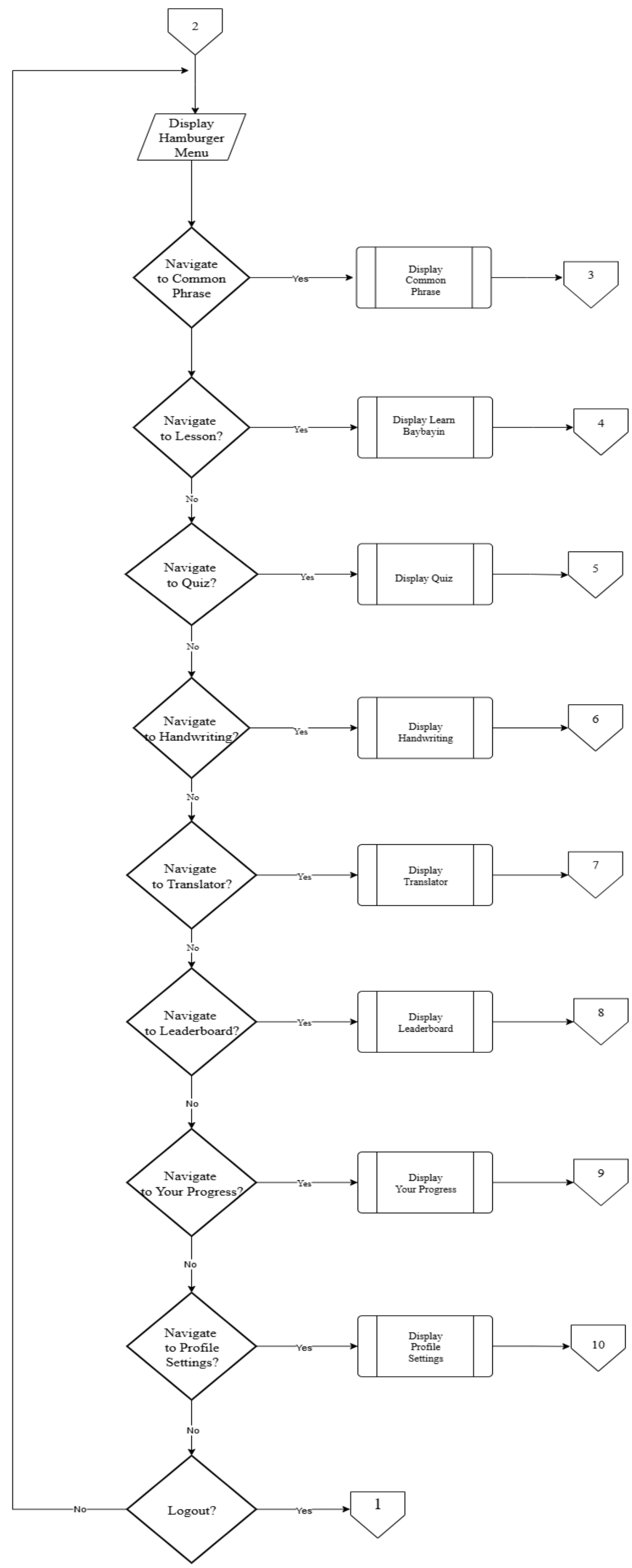


Figure 9

Flowchart of the Dashboard feature of TalaBaybayin app

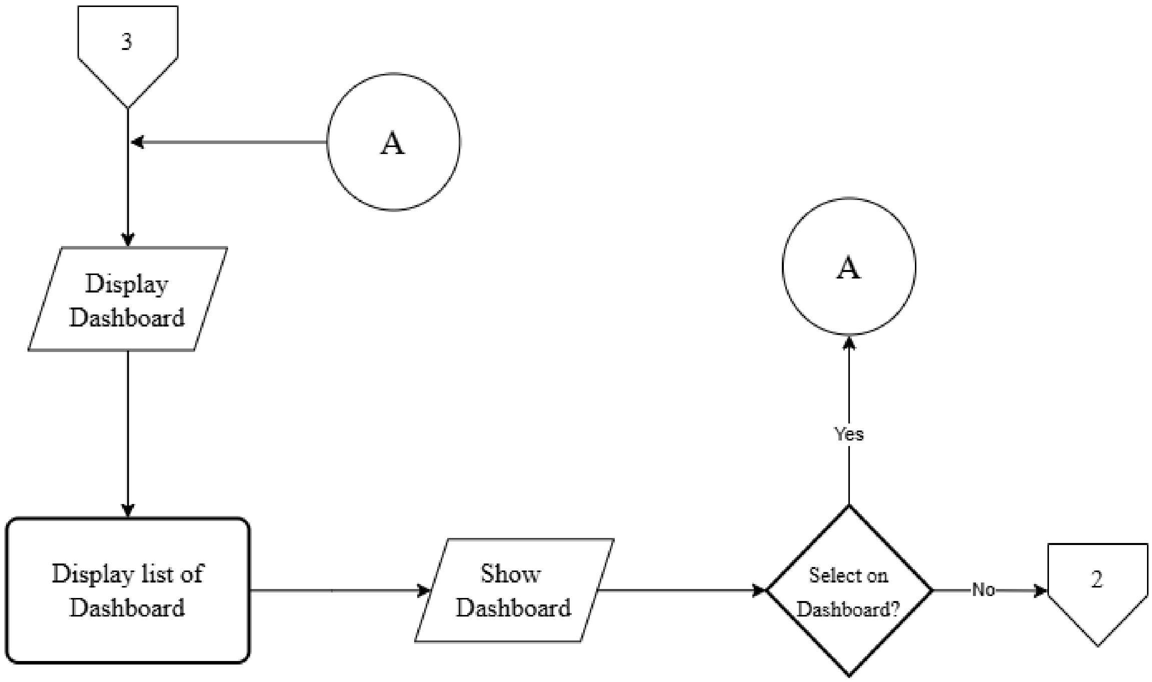


Figure 10

Flowchart of the Common Phrase feature of TalaBaybayin app

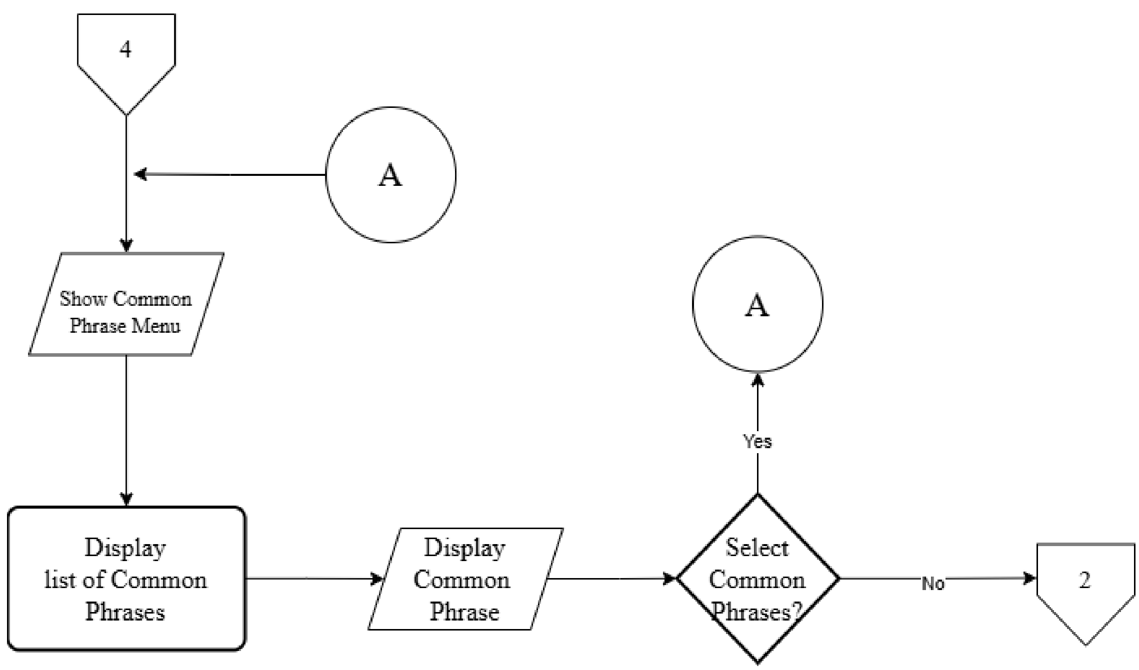


Figure 11

Flowchart of the Learn Baybayin of TalaBaybayin app

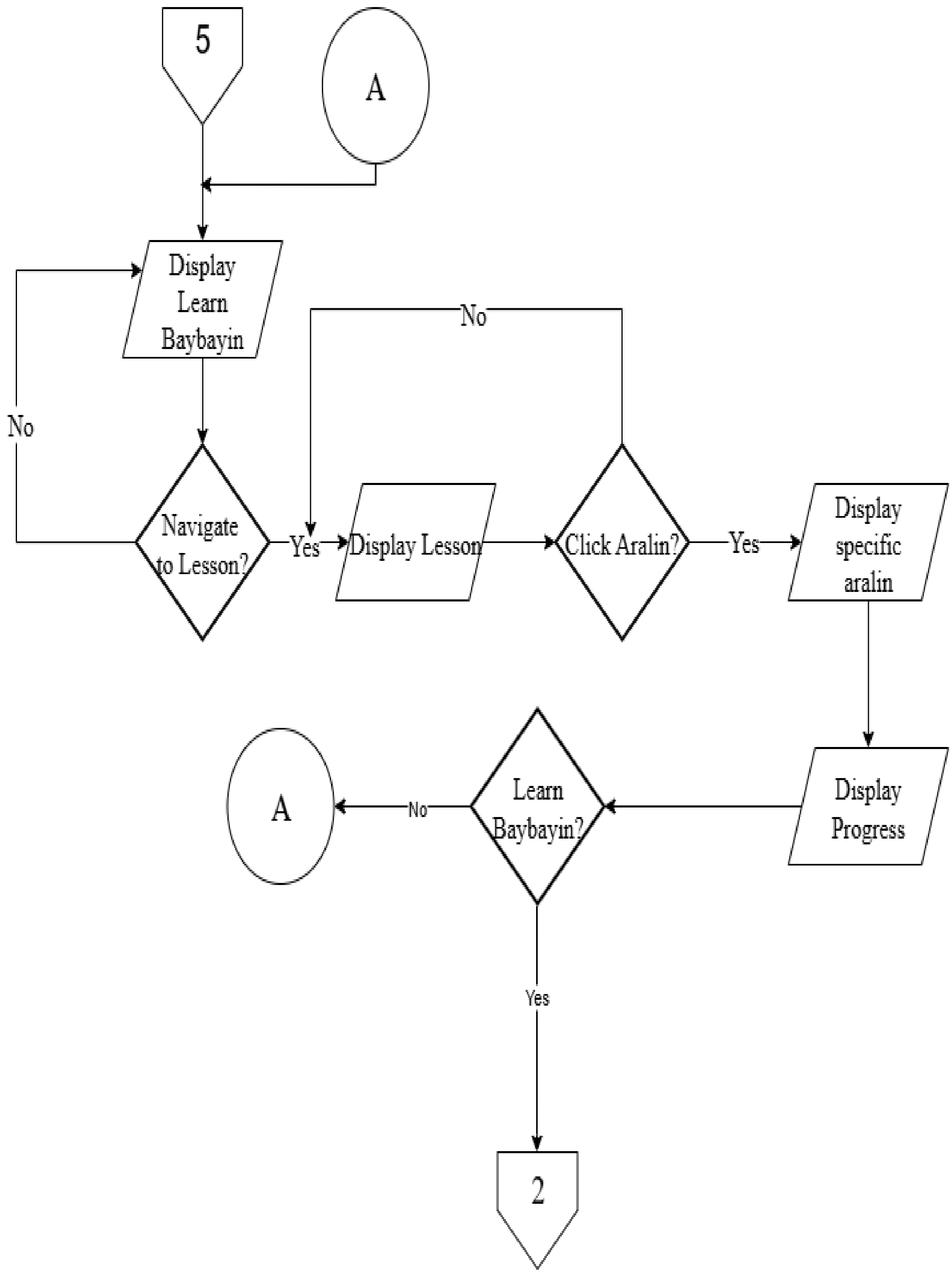


Figure 12

Flowchart of the Quiz of TalaBaybayin app

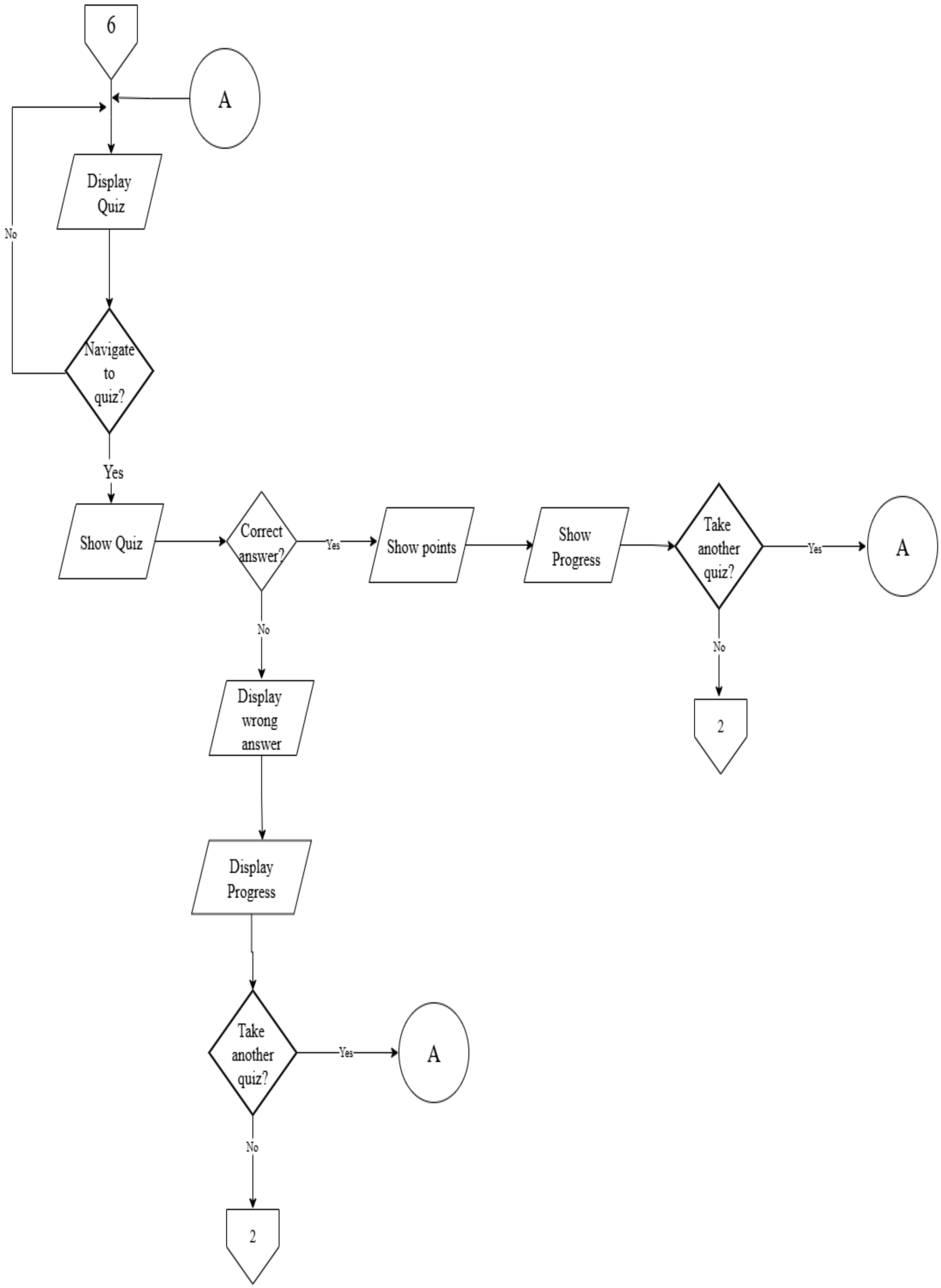


Figure 13

Flowchart of the Handwriting feature of TalaBaybayin app

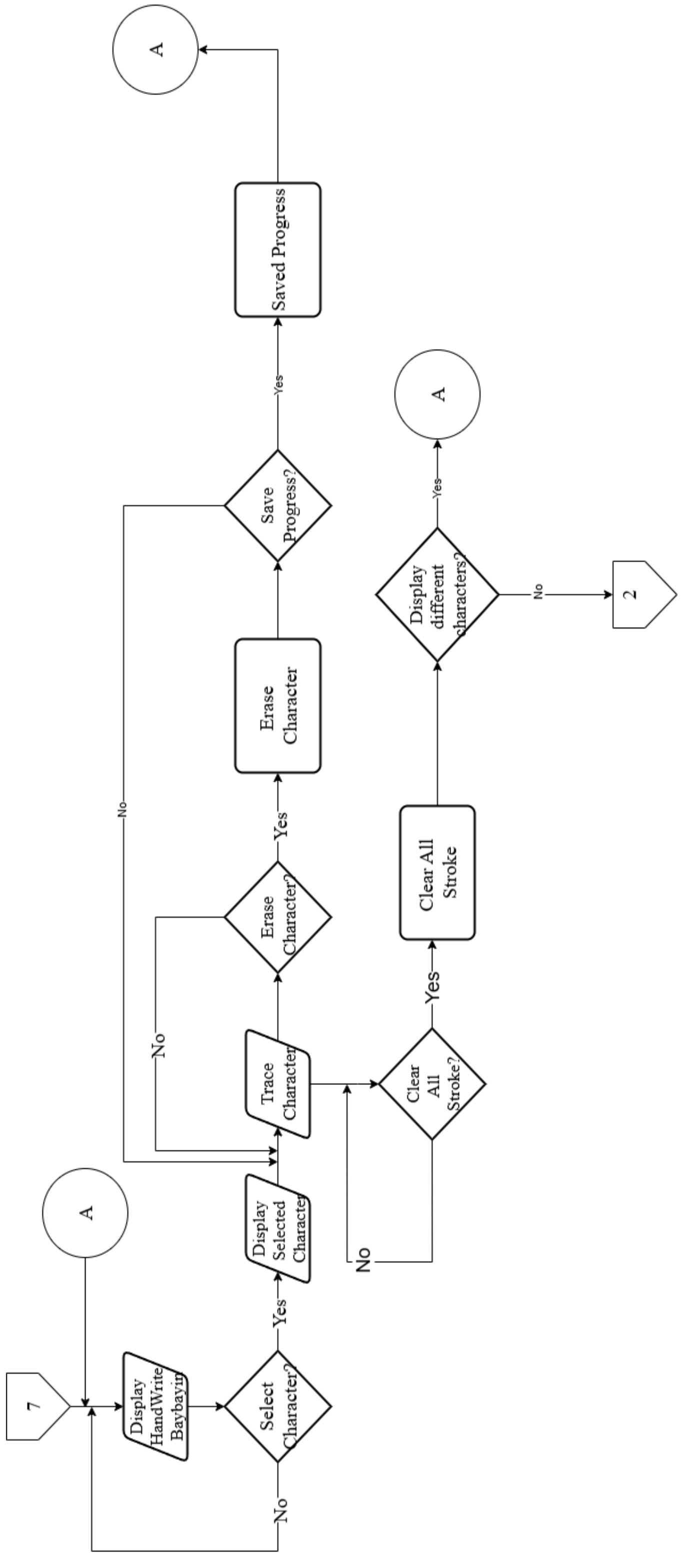


Figure 14

Flowchart of the Translator of TalaBaybayin app

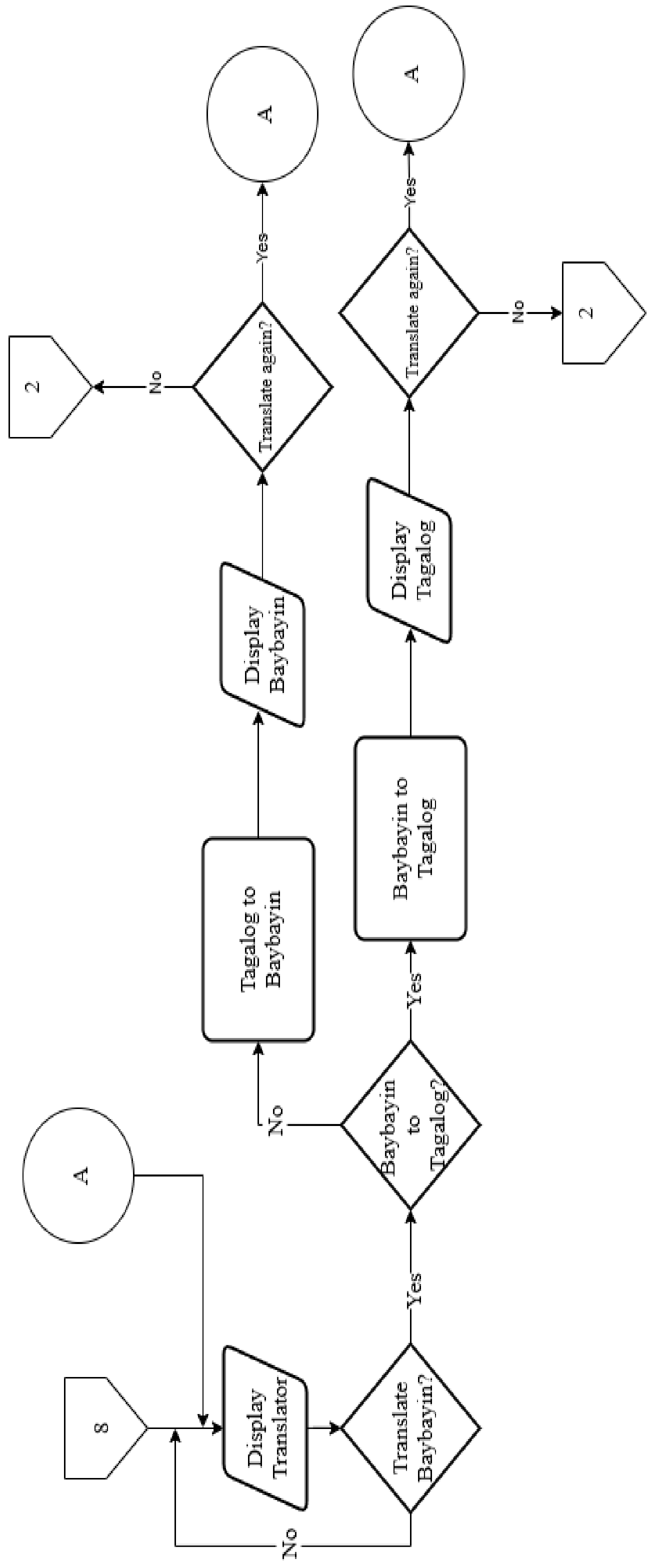


Figure 15

Flowchart of the Leaderboard of TalaBaybayin app

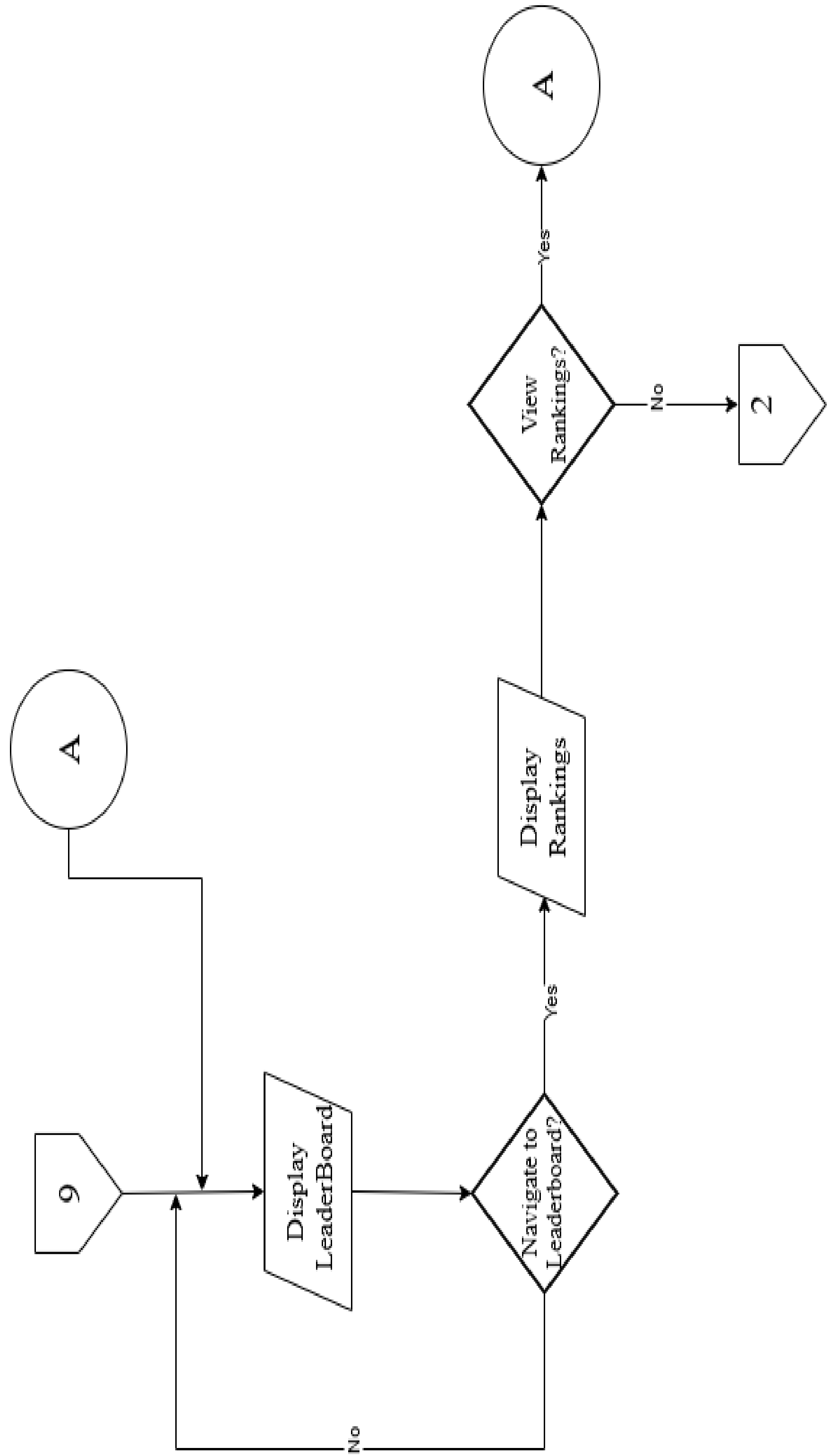


Figure 16

Flowchart of the Progress of TalaBaybayin app

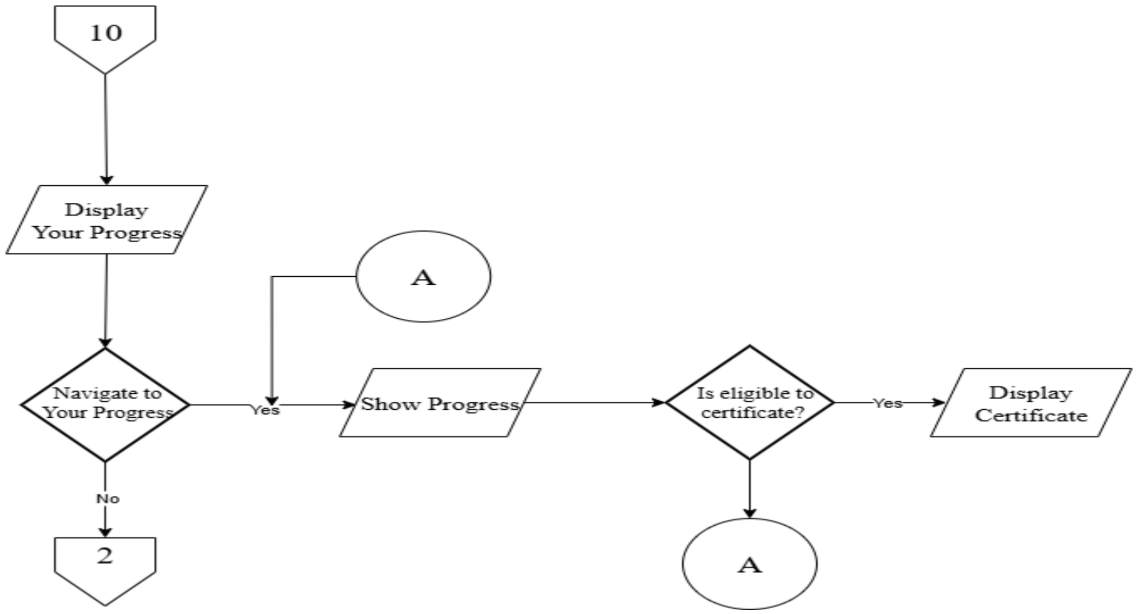


Figure 17

Flowchart of the Leaderboard of TalaBaybayin app

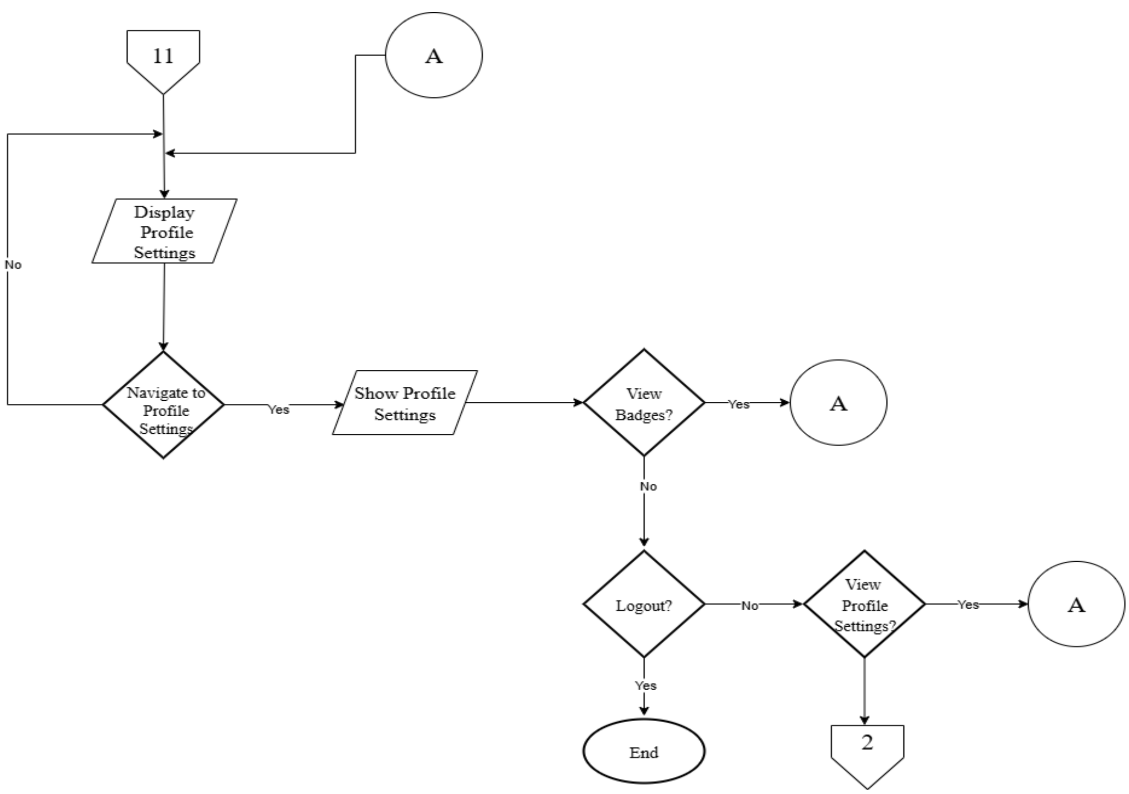
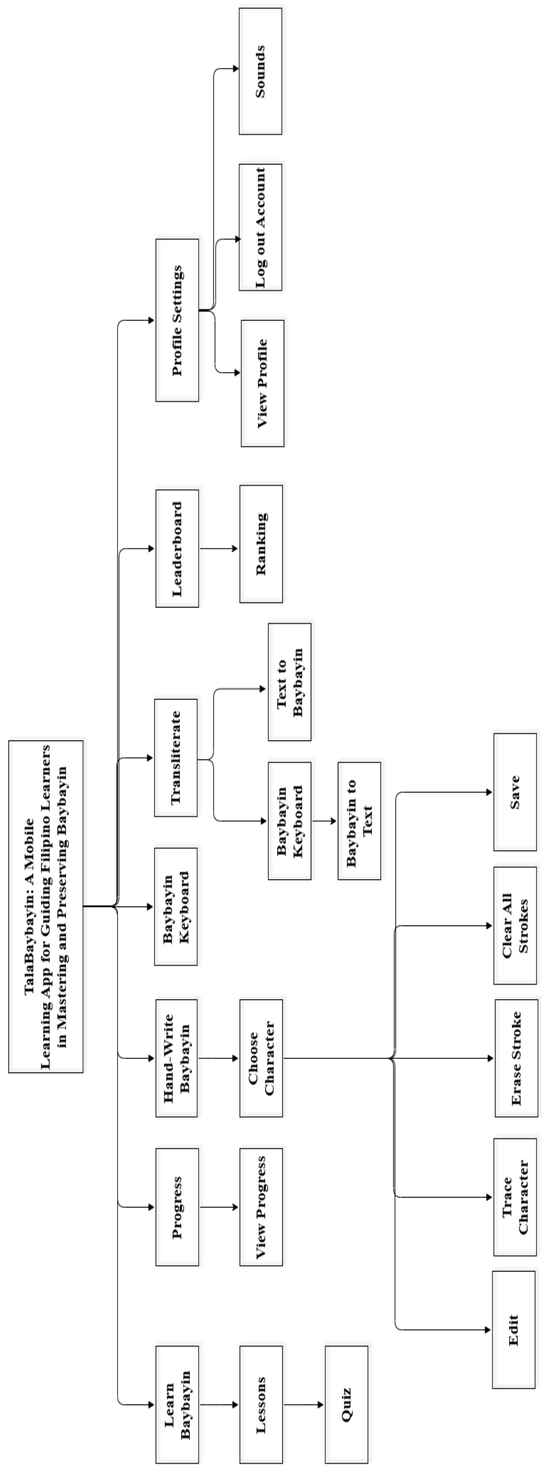


Figure 18

Visual Table of Content of TalaBaybayin: A Mobile Learning Application for Guiding Filipino Learners in Mastering and Preserving Baybayin



Testing

To guarantee the stability, performance, security, and usability of TalaBaybayin, a mobile learning application in mastering and maintaining the Baybayin script, the Black Box Testing approach was utilized. This testing methodology systematically evaluated the application's functionality, interface behavior, and user experience without requiring any knowledge of the internal code structure.

According to Ostrand (2002), Black Box Testing focuses on assessing the outputs generated by the system in response to specific inputs, ensuring that all features operate according to their intended purpose. This method is implemented across the development lifecycle to confirm that TalaBaybayin satisfies all functional and non-functional requirements. In the requirement analysis phase. During the requirement analysis phase, user expectations, such as an interactive writing system, Baybayin keyboard, translation capability, and gamified learning modules, were clearly defined and later verified through User Acceptance Testing (UAT). This validation aligned with the Technology Acceptance Model (TAM) to assess usability, user satisfaction, and perceived effectiveness.

During the system design phase, the functional requirements are clearly defined and later assessed through system and functional testing, guaranteeing compliance with the ISO/IEC 25010:2023 quality criteria, including usability, dependability, and security. Additionally, comprehensive integration testing is performed to ensure smooth data flow and seamless interaction among the system’s core modules, encompassing the writing system, keyboard functionality, and gamified learning components.

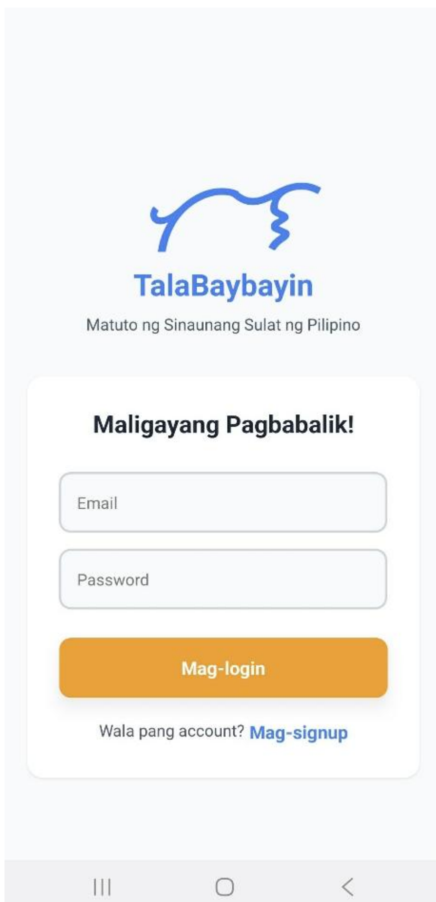
Description of the Prototype

This part of the study illustrates the UI design of TalaBaybayin, highlighting key features like the writing interface, Baybayin keyboard, and translation tool. Gamified elements such as challenges and rank progression boost engagement and motivation.

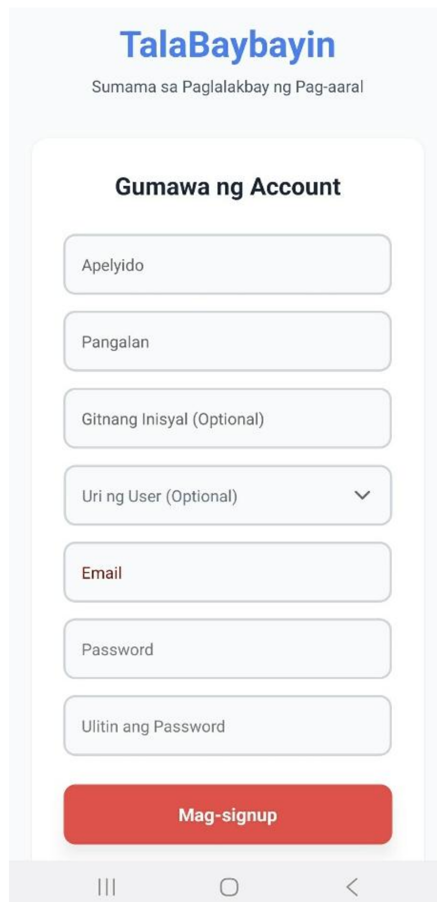
The user is given the option to either log in if they already have an account or sign up if they are a new user

Figure 19

Log-in and Sign-up Page



(a)

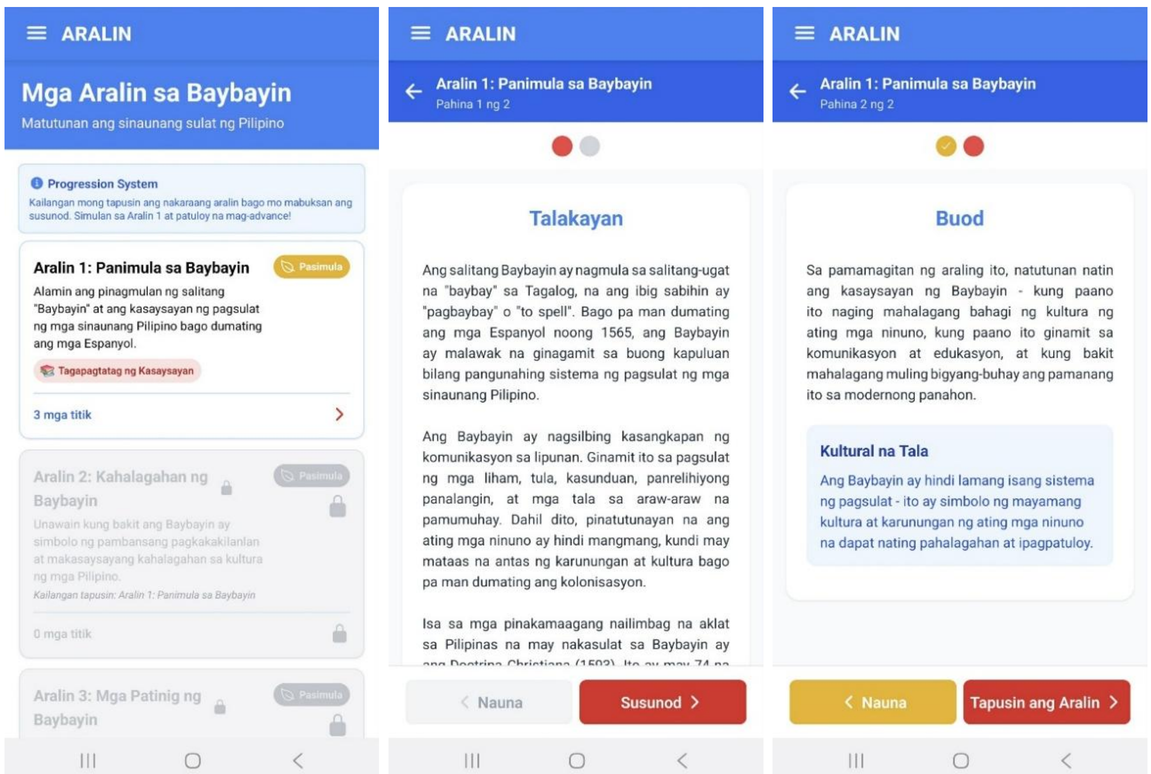


(b)

On the lessons page, users find an interactive setup that makes learning Baybayin fun and engaging. This keeps them motivated and helps information stick better. Research shows that interactive learning really boosts how well students stay involved and learn, so the lessons are designed to be clear, immersive, and easy to follow.

Figure 20

Lessons Page



(a) (b) (c)

On the leaderboard page, users can see how they rank compared to others, encouraging friendly competition and celebrating their progress. It tracks quiz scores, lesson completions, and writing accuracy in Baybayin, motivating learners to stay active and keep improving. Seeing their achievements displayed alongside peers gives students a tangible sense of accomplishment and belonging. This collaborative yet motivating environment inspires consistent participation and helps create a community of engaged Baybayin learners.

Figure 21

Leaderboard Page

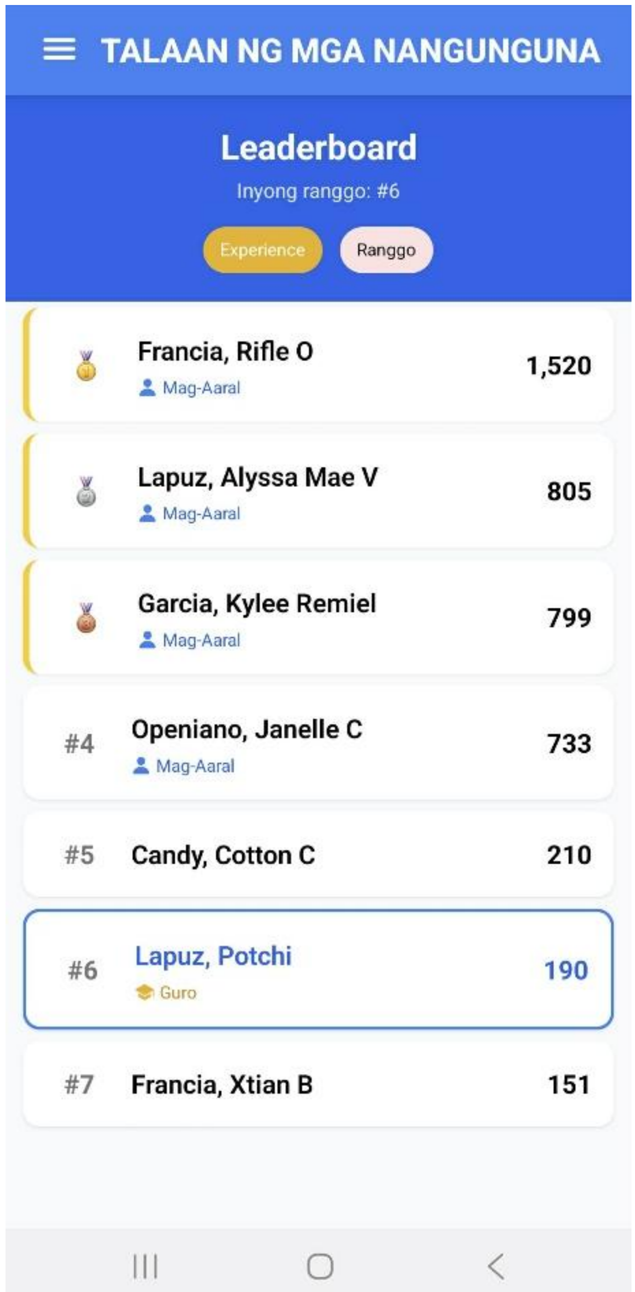
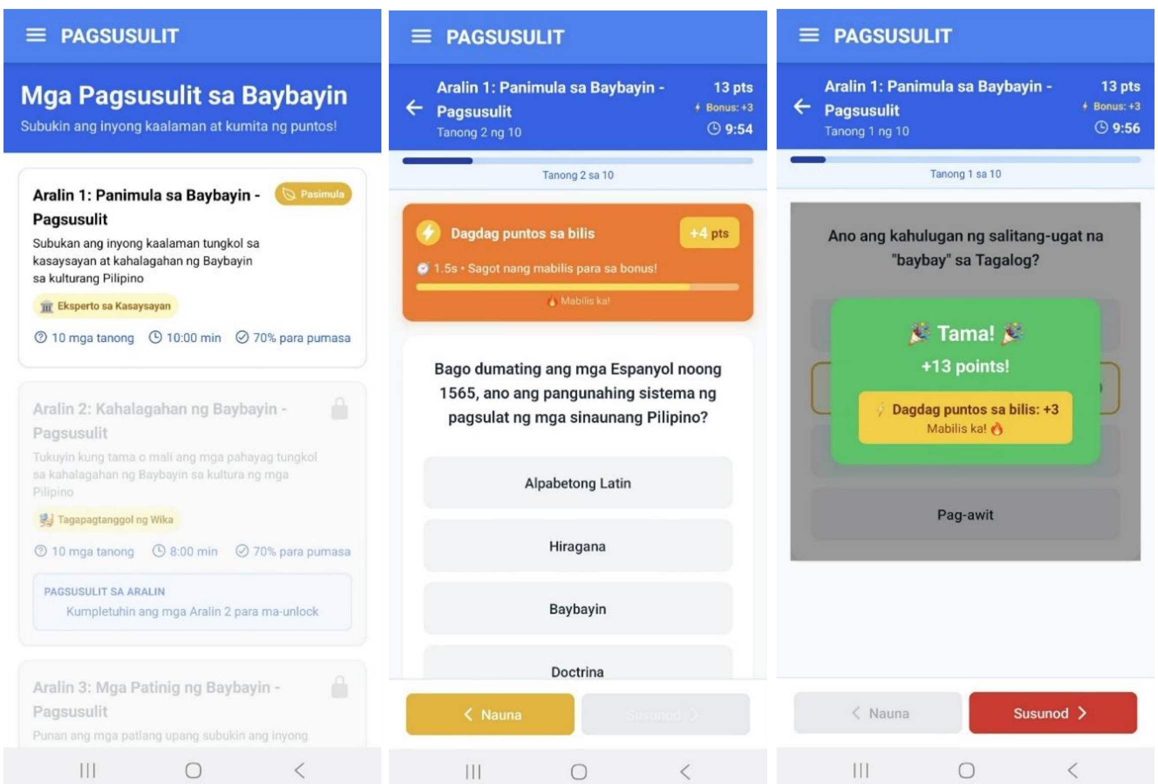


Figure 22 displays the TalaBaybayin app's quiz interface, showcasing its time-based scoring feature. Panel (a) introduces the quiz, with a progress bar and timer at the top. Panel (b) presents the user answering a question, with an option to earn bonus points for quick responses. Panel (c) highlights the feedback screen after a correct answer, displaying the score increase and bonus points earned, encouraging users to answer quickly for higher rewards. This interactive feature is designed to engage users by motivating them to improve their speed and accuracy while learning Baybayin. It encourages consistent practice and enhances their mastery of the script through active participation.

Figure 22

Quiz Interface Showing Time-Based Scoring Feature

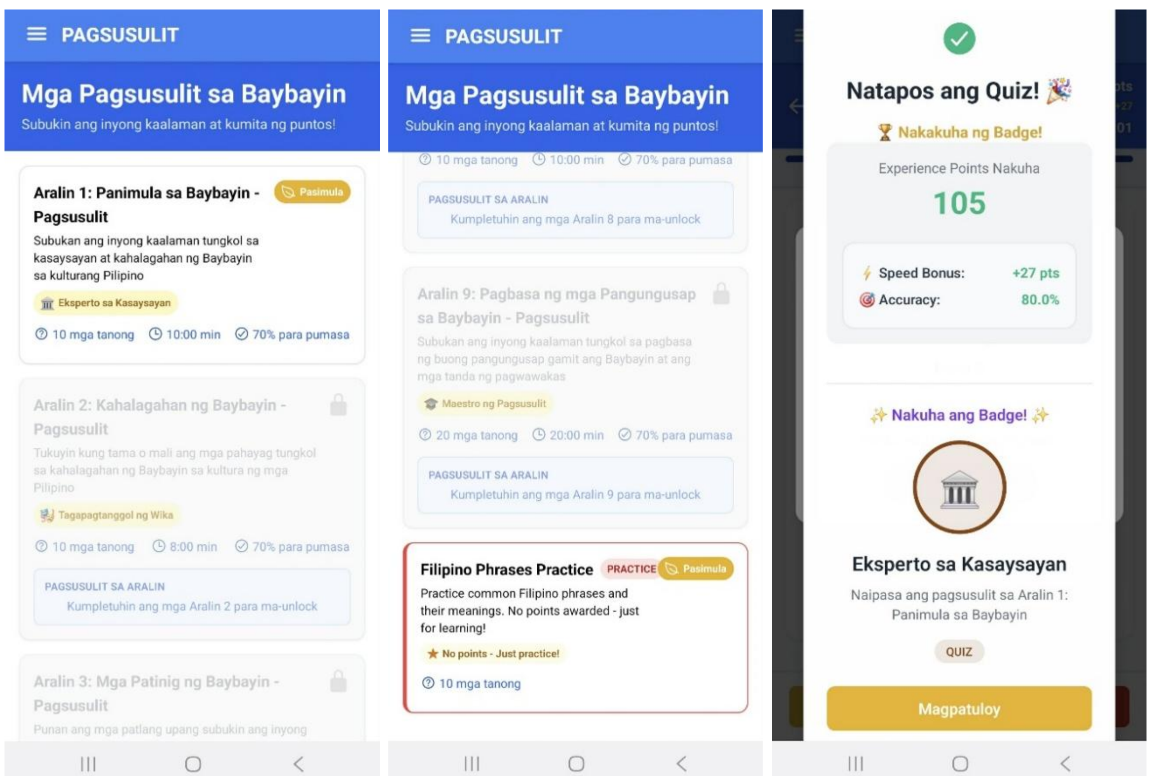


(a) (b) (c)

Figure 23 illustrates the quiz completion flow in the TalaBaybayin app. Panel (a) shows the initial quiz page with the available lesson and progress status. Panel (b) displays the "Filipino Phrases Practice" section, which offers users an opportunity to practice and earn experience points. Panel (c) demonstrates the final screen after quiz completion, where users are shown their experience points, speed bonus, accuracy score, and the badge earned for completing the quiz. This feature enhances user engagement by rewarding progress and performance. This feature not only motivates users by tracking their performance but also provides a sense of accomplishment with badges and experience points, encouraging continued learning.

Figure 23

Quiz Interface



(a) (b) (c)

Figure 24 shows the user dashboard of the TalaBaybayin app, displaying key progress metrics. The top section greets the user, showing their current progress with icons indicating completed lessons, quizzes, and experience points (XP). Below, users can select their next action, with options to continue learning, take a quiz, or engage in practice.

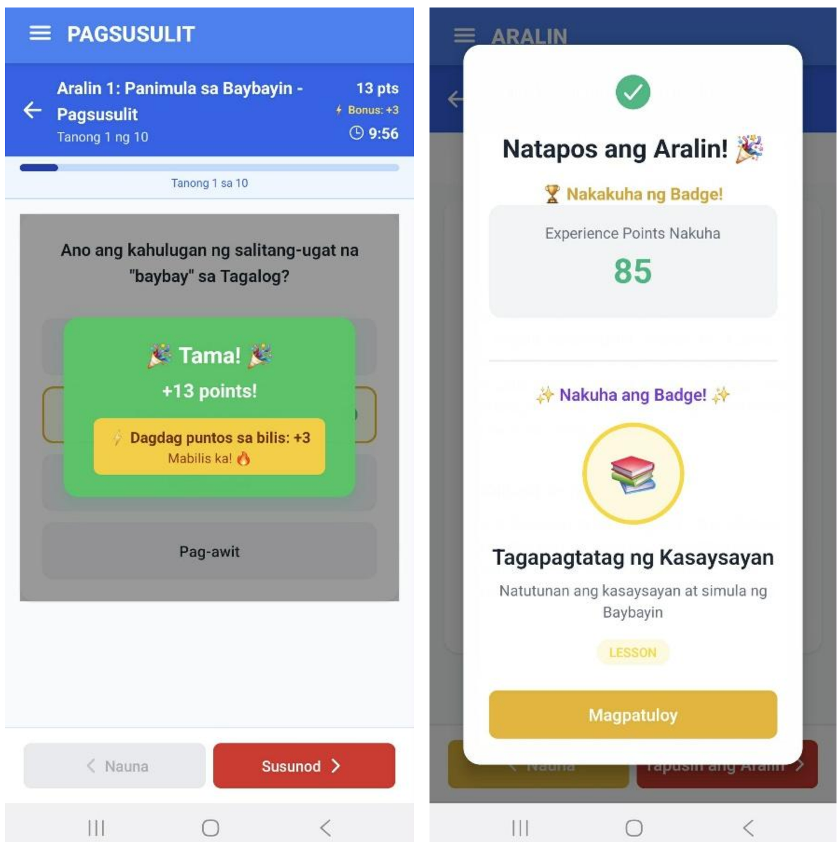
Figure 24

User Dashboard with Progress Overview

Figure 25 illustrates the quiz experience in the TalaBaybayin app. Panel (a) shows the user answering a question, with feedback indicating they’ve answered correctly and earned additional points for speed. Panel (b) shows the completion screen, where users are awarded experience points and a badge for completing the lesson. The badge display is prominently featured on the completion screen, allowing users to easily see their accomplishments. This visual recognition of their efforts not only boosts motivation but also provides a sense of progression as they unlock new badges with each achievement.

Figure 25

Quiz Completion and Badge Achievement

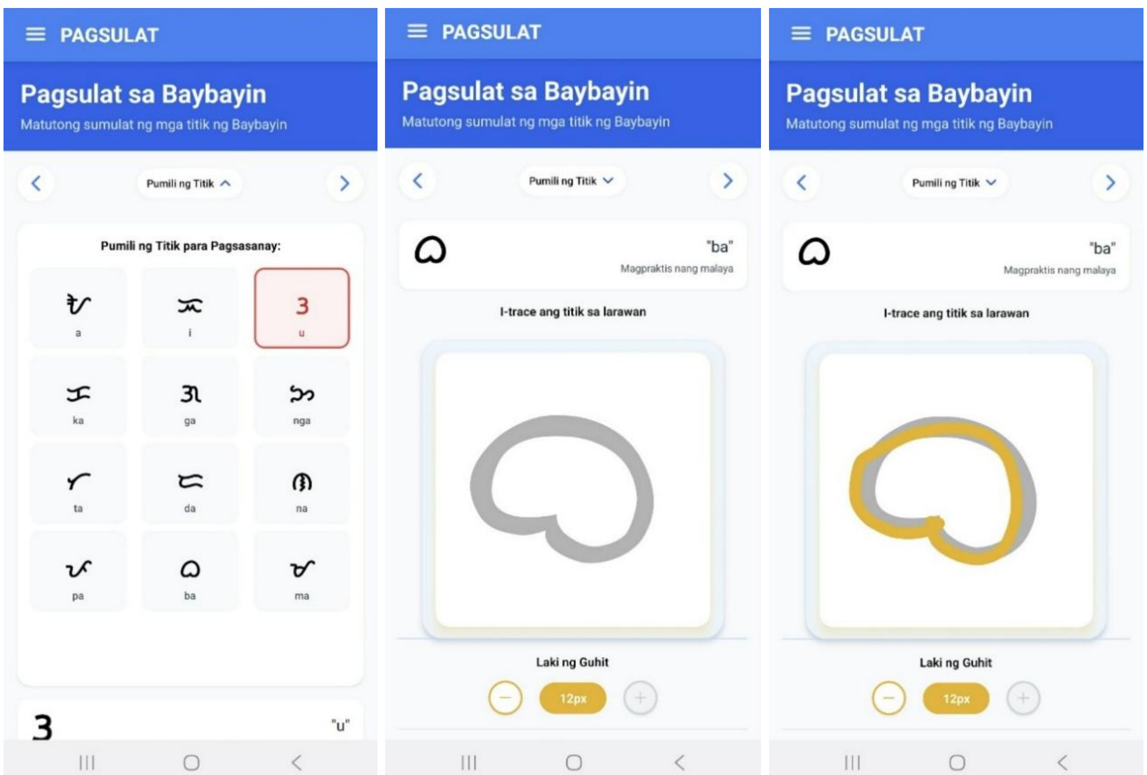


(a) (b)

Figure 26 illustrates the handwriting interface in the TalaBaybayin app. Panel (a) displays the character selection screen, where users choose a Baybayin symbol to practice writing. Panel (b) presents the handwriting input area, allowing users to draw the selected character, which is shown in gray for their reference. Panel (c) demonstrates the completed handwriting input, with the user's traced character highlighted in yellow for feedback. This step-by-step interface guides users in practicing correct stroke order and form in Baybayin writing. It provides an interactive and visual approach that helps users improve their handwriting accuracy. Through this feature, learners can develop better familiarity and confidence in writing Baybayin characters correctly.

Figure 26

Handwriting Interface



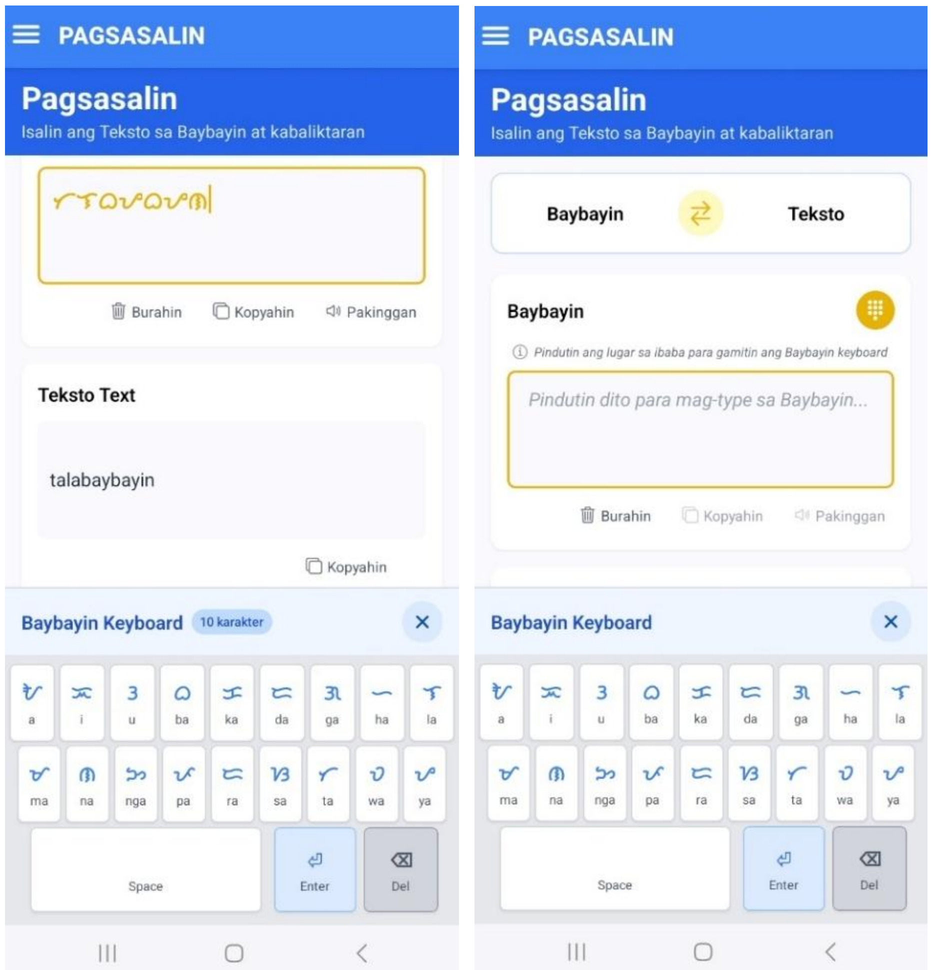
(a) (b) (c)

Figure 27 illustrates the Baybayin Translator and Keyboard feature in the TalaBaybayin app. Panel (a) shows the interface where users type text using the Baybayin keyboard, with the Baybayin script appearing in the input box above as they type. The transliterated text also appears below for reference, helping users see both Baybayin and

Latin script representations. Panel (b) demonstrates the mode selection screen, where users can switch between Baybayin and Teksto input modes.

Figure 27

Baybayin Translator and Keyboard

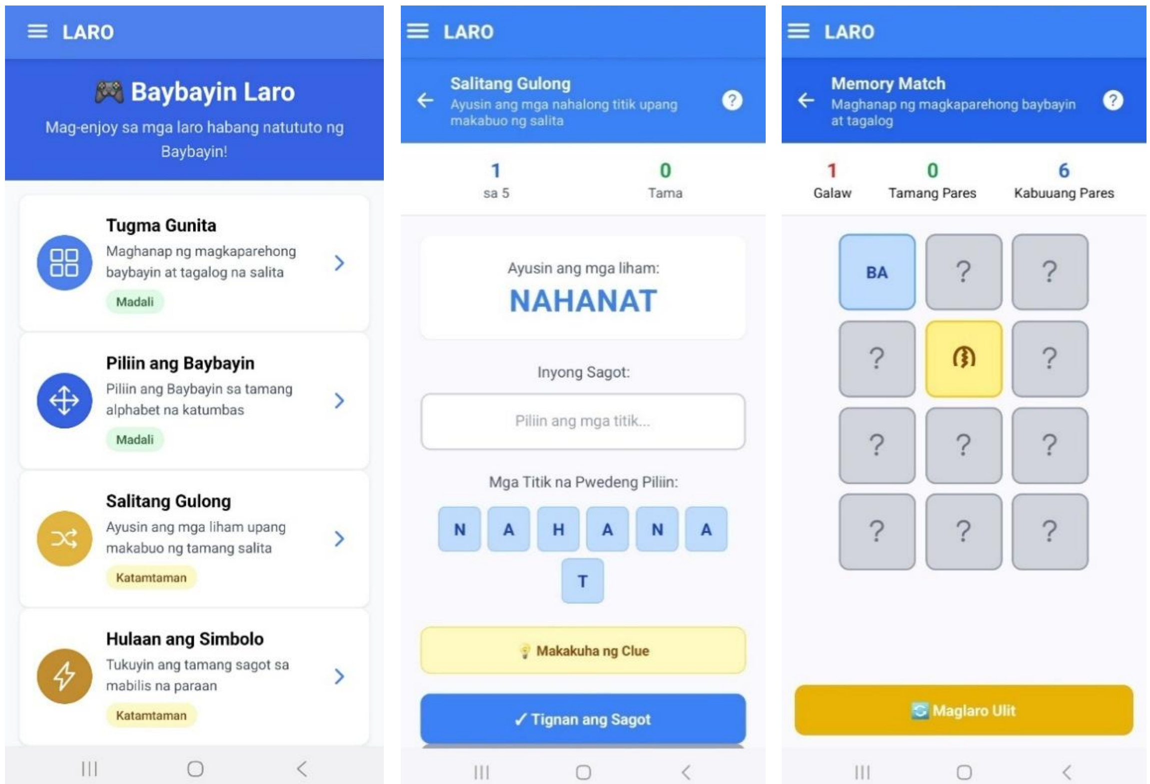


(a) (b)

Figure 28 showcases the interactive learning games in the TalaBaybayin app. Panel (a) presents the main game menu, offering various activities like "Tugma Gunita" (Word Match) and "Salitang Gulong" (Wheel of Words). Panel (b) shows the "Salitang Gulong" game, where users must rearrange letters to form a word, offering clues if needed. Panel (c) displays the "Memory Match" game, where players match Baybayin characters with their corresponding Tagalog letters. These engaging games aim to reinforce learning while making the process enjoyable and interactive.

Figure 28

Mini-games



(a) (b) (c)

Figure 29 displays the profile settings page of the TalaBaybayin app, where users can update their personal details, such as name, email, and user type. The bottom section highlights the "Background Music" option, allowing users to toggle music on or off during their learning sessions. This feature enhances the learning experience by providing an option for calming background music while studying Baybayin, helping users focus or create a more enjoyable environment. The settings page allows for easy navigation, ensuring that users can quickly make changes to suit their preferences. Giving users control over background music and personal details empowers them to customize their study environment, making each session uniquely theirs.

Figure 29

Profile Settings with Background Music Option

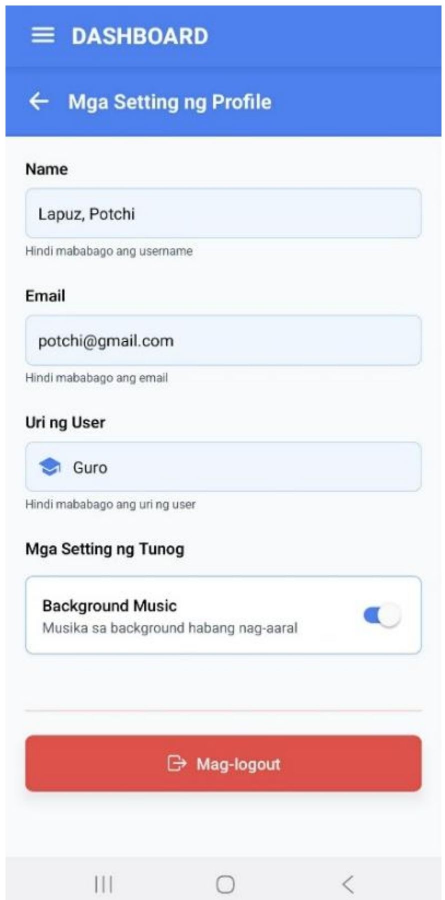
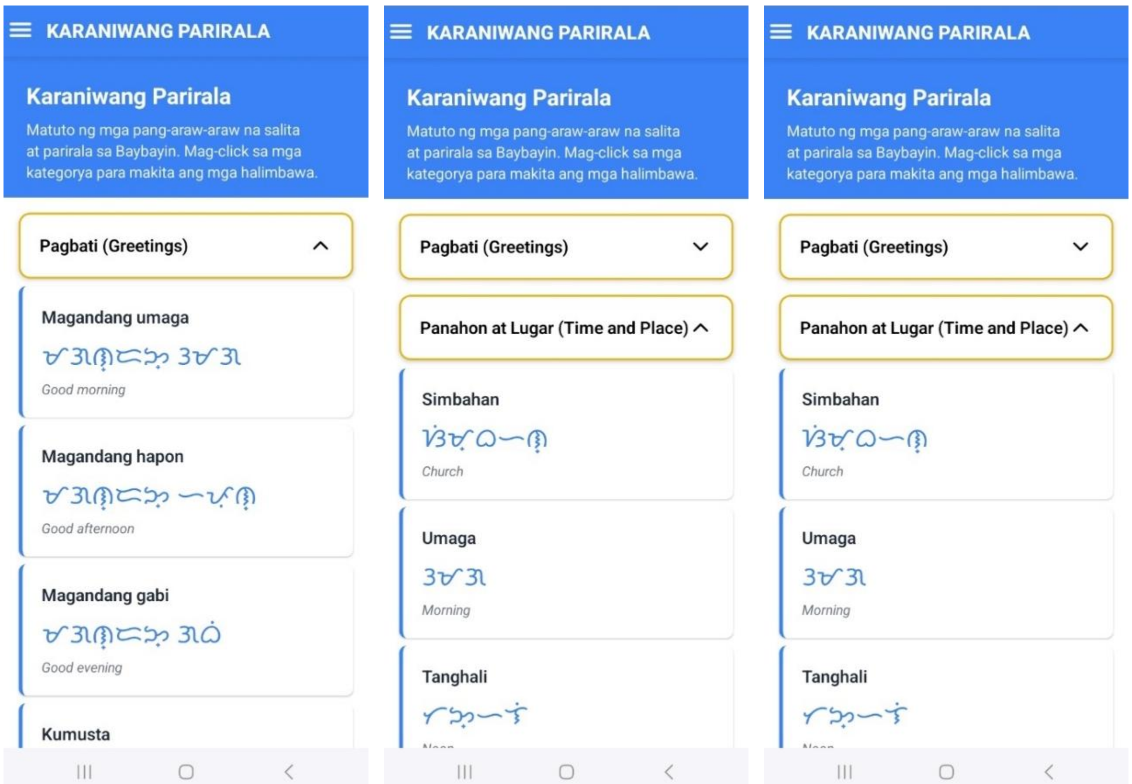


Figure 30 displays a section of the TalaBaybayin app focused on learning common phrases in the Baybayin script. Panel (a) shows the "Pagbati" (Greetings) category, where users can see phrases like "Magandang umaga" (Good morning) in both Romanized Filipino and its Baybayin representation. Panel (b) displays the same category with additional greetings like "Magandang hapon" (Good afternoon). Panel (c) introduces the "Panahon at Lugar" (Time and Place) category, which includes words like "Simbahan"

(Church) and "Umaga" (Morning). Learners can easily switch between categories, making it simple to find phrases relevant to various real-life scenarios.

Figure 30

Common Phrases in Baybayin Script

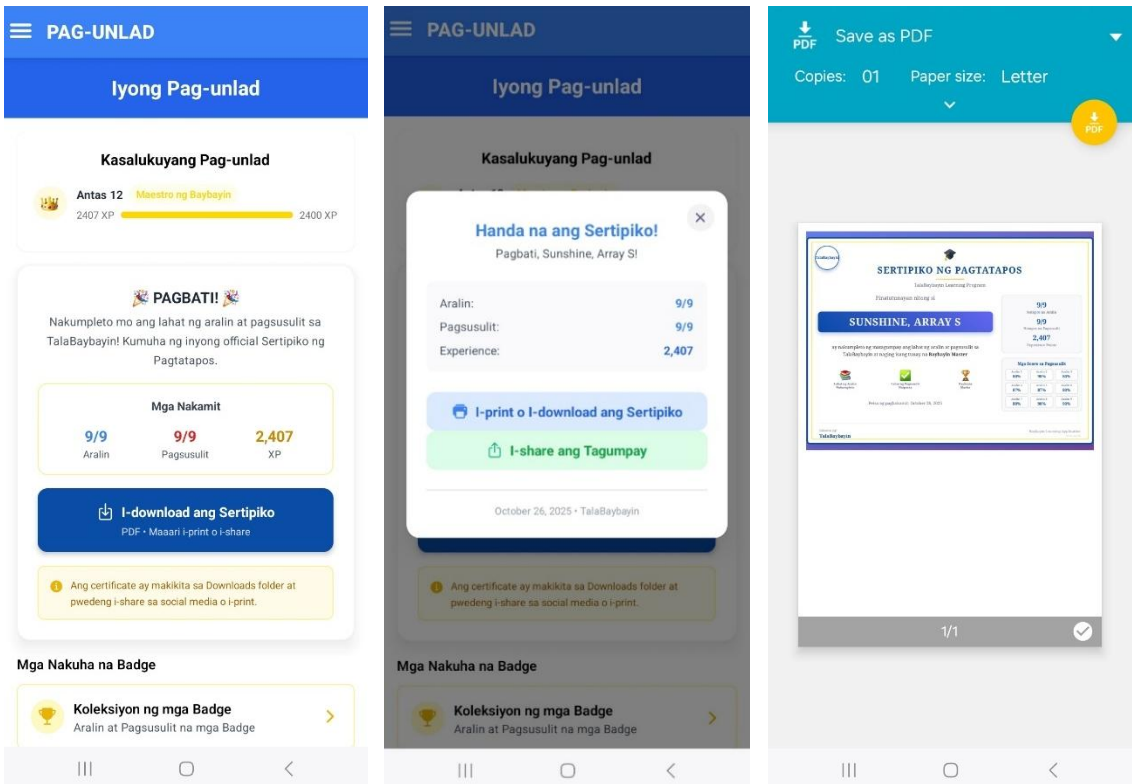


(a) (b) (c)

Figure 31 presents the Completion Certificate feature of the TalaBaybayin app, which recognizes users' achievements upon finishing all lessons and quizzes. Panel (a) shows the progress overview, including completed lessons and available certificates for download. Panel (b) displays the notification indicating a certificate is ready, with details of the user's accomplishments and options to print or share the certificate. Panel (c) illustrates the actual digital certificate, which can be saved as a PDF. This feature motivates learners by formally acknowledging their progress and providing a tangible record of their success. Having access to a certificate not only marks the learner's dedication and growth, but also offers a sense of pride that extends beyond the app itself.

Figure 31

Completion Certificate



(a) (b) (c)

CHAPTER IV

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter discusses and interprets the data collected from interviews, surveys, and evaluations, emphasizing the development of TalaBaybayin, a mobile learning application designed to support Filipino learners in studying and preserving the Baybayin writing system. The purpose of this analysis is to evaluate the app's performance as an innovative platform for promoting Baybayin literacy, while also examining its usability and the level of acceptance among its target users.

In addition, this section discusses how each research objective was addressed through the findings and user assessments. It evaluates how well the app supports learners in understanding and practicing Baybayin through its interactive and engaging features.

1. To determine the current state of studying Baybayin and its preservation among Filipino learners, particularly in terms of accessibility, interest, and the effectiveness of available materials through surveys and interviews:

The first objective of this study was to examine the current state of Baybayin instruction and its preservation, with a particular focus on the accessibility of resources, students' interest in learning, and the effectiveness of traditional teaching methods. To achieve this goal, the researchers conducted interviews with three Filipino language teachers and seventy (70) Grade 7 students from Caniogan National High School. These participants were carefully chosen to represent both the teaching and learning perspectives of Baybayin instruction within a public secondary school setting. The insights obtained from these participants were crucial in identifying the existing challenges encountered by both educators and students, along with the factors that limit effective teaching and appreciation of the Baybayin script.

The survey revealed that 84.5% of respondents use Android phones, 5.6% use iPhones, and 1.4% use other Android devices or do not use a mobile phone. This data indicates that Android devices are the most common among students, making them the primary tool for both communication and learning. Since TalaBaybayin was developed exclusively for Android, the extensive use of these devices among learners guarantees the app's accessibility and continued relevance to its target users.

A common theme that emerged from the interviews was the shortage of interactive and engaging learning materials available for Baybayin education. Mrs. Jen Oliquino-Cabantog from Caniogan National High School shared, "*Talaga namang may struggle sa pagturo ng Baybayin kasi kulang tayo sa gamit na pangturo. Kailangan ko pa maghanap sa* internet *para isalin sa Baybayin yung mga tanong sa* exam." (There is indeed a struggle in teaching Baybayin because we lack teaching materials. I still have to search the internet to translate the exam questions into Baybayin.) This highlights how traditional pen-and-paper methods fall short in keeping students engaged and motivated, This made it challenging for learners to sustain their interest and retain knowledge beyond classroom instruction.

The absence of interactive learning resources proved to be a major obstacle to consistent practice and mastery of Baybayin among students. All three teachers emphasized the need for a more dynamic and continuous learning environment. They agreed that modern tools, especially digital ones, could address this issue. Mr. Brian Manansala highlighted the potential impact of such tools, stating, "*Malaking tulong talaga* *ang ganitong* application*, lalo na para sa mga kabataang mas* comfortable *na gumamit ng teknolohiya sa pag-aaral*".

These challenges are not exclusive to Baybayin instruction but are also reflected in Sother indigenous language programs, where limited teaching resources and outdated methods often result in low student engagement and poor retention (Tagal-Bustillo et al., 2024). In the case of Baybayin, the absence of interactive reinforcement activities and digital practice tools contributes to the gradual decline of learners' interest and knowledge over time.

The collective insights from teachers and students highlight the urgent need to modernize Baybayin instruction by merging traditional approaches with interactive digital tools. Teachers unanimously agreed that this blended approach could significantly enhance learning outcomes, strengthen student motivation, and encourage continuous practice beyond the classroom environment. Furthermore, students indicated a preference for interactive and visually engaging learning materials, which would enhance the accessibility and enjoyment of Baybayin. The integration of ancient methods with new technology facilitates the reconciliation of cultural preservation and contemporary educational practices, thereby preserving the relevance of Baybayin in the digital era.

To support these insights, a survey was conducted with 70 Grade 7 students from Caniogan National High School to identify common challenges in learning Baybayin. The questionnaire, which was carefully reviewed and validated by a licensed psychometrician, ensured that the data collected were reliable and valid. The survey focused on students' familiarity with Baybayin, their perceived difficulties, and their attitudes toward using digital learning tools. The summarized results of their responses, based on their learning experiences and perceptions, are presented in Table 7.

Table 7

Level of Difficulty in Learning Baybayin Among Grade 7 Students of Caniogan National High School (n = 70).

Items	Mean	Standard Deviation	Descriptive Interpretation
1. Bumasa (makilala) ng isang titik sa Baybayin.	2.34	1.19	Slightly Difficult
2.Bumasa ng Baybayin na may kudlit (hal. kudlit sa itaas o ibaba).	2.23	0.96	Slightly Difficult
3.Bumasa ng isang salita sa Baybayin.	2.49	1.11	Slightly Difficult
4.Bumasa ng isang pangungusap na nakasulat sa Baybayin.	2.57	0.93	Moderately Difficult
5. Makilala ang magkakahawig na titik sa Baybayin (hal. *ba–pa*, *ka–ga*).	2.49	1.11	Slightly Difficult
6.Bumasa ng mahahabang salita sa Baybayin.	2.71	1.07	Moderately Difficult
7. Bumasa ng Baybayin nang walang kopya o gabay	3.10	0.97	Moderately Difficult
8. Pagbasa ng Baybayin nang mabilis at tama.	2.61	1.05	Moderately Difficult
9. Pag-unawa sa isang salitang nakasulat sa Baybayin.	2.44	1.09	Slightly Difficult
10. Pag-unawa sa isang pangungusap na nakasulat sa Baybayin.	2.43	1.05	Slightly Difficult
11. Magsulat ng isang titik sa Baybayin.	2.10	1.21	Slightly Difficult
12. Magsulat ng isang titik sa Baybayin na may kudlit.	2.31	1.08	Slightly Difficult
13. Magsulat ng isang salita sa Baybayin.	2.27	1.05	Slightly Difficult
14. Magsulat ng isang pangungusap sa Baybayin.	2.49	0.95	Slightly Difficult
15. Isulat ang iyong pangalan sa Baybayin.	2.29	1.06	Slightly Difficult
16. Madali kong naisusulat ang aking pangalan gamit ang Baybayinsalita sa Baybayin.	2.23	0.97	Slightly Difficult
17. Madali kong natutukoy ang tamang mga simbolo ng Baybayin.	2.57	0.93	Moderately Difficult
18. Madali kong natutukoy ang tamang mga simbolo ng Baybayin.	2.46	1.0	Slightly Difficult
19. Natutukoy ko agad ang tamang simbolo kapag nagsusulat ng isang salita.	2.59	1.09	Moderately Difficult
20. Naisusulat ko nang tama ang kombinasyon ng mga patinig at katinig sa Baybayin.	2.64	1.02	Moderately Difficult
Overall Mean	2.47	1.04	Moderately Difficult

While most participants found the tasks moderately challenging, perceptions varied. Writing a letter in Baybayin showed the widest range of difficulty (SD = 1.21), while reading sentences or identifying symbols had more consistent ratings with SD values around 0.93 to 1.0. The overall mean score for the preliminary survey on the difficulty of learning Baybayin among grade 7 students at Caniogan National High School is 2.47, which falls within the moderately difficult range, with a standard deviation of 1.04. As noted by Darling (2022), standard deviation represents how dispersed the data are from the mean, where smaller values suggest higher consistency and reliability in the responses. A standard deviation of 0 suggests perfect consistency, while higher values indicate greater variation in responses.

The results show that most respondents find Baybayin reading and writing tasks moderately difficult, with variations based on familiarity and skill. Thematic coding identified three main challenges: lack of learning materials, difficulty retaining characters, and low student engagement. These issues emphasize the need for a more interactive, digital approach to teaching Baybayin. Huang et al. (2020) suggest that features like leaderboards, badges, and collaborative activities can boost motivation and learning outcomes.

2. To design and develop a comprehensive platform for learning Baybayin, incorporating features that enhance learner engagement, facilitate progress tracking, and improve knowledge retention.

Several significant elements have been included in the TalaBaybayin mobile application based on ideas gained from teacher interviews and a student survey. Specifically, these elements were expressly created to solve data collection issues such as limited learning materials, low retention, and a lack of interesting practice tools. As a result, the following elements reflect the system's reaction to these needs and are intended to improve the overall Baybayin learning experience.

2.1 Log-in/Sign-up

The TalaBaybayin app features a secure Login and Sign-Up system powered by Firebase authentication. It safely manages user credentials, logins, and password recovery, ensuring data protection and privacy.

The Log-in and Sign-up interface of the TalaBaybayin Mobile Learning Application, shown in Figure 32, provides users with a secure and seamless experience when accessing or creating their accounts. On the Log-in page, users can enter their registered email and password to safely access their personalized learning environment, which directs them to the main dashboard where they can begin their Baybayin lessons. For new users, the Sign-up page offers a simple registration process by requiring their full name, email address, and password.

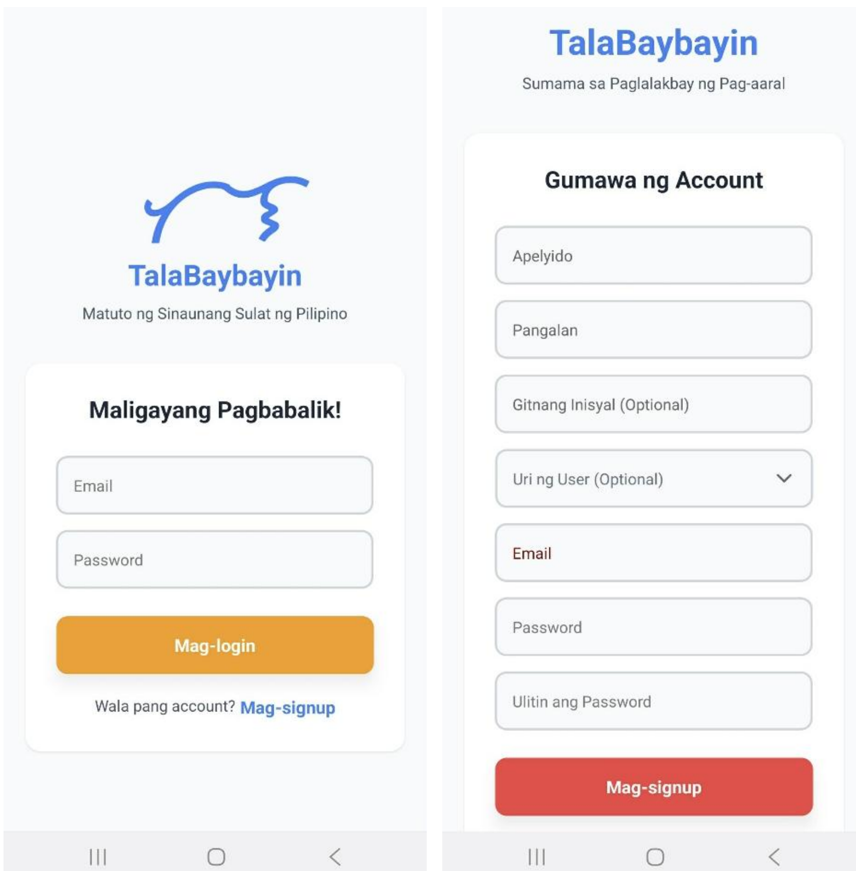
Upon submission, the system verifies the provided information to ensure that no two accounts share the same email, maintaining data integrity and preventing duplicate registrations. If an already registered email is used, an error message is displayed to notify the user. Firebase handles the authentication process, ensuring data security and consistency across all user accounts.

According to Ma'roef et al. (2024), login functionalities typically require users to verify their identity through a username and password, ensuring a straightforward yet secure authentication process. This implementation reflects best practices in mobile app development, providing both reliability and user convenience.

Firebase provides several methods of secure sign-ins, including email/password-based authentication and social login options. In incorporating Firebase, the learning app minimizes user barriers, making the login process both secure and user-friendly. This allows learners to quickly access their accounts without complicated setup steps, encouraging more consistent use of the app. By simplifying account management, users can focus more on learning Baybayin rather than dealing with technical difficulties.

Figure 32.

Log-in/Sign-up Page



(a) (b)

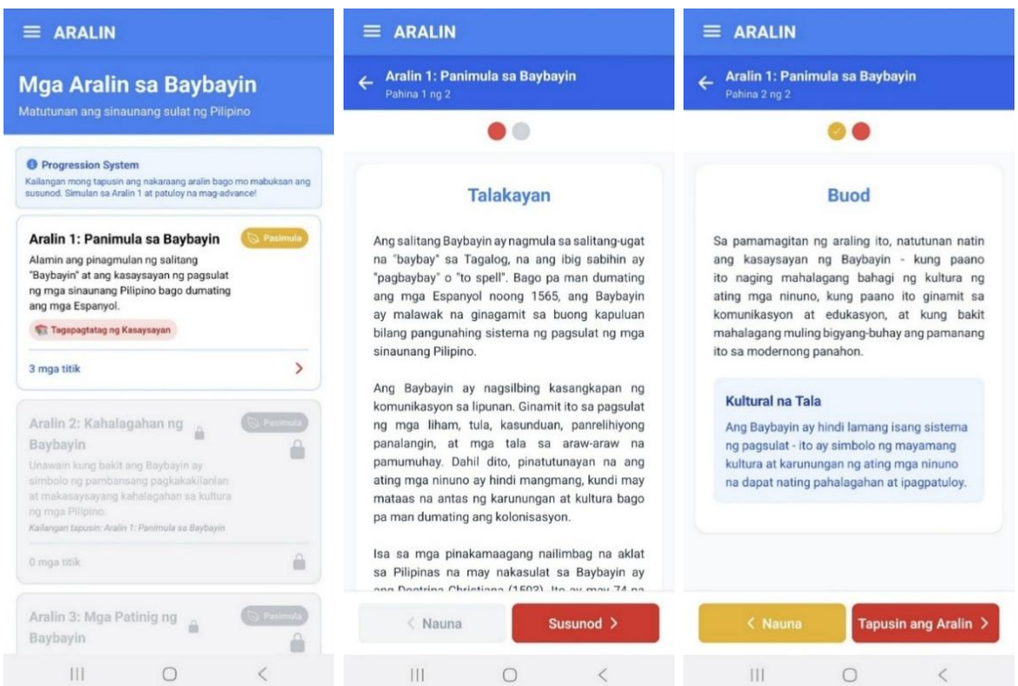
2.2 Lessons

Three Filipino Baybayin teachers validated the TalaBaybayin Mobile Learning Application's lessons for content accuracy, cultural relevance, and teaching suitability, providing feedback to improve clarity. This validation ensures the lessons are educationally sound and authentic.

Figure 33 shows the interactive lesson interface designed to enhance engagement and learning retention. Consistent with Barnett-Itzhaki et al. (2023), interactive learning improves student outcomes. Survey results revealed that 50 out of 70 respondents (70.4%) considered interactive lessons the most important feature in a Baybayin learning app.

Figure 33

Lessons Page



(a) (b) (c)

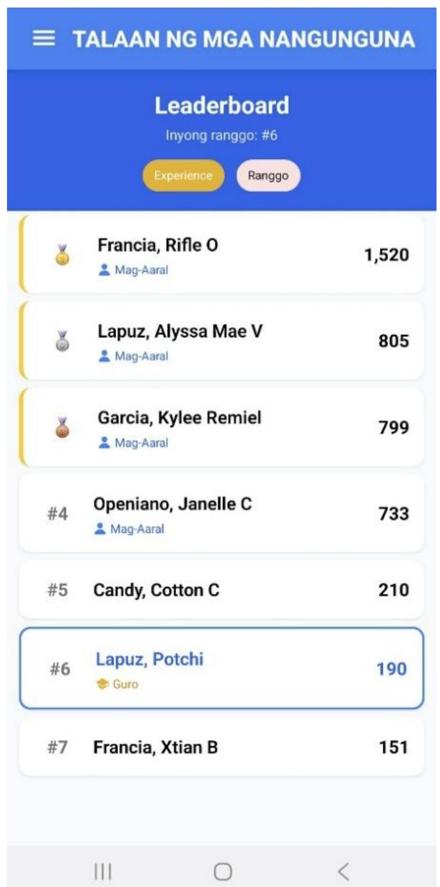
2.3 Leaderboard

The TalaBaybayin app features a Leaderboard that promotes friendly competition and recognizes top performers based on quiz results, lesson completion, and writing accuracy. This motivates learners to stay active and consistent in their studies.

Steiner (2024) noted that gamified systems like Duolingo use points, badges, and rankings to boost motivation and engagement. Inspired by this, TalaBaybayin uses Firebase's real-time database to update scores instantly, making the leaderboard accurate and fair. This feature adds a fun, competitive element that keeps learners motivated and excited to master Baybayin.

Figure 34

Leaderboard Page



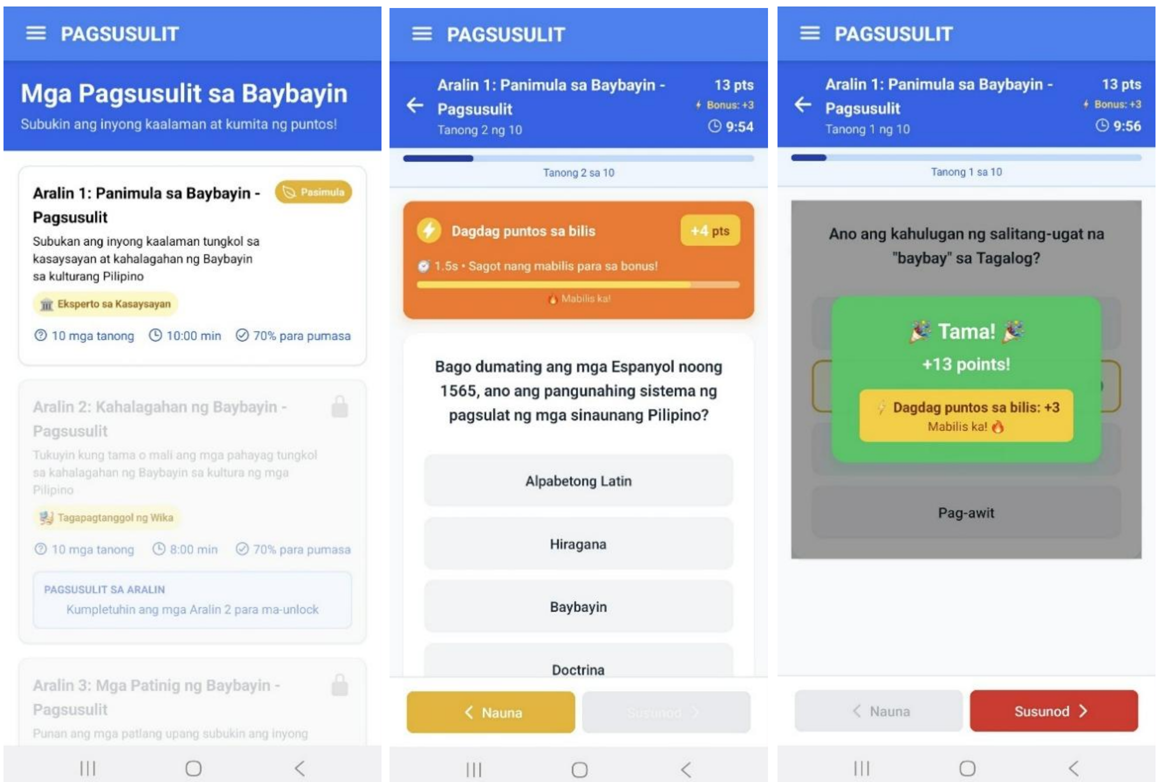
2.4 Time-Based Scoring

The TalaBaybayin app uses a time-based scoring system that rewards accuracy and speed, making quizzes more engaging and encouraging learners to improve their Baybayin recall.

According to Ram et al. (2025), instant feedback enhances motivation, while Steiner (2024) noted that points and rankings sustain engagement. Based on these ideas, TalaBaybayin scores users by response speed with real-time Firebase updates, turning quizzes into fun challenges that motivate learners to improve their Baybayin writing speed and accuracy.

Figure 35

Quiz Interface Showing Time-Based Scoring Feature



(a) (b) (c)

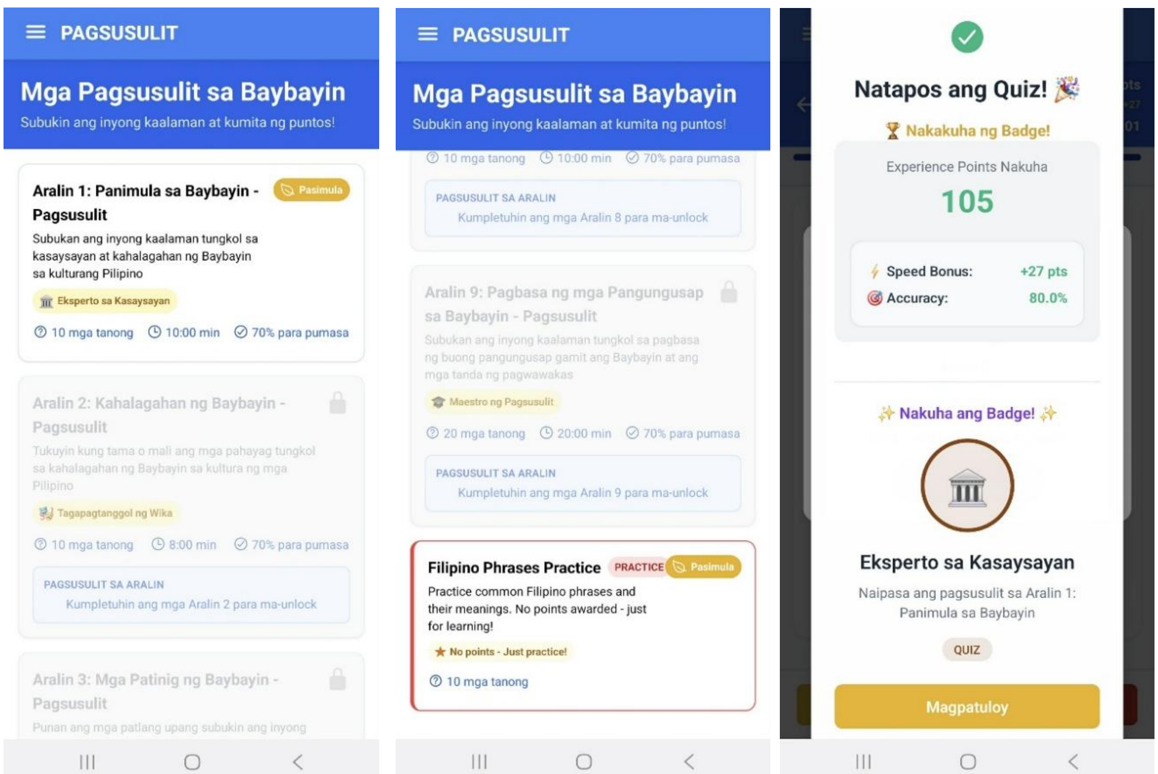
2.5 Quizzes

The quiz feature lets learners test their knowledge of Baybayin characters and words after each lesson. It offers quick, interactive assessments with instant feedback, helping users monitor their progress and recognize areas that require improvement.

Quizzes play an essential role in the learning process as they act as valuable instruments for assessing learners' comprehension and mastery of a subject. Furthermore, they promote knowledge retention and reinforce the understanding of key concepts through active recall. This further improves learning outcomes, encourages consistent practice, and keeps learners motivated to progress.

Figure 36

Quizzes Page



(a) (b) (c)

2.6 Progress Tracking

The Progress Tracking feature records and displays the learner's accomplishments throughout their study. It shows which lessons have been completed, quiz scores, and overall performance.

Gamification plays a crucial role in increasing learners' motivation and engagement in the TalaBaybayin app. It integrates elements such as points, badges, achievements, levels, and rankings to make learning more fun and rewarding. The system encourages learners to participate actively and consistently. These features align with the concepts discussed in Chapter 2, where Akther et al. (2024) highlighted gamification as a key strategy for fostering user engagement, which is further supported by the data and findings.

Figure 37

Gamification Interface Showing Progress Tracking and Achievement Rewards

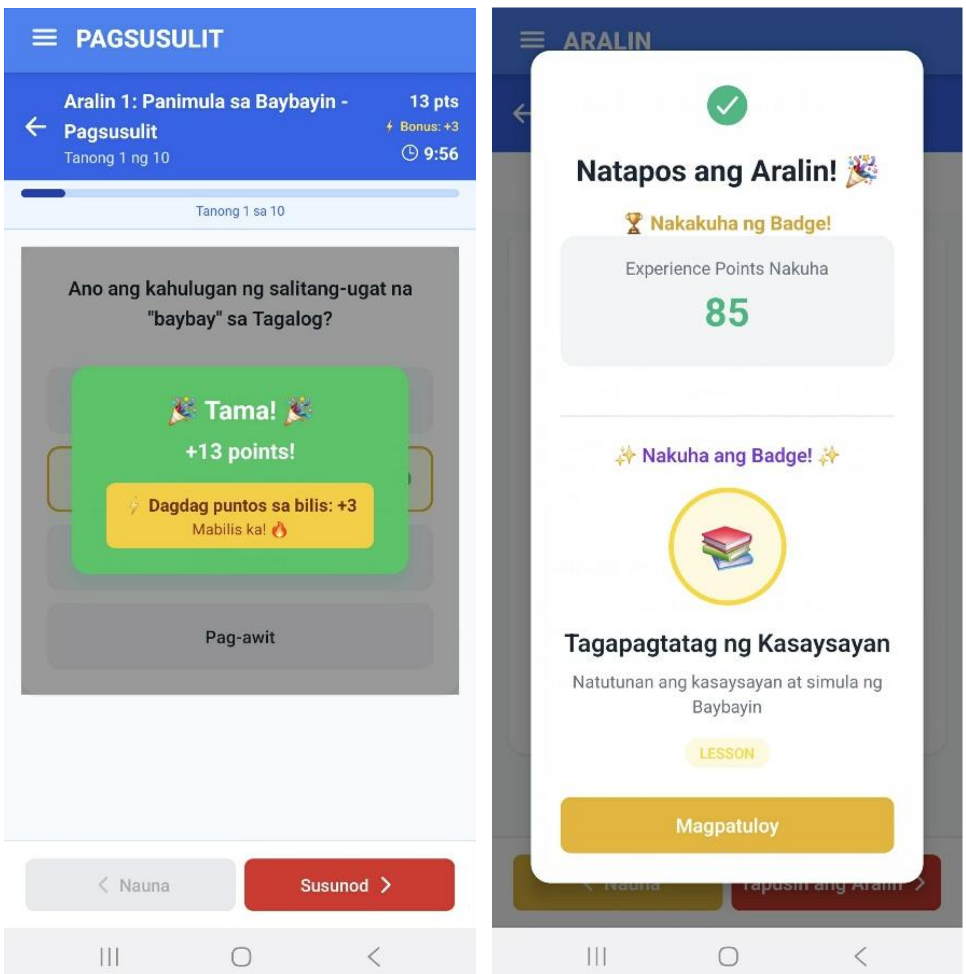
2.7 Gamification Features

The gamification aspect improves the user experience as it incorporates the use of game elements of points, badges and ranking system into the learning system.

The survey revealed that students value gamification for boosting engagement and motivation. Elements such as points and leaderboards make the learning experience more interactive, enjoyable, and goal-oriented. As Steiner (2024) emphasized, these elements can effectively encourage consistent participation, like Duolingo. Additionally, students shared that these game-based rewards give them a sense of progress, making the learning process feel more satisfying.

Figure 38

Game-Based Learning Features Integrated into the TalaBaybayin



(a) (b)

2.8 Mini-games

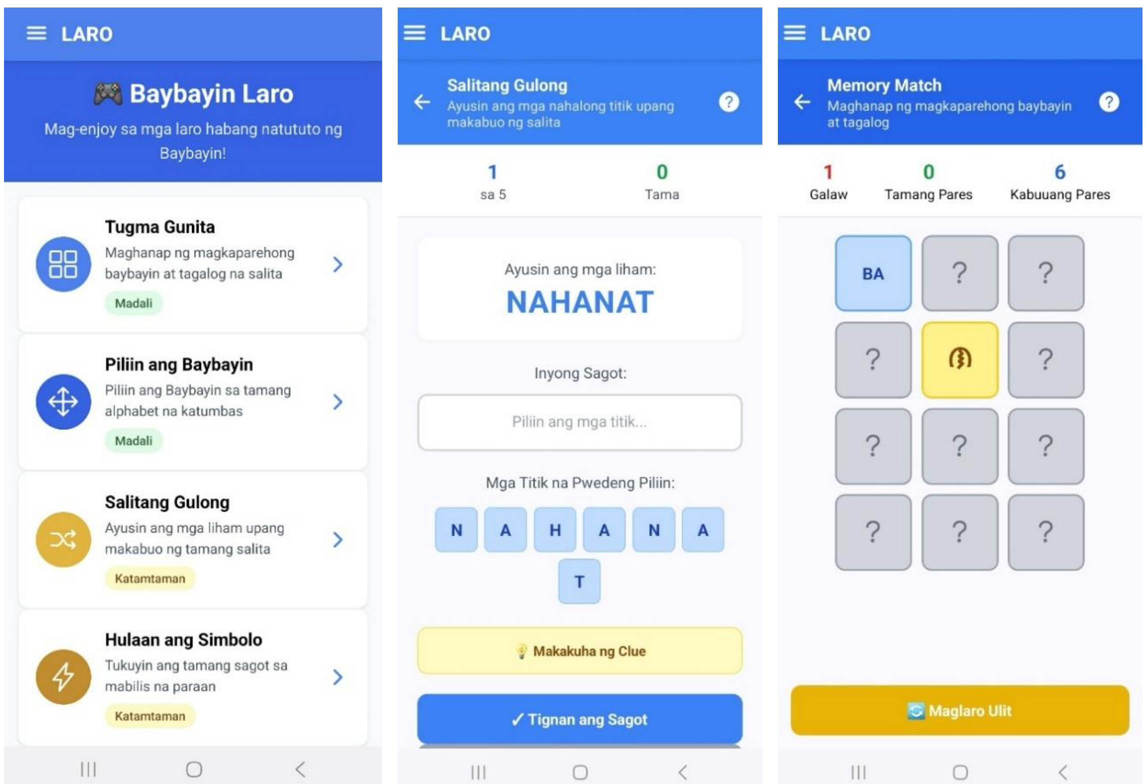
The mini-games feature offers fun activities like memory matching and word scrambling to help users practice Baybayin in an enjoyable way. These games provide extra learning without the pressure of tests, promoting active learning and improving recognition and writing skills. The mini-games interface of the TalaBaybayin Mobile Learning Application is shown in Figure 39, which illustrates how the system teaches and tests users on the Baybayin writing system. The app provides several mini-games designed to make the learning process interactive and enjoyable. In Figure 39 (a), users can choose from multiple games, including *Tugma Gunita* (Memory Game), where players match Baybayin characters with their Tagalog counterparts; *Piliin ang Baybayin* (Pick the Correct Baybayin), which requires selecting the correct Baybayin symbol for a given letter; *Salitang Gulong* (Word Wheel), where users arrange jumbled letters to form a correct Baybayin word; and *Hulaan ang Simbolo* (Guess the Symbol), a fast-paced game that involves identifying the correct Baybayin symbol. In Figure 39 (b), the interface displays the *Salitang Gulong* game, where the player arranges mixed letters to form a word, with a progress bar and timer indicating game completion and remaining time. Lastly, Figure 39 (c) shows the *Memory Match* game, where users pair Baybayin symbols with their corresponding Tagalog translations.

The app effectively integrates interactivity and education, making the process of learning Baybayin both engaging and challenging. Practicing activities need not be as formal and boring as the traditional exercises, and the use of mini-games can make lessons more fun to learn. Phadke et al. (2024) supported this observation by stating that mini-games are both highly effective as learning and engagement tools.

Their study found that mini-games help learners grasp complex ideas, use intuition and pattern recognition, and practice interactively. Including mini-games in the TalaBaybayin app will strengthen learners' mastery of Baybayin and make learning more engaging and motivating. By blending playful experiences with essential skills, the app transforms practice sessions into something students look forward to each day. By blending playful experiences with essential skills, the app transforms practice sessions into something students look forward to each day. This fusion of fun and learning helps build confidence and encourages regular participation, leading to more meaningful progress in mastering Baybayin.

Figure 39

Mini-games



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(a) (b) (c)

2.9 Multimedia sounds

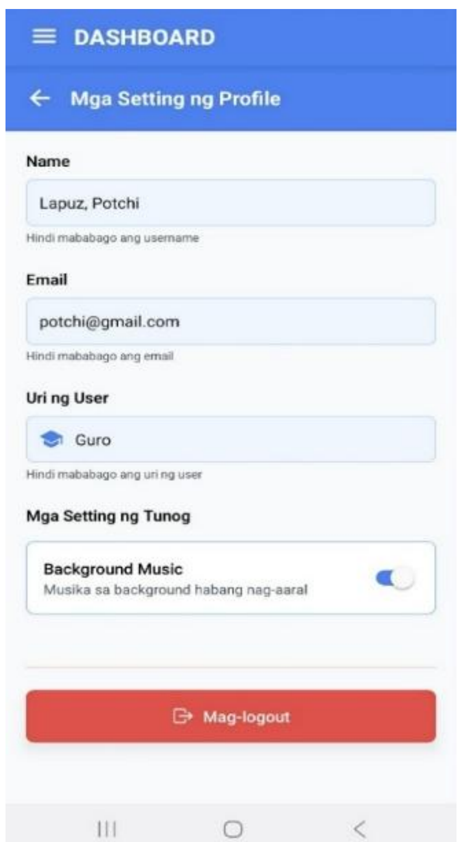
The multimedia sounds feature includes background music and sound effects to make learning more engaging and immersive. Music creates a focused and enjoyable atmosphere, while sound effects provide feedback that supports different learning styles and makes the experience more dynamic and inspiring.

Figure 40 shows the multimedia sound interface of the TalaBaybayin app, where learners can conveniently toggle the sound on or off depending on their preference. The inclusion of background music and sound effects not only enhances engagement and enjoyment but also helps reinforce correct responses and guide learners during activities or quizzes.

Adding background music and sound effects enhances engagement and effectiveness. Studies by Abdulrahaman et al. (2020) and Jian-hua and Hong (2012) found that combining audio, visuals, and text improves understanding, engagement, and memory.

Figure 40

Profile Settings Page



2.10. Common Phrases

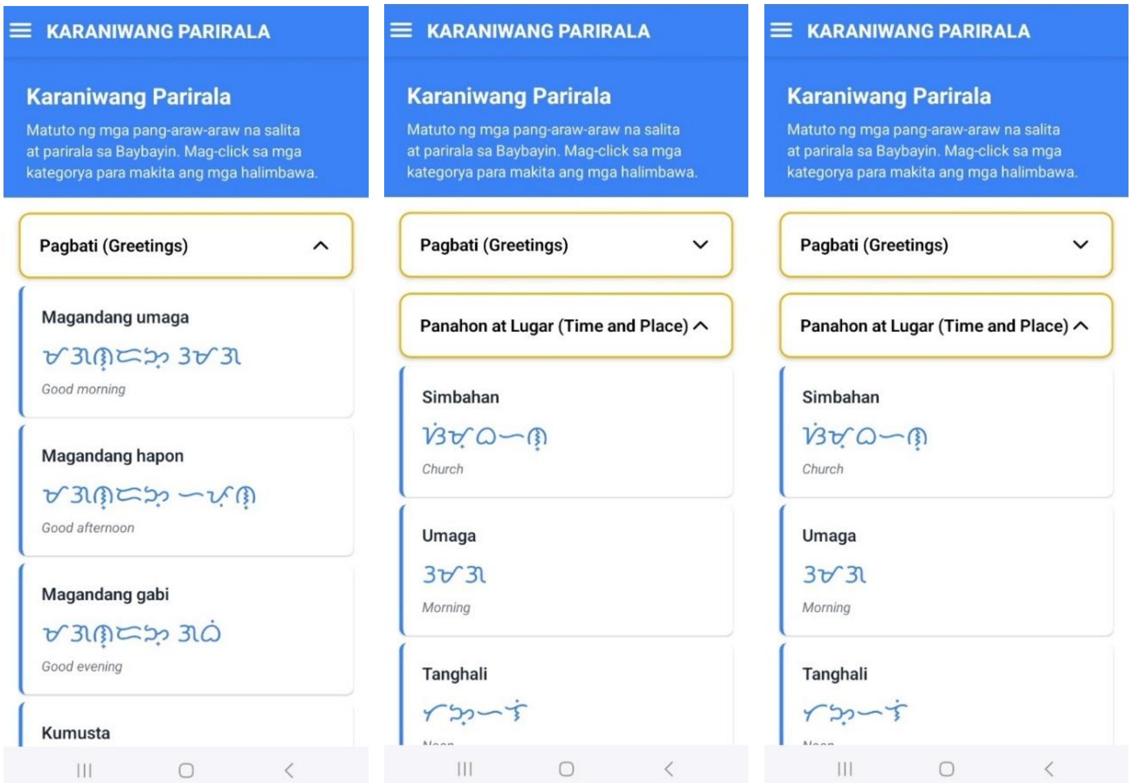
Presentation of common Tagalog phrases that are translated to Baybayin would make the learning process more comprehensive and interactive. Instead of isolating characters or words, this feature centers on entire expressions which are representative of real language usage.

Figure 41 shows the Common Phrases interface of the TalaBaybayin Mobile Learning Application, where learners can view and practice frequently used Tagalog expressions translated into Baybayin. This feature helps learners apply the script in meaningful language contexts, improving their fluency and cultural understanding.

Alcantara et al. (2024) follow this method; they created a machine learning model to translate Baybayin into Tagalog by transliteration at a phrase level. Their experiment indicated that phrase-learning can be actively incorporated into digital tools to allow script training to become more realistic and active. The use of frequent phrases enables learners to apply Baybayin in full language units rather than individual characters, making the learning process more relevant, natural, and easier to remember. This approach highlights the potential of technology-driven language learning to bridge traditional scripts with modern communication practices. It also enhances contextual understanding by showing how Baybayin functions in real communication. Moreover, it strengthens cultural appreciation by connecting learners with authentic Filipino expressions. This method promotes fluency through repetition and pattern recognition, making learning more intuitive. Lastly, it encourages learners to use Baybayin in daily communication, reinforcing both mastery and cultural pride and also encourage sense of identity and belonging by empowering learners.

Figure 41

Common Phrases Page



(a) (b) (c)

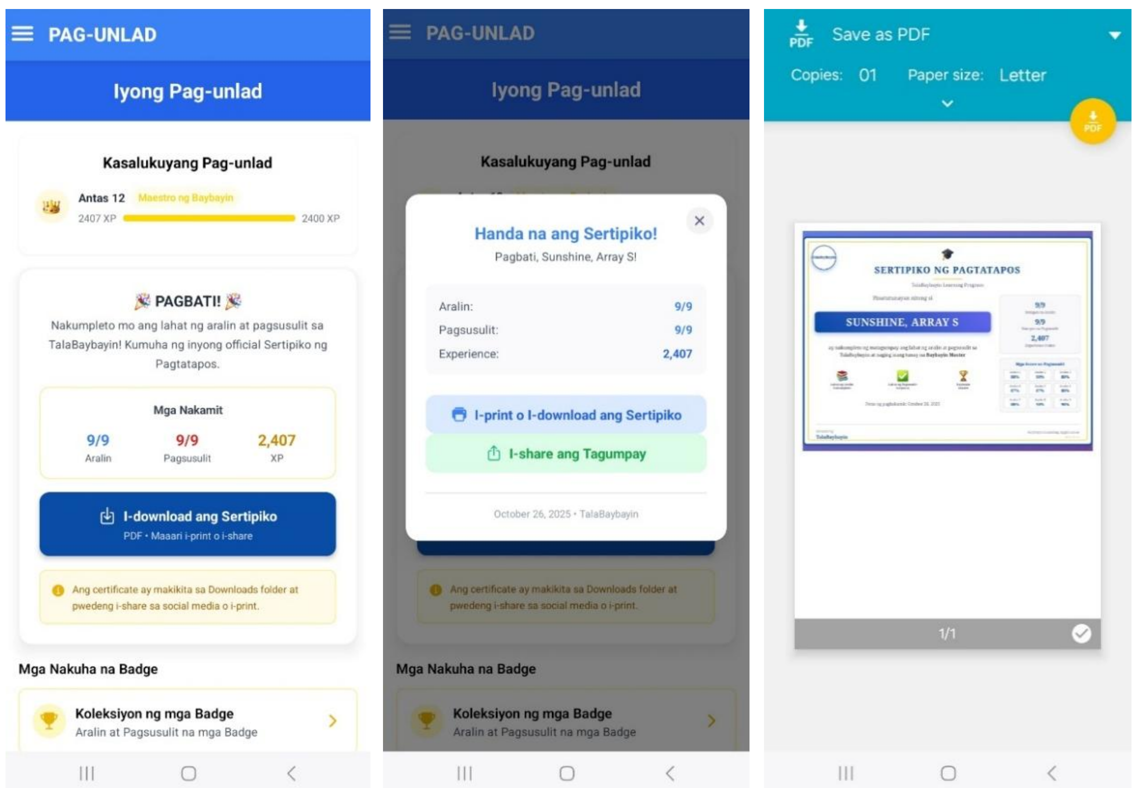
2.11. Completion Certificate

The Completion Certificate in our mobile learning application is conferred to learners upon the successful fulfillment of all lessons and exams. This certificate formally acknowledges their dedication and achievement, underscoring their commitment to ongoing learning and academic advancement.

Figure 42 (a-c) illustrates the Completion Certificate feature in the mobile application. In (a), users can view their progress and see available rewards after completing lessons and quizzes. In (b), the app highlights when a certificate has been earned, reinforcing a user’s sense of achievement. In (c), the completion certificate is displayed and can be saved, signifying official recognition of the user's accomplishment. As highlighted by Shaikh, Rehman, and Rehman (2020), the issuance of a certificate significantly motivates learners by reinforcing accomplishment and encouraging continued engagement.

Figure 42

Completion Certificate



(a) (b) (c)

3. To include a specialized feature that improves the user experience and learning process

The program includes specific functions that were carefully planned to make the user experience better and make learning easier by meeting users' needs and concerns. These features are designed to make the learning environment more engaging and effective, allowing students to learn at their own pace while promoting interaction and collaboration with others.

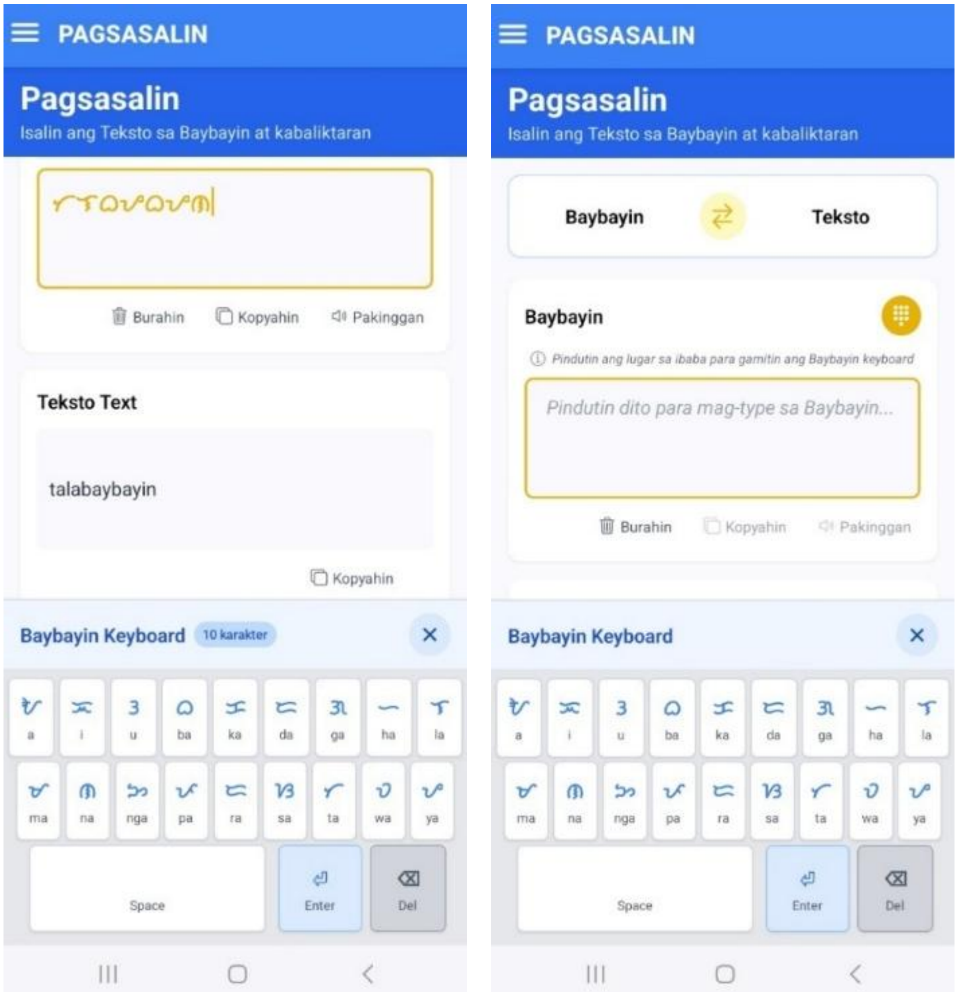
3.1. Baybayin Keyboard

The TalaBaybayin app features a Baybayin Keyboard that lets users type and compose text in Baybayin, enhancing accessibility and allowing interactive practice of what they've learned. This feature enables learners to apply their knowledge in real-time by forming words and sentences using authentic Baybayin characters. Through continuous typing practice, users can develop familiarity with symbol recognition, proper stroke order, and contextual usage, which strengthens retention and writing fluency. The keyboard also provides an avenue for creative expression, allowing learners to personalize messages, names, and phrases using the ancient script.

Figure 43 shows the Baybayin keyboard interface in the TalaBaybayin app, where users can type symbols directly on-screen. The layout is designed with simplicity and responsiveness in mind, ensuring that users can easily access each character without confusion. As Weerakoon (2023) noted, React Native custom keyboards enhance usability through smooth input transitions, contributing to a seamless user experience. This technological integration highlights how modern mobile frameworks can preserve and promote cultural heritage through interactive tools. By combining digital accessibility with traditional writing systems, the Baybayin keyboard not only supports learning but also strengthens cultural appreciation and identity among users. Ultimately, it serves as a bridge between traditional Filipino culture and contemporary digital communication. It encourages learners to actively use Baybayin in daily communication, fostering continuous practice and familiarity.

Figure 43

Baybayin Keyboard



(a) (b)

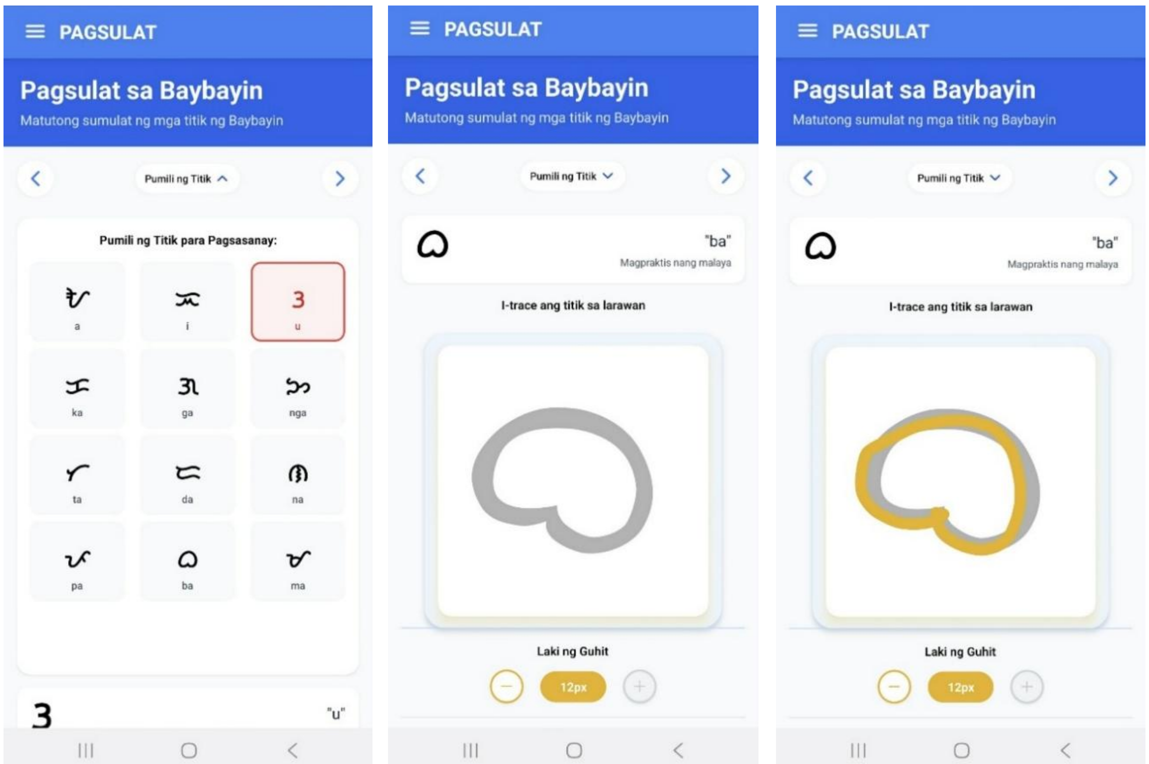
3.2 Handwriting Feature

The handwriting feature allows users to write and practice Baybayin characters directly within the app, promoting better retention and familiarity with the script. Using a touchscreen, learners can trace or freely draw the characters in the provided writing area. The system displays the correct form of each Baybayin symbol as a guide, and saves the strokes so users can go back to it later.

Figure 44a, b, and c shows the handwriting interface of the TalaBaybayin Mobile Learning Application, where learners can trace and write Baybayin symbols directly on the screen.

Figure 44

Handwriting Interface



(a) (b) (c)

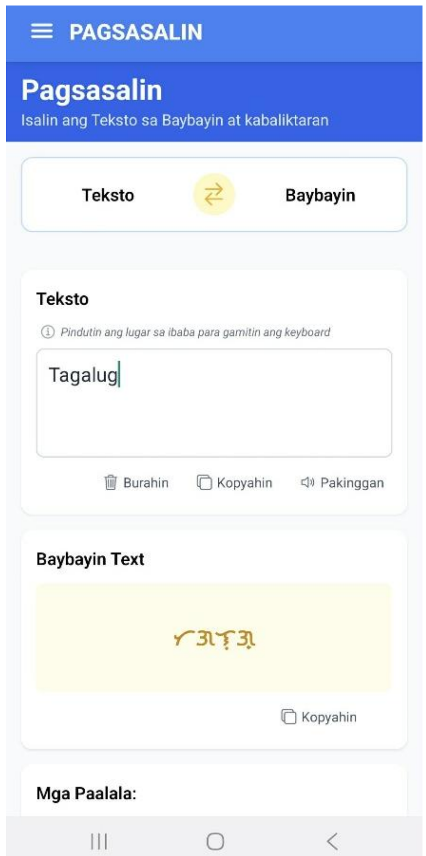
3.4 Text-to-Speech

The text-to-speech feature enables learners to listen to accurate pronunciations of Baybayin characters and words, helping them develop correct pronunciation and a deeper understanding of the script's sound patterns. By providing clear audio output, it supports auditory learners and reinforces the connection between written symbols and their corresponding sounds.

Figure 45 shows the text-to-speech feature of the TalaBaybayin Mobile Learning Application, where learners can listen to the correct pronunciation of Baybayin characters and words by tapping the "*Pakinggan*" button.

Figure 45

Text-to-speech Feature



4. To evaluate the acceptability of the system using ISO/IEC 25010 product quality evaluation.

A total of 100 respondents participated in evaluating the system's acceptability using an ISO/IEC 25010 questionnaire. The respondents included 70 Grade 7 students from Caniogan National High School, 10 teachers, 10 IT experts from the field, and 10 IT professionals from Bulacan State University (BulSU). This diverse group provided valuable feedback on the app's usability, functionality, and overall effectiveness in enhancing the learning of Baybayin.

ISO/IEC 25010:2023, developed by ISO/IEC JTC 1, Subcommittee SC **7** (ISO, 2023), is an international standard for evaluating software quality. It defines nine product quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, portability, and safety, each with detailed sub-characteristics. This framework allows for a comprehensive assessment of both functional and non-functional aspects, guiding the specification, measurement, and evaluation of software throughout its development and use.

4.1 FUNCTIONAL SUITABILITY

The TalaBaybayin app ensures Functional Suitability by providing reliable features that effectively support users' learning goals. Following ISO/IEC 25010:2023 standards, it delivers the necessary functions to meet user needs and create a smooth, goal-oriented learning experience.

Table 8 shows the mean distribution of respondents' responses in terms of functional suitability.

Table 8

Mean Distribution of Respondents' Responses in terms of Functional Suitability

Functional Suitability	Mean	Standard Deviation	Descriptive Interpretation
1. Functional completeness	3.58	0.78	Highly Acceptable
2. Functional correctness	3.70	0.59	Highly Acceptable
3. Functional appropriateness	3.58	0.86	Highly Acceptable
Mean	3.62	0.74	Highly Acceptable

The table illustrates how respondents assessed the Functional Suitability of the TalaBaybayin Mobile Learning Application, evaluated through three main criteria: Functional Completeness, Functional Correctness, and Functional Appropriateness. For Functional Completeness (Naibibigay ng TalaBaybayin Mobile Learning Application ang lahat ng kinakailangang function para magawa ang mga nakatakdang gawain at layunin ng mga gumagamit), the application obtained a mean score of 3.58 and a standard deviation of 0.78. This result implies that users generally agree the app offers the essential functions required to accomplish tasks and goals, though there is a moderate level of response variation. While most respondents perceive the app as sufficiently complete, a few expressed differing opinions, indicating minor areas for improvement. Regarding Functional Correctness (Nagbibigay ang TalaBaybayin Mobile Learning Application ng tamang resulta kapag ginamit ng mga nakatakdang gumagamit), the app achieved a mean score of 3.70 and a standard deviation of 0.59, reflecting strong agreement that the system provides accurate and reliable outputs.

The relatively lower standard deviation suggests that users' responses were more consistent, indicating a shared perception that the application performs its functions accurately and dependably. Meanwhile, for Functional Appropriateness (Naibibigay ng TalaBaybayin Mobile Learning Application ang mga function na nakakatulong para matapos ang mga gawain at layunin), the app obtained a mean of 3.58 and a standard deviation of 0.86. This implies that most respondents believe the app's features effectively support them in completing their tasks and objectives. However, the slightly higher standard deviation reflects a wider range of opinions, suggesting that user experiences with this aspect of the app may vary.

This explains that while many users find the app's functions useful, some may feel that it could be more tailored to their needs or more efficient in supporting task completion. The Mean for Functional Suitability is 3.62, which falls under the "Highly Acceptable" category, with a standard deviation of 0.74. This indicates that the *TalaBaybayin* Mobile Learning Application is generally seen as highly functional, meeting the needs of users in terms of completeness, correctness, and appropriateness.

4.2 PERFORMANCE EFFICIENCY

The TalaBaybayin Mobile Learning Application upholds Performance Efficiency by ensuring smooth, fast, and reliable operation throughout its use, allowing learners to navigate lessons and activities without delay or interruption. It is designed to perform essential functions within an optimal time frame while efficiently managing device resources such as memory, processing power, and storage. This balance of speed and stability ensures that the app remains responsive even when multiple features are in use, providing users with a fluid and uninterrupted learning experience. Moreover, the app is optimized to function efficiently across a wide range of Android devices, maintaining stable performance even with varying hardware specifications and internet connectivity levels, making it accessible and reliable for all learners. The system also incorporates efficient caching and data loading mechanisms to reduce waiting times, allowing users to access content faster and continue their learning without unnecessary delays.

In accordance with the ISO/IEC 25010:2023 software quality standard, Performance Efficiency measures how effectively a system utilizes resources and maintains stable performance under varying conditions. The TalaBaybayin Mobile Learning Application adheres to this standard by implementing optimized code execution, streamlined data processing, and adaptive performance strategies that adjust to different device capabilities. Additionally, the app undergoes continuous performance monitoring and testing to ensure that updates and new features do not negatively affect its speed or responsiveness. These measures minimize latency, prevent system overload, and enhance responsiveness, ensuring that users experience consistent speed and reliability. By prioritizing efficiency and stability.

Table 9 shows the mean distribution of respondents' responses in terms of performance efficiency.

Table 9

Mean Distribution of Respondents' Responses in terms of Performance Efficiency

Performance Efficiency	Mean	Standard Deviation	Descriptive Interpretation
1. Time behavior	3.57	0.76	Highly Acceptable
2. Resource utilization	3.48	0.83	Highly Acceptable
3. Capacity	3.53	0.85	Highly Acceptable
Mean	3.53	0.81	Highly Acceptable

The table presents the respondents' assessment of the Performance Efficiency of the TalaBaybayin Mobile Learning Application, which was evaluated based on three dimensions: Time Behavior, Resource Utilization, and Capacity. For Time Behavior (Ginagawa ng TalaBaybayin Mobile Learning Application ang tungkulin nito sa tamang oras at bilis ayon sa kailangan), the application obtained a mean score of 3.57 and a standard deviation of 0.76. This indicates that most respondents agree the app carries out its functions promptly and at an appropriate speed. However, the moderate variation in responses implies that a few users experienced differences in the app's responsiveness or loading time.

In terms of Resource Utilization (*Hindi gumagamit ang TalaBaybayin* Mobile Learning Application *ng sobra sa itinakdang dami ng resources kapag gumagana*), the app earned a mean of 3.48 and a standard deviation of 0.83, showing that most users find the app efficient in using resources. However, the higher standard deviation signifies there is some divergence in user opinions regarding the app's resource efficiency.

For Capacity (*Natutugunan ng TalaBaybayin* Mobile Learning Application *ang mga pangangailangan kaugnay ng pinakamataas na hangganan ng sistem*), the app scored a mean of 3.53 and a standard deviation of 0.85. This indicates that the app generally meets the system's maximum capacity requirements, but the higher standard deviation points to varying experiences among users regarding its performance under maximum conditions.

The mean score for Performance Efficiency is 3.53 (Highly Acceptable) with a standard deviation of 0.81. This indicates that the app is efficient in terms of time, resource usage, and capacity. However, the moderate variation suggests that some users experienced inconsistencies, pointing to areas where performance can still be improved.

4.3 COMPATIBILITY

The TalaBaybayin mobile app is developed exclusively for Android devices to ensure smooth functionality, accessibility, and reliable performance. The minimum software requirement is Android 8.1 (Oreo), allowing users with older yet stable devices to run the app efficiently. The recommended version is Android 16, providing enhanced speed, security, and improved visual performance for a smoother learning experience.

Table 10 shows the mean distribution of respondents' responses in terms of compatibility.

Table 10

Mean Distribution of Respondents' Responses in terms of Compatibility

Compatibility	Mean	Standard Deviation	Descriptive Interpretation
1.Co-existence	3.67	0.63	Highly Acceptable
2.Interoperability	3.58	0.78	Highly Acceptable
Mean	3.63	0.71	Highly Acceptable

The table outlines the respondents' assessment of the Compatibility of the TalaBaybayin Mobile Learning Application, focusing on two primary aspects: Co-existence and Interoperability.

For Co-existence (Kayang gumana ng TalaBaybayin Mobile Learning Application ng maayos habang kasama o ka-share ng ibang produkto nang hindi nakakasira dito), the application obtained a mean score of 3.67 and a standard deviation of 0.63. This suggests that users generally agree the app operates effectively alongside other software or applications without causing conflicts or malfunctions. The relatively low standard deviation further implies that the respondents' experiences were largely consistent, showing minimal differences in their assessments.

In terms of Interoperability (*Kayang makipagpalitan ng impormasyon ng TalaBaybayin* Mobile Learning Application *sa ibang produkto at magamit ito nang sabay*), the app earned a mean of 3.58 and a standard deviation of 0.78. This shows that most users find the app capable of exchanging information with and functioning alongside other products. However, the slightly higher standard deviation indicates some variation in user experiences, suggesting that while many users are satisfied, a few may have encountered challenges with the app's interoperability.

The Mean for Compatibility is 3.63, categorized as “Highly Acceptable,” with an average standard deviation of 0.71. The findings suggest that respondents perceive the TalaBaybayin Mobile Learning Application as highly compatible with other applications, facilitating seamless integration and efficient information exchange, thereby ensuring a dependable and user-friendly experience on supported Android devices. This level of compatibility helps reduce technical issues that could interfere with learning. It also reassures users that the app will function smoothly across different device setups, enhancing overall reliability and convenience.

4.4 INTERACTION CAPABILITY

The TalaBaybayin Mobile Learning Application upholds Interaction Capability in accordance with the ISO/IEC 25010:2023 software quality standard, ensuring that users can efficiently and meaningfully engage with the system. It provides a user-friendly interface that supports smooth navigation, clear feedback, and accessible functions designed to meet diverse user needs. Simple and intuitive, the app helps users of all levels find their way with ease.

Table 11 shows the mean distribution of respondents’ responses in terms of interaction capability. It presents how users evaluated the system’s ability to provide responsive, intuitive, and user-friendly interactions that enhance the overall learning experience. The results indicate that users found the system easy to navigate and engaging to use. Interactive elements were effective in maintaining learners’ interest and promoting active participation. Overall, the system’s interactive design contributed to a more immersive and enjoyable learning process. This shows that effective interaction significantly supports user satisfaction and learning retention.

Table 11

Mean Distribution of Respondents' Responses in terms of Interaction Capability

Interaction Capability	Mean	Standard Deviation	Descriptive Interpretation
1. Appropriateness Recognizability	3.63	0.78	Highly Acceptable
2. Learnability	3.62	0.72	Highly Acceptable
3. Operability	3.60	0.73	Highly Acceptable
4. User Error Protection	3.49	0.84	Highly Acceptable
5. User Engagement	3.88	0.63	Highly Acceptable
6. Inclusivity	3.84	0.62	Highly Acceptable
7. User Assistance	3.81	0.68	Highly Acceptable
8. Self-Descriptiveness	3.85	0.78	Highly Acceptable
Mean	3.85	0.72	Highly Acceptable

The table illustrates the respondents' assessment of the Interaction Capability of the TalaBaybayin Mobile Learning Application, which was evaluated across eight essential dimensions: Appropriateness Recognizability, Learnability, Operability, User Error Protection, User Engagement, Inclusivity, User Assistance, and Self-Descriptiveness.

For Appropriateness Recognizability (Nakikita ng mga gumagamit na ang TalaBaybayin Mobile Learning Application ay bagay sa kanilang pangangailangan), the application obtained a mean score of 3.63 and a standard deviation of 0.78. This suggests that users generally perceive the app as suitable and relevant to their needs. The moderately low standard deviation indicates a shared agreement among most respondents, although slight variations in perception were also observed.

In terms of Learnability (*Madaling matutunan ng mga gumagamit kung paano gamitin ang mga function ng TalaBaybayin App sa takdang oras*), the app earned a mean of 3.62 and a standard deviation of 0.72. This indicates that users find the app easy to learn and use within the required time. The relatively low standard deviation shows consistent user experience, with minimal variation in how easy users find the app to learn.

For Operability (*May mga function at katangian ang TalaBaybayin App na madaling kontrolin at patakbuhin*), the app scored a mean of 3.60 and a standard deviation of 0.73. This implies that users generally find the app's functions and features easy to control and operate. The moderate standard deviation signifies that while most users are satisfied, there is some variability in their experiences with the app's operability.

Regarding User Error Protection (*Kayang pigilan ng TalaBaybayin* Mobile Learning Application *ang maling pagpapatakbo*), the app received a mean of 3.49 and a standard deviation of 0.84. This indicates that while most users find the app capable of preventing user errors, the higher standard deviation points to more variation in experiences. Some users may have faced challenges with error protection, suggesting that further improvements could be made in this area.

For User Engagement (Kaaya-aya at nakakaengganyong gamitin ang TalaBaybayin App para ipagpatuloy ang pakikipag-ugnayan), the application obtained a mean score of 3.88 and a standard deviation of 0.63. This relatively high rating indicates that respondents find the app enjoyable and motivating to use. The low standard deviation further signifies that user feedback was largely consistent, showing general agreement about the app's engaging nature.

In terms of Inclusivity (Maaaring gamitin ang TalaBaybayin Mobile Learning Application ng iba't ibang uri ng tao), the app received a mean of 3.84 and a standard deviation of 0.62. This suggests that most users believe the app is inclusive and can be accessed by individuals with diverse backgrounds and characteristics. The minimal variation in responses implies a strong consensus regarding the app's accessibility and inclusivity.

For User Assistance (Magagamit ng mga tao na may iba't ibang kakayahan at katangian ang TalaBaybayin App para maabot ang kanilang layunin), the app earned a mean score of 3.81 with a standard deviation of 0.68. This result reflects that users generally perceive the app as supportive and helpful in guiding individuals with different abilities toward achieving their learning goals. The relatively low standard deviation indicates agreement among most respondents, though minor variations in user experiences were noted.

Lastly, regarding Self-Descriptiveness (Kayang magpakita ng tamang impormasyon ang TalaBaybayin App upang madaling maintindihan ng gumagamit kung paano ito gagamitin), the app achieved a mean of 3.85 and a standard deviation of 0.78. This implies that users find the app's information clear, informative, and easy to follow. However, the moderate standard deviation suggests that some respondents differed slightly in how intuitively they understood the app's instructions and functions.

The Mean for Interaction Capability is 3.85, categorizing it as "Highly Acceptable," with a mean standard deviation of 0.72. The app is regarded as highly effective in interaction, particularly in user engagement, inclusivity, and self-descriptiveness. The moderate variation in standard deviations indicates that, although the majority of users report positive experiences, there remain aspects of the app that require enhancement to achieve a more uniform user experience across the board.

4.5. RELIABILITY

The TalaBaybayin Mobile Learning Application upholds Reliability to ensure consistent performance, data stability, and dependable functionality throughout use. It is designed to operate smoothly even under varying network conditions, maintaining a steady connection and preventing unexpected crashes, data loss, or interruptions. Through the integration of Firebase's real-time database, the system synchronizes user progress, quiz results, and other essential data across sessions, ensuring that learners' information remains accurate, secure, and continuously updated. This enables users to resume their learning journey anytime without losing progress, promoting both convenience and trust in the app's performance.

In alignment with the ISO/IEC 25010:2023 software quality standard, Reliability measures a system's ability to maintain its performance and recover from failures under defined conditions. The TalaBaybayin Mobile Learning Application follows this principle by employing efficient data handling, automated backups, and thorough system testing to minimize errors and downtime. These measures ensure that users experience a dependable and uninterrupted platform for studying Baybayin, fostering confidence, consistency, and an optimal digital learning environment, while also promoting cultural appreciation and engagement through modern technology.

Table 12 shows the mean distribution of respondents' responses in terms of reliability.

Table 12

Mean Distribution of Respondents' Responses in terms of Reliability

Reliability	Mean	Standard Deviation	Descriptive Interpretation
1. Faultlessness	3.66	0.75	Highly Acceptable
2. Availability	3.6	0.82	Highly Acceptable
3. Fault Tolerance	3.46	0.92	Highly Acceptable
4. Recoverability	3.63	0.74	Highly Acceptable
Mean	3.59	0.81	Highly Acceptable

The table presents an analysis of respondents' evaluations regarding the reliability of the TalaBaybayin Mobile Learning Application, emphasizing four primary dimensions. Faultlessness, Availability, Fault Tolerance, and Recoverability.

For Faultlessness (*Gumagana nang tama ang TalaBaybayin App sa normal na paggamit*), the app received a mean of 3.66 and a standard deviation of 0.75. This refer to users generally find the app to work correctly during normal use. The relatively low standard deviation indicates that most respondents share similar positive experiences, with minimal variation in their perceptions of the app's faultlessness.

In terms of Availability (*Naa-access at nagagamit ang TalaBaybayin* Mobile Learning *Application kapag kinakailangan*), the app earned a mean of 3.60 and a standard deviation of 0.82. This shows that users feel the app is generally available and accessible when needed. However, the slightly higher standard deviation indicates some variation in experiences, meaning that while most users are satisfied, a few may have encountered issues with availability.

For Fault Tolerance (*Gumagana pa rin ang TalaBaybayin* Mobile Learning Application *kahit may error sa hardware o software*), the app scored a mean of 3.46 and a standard deviation of 0.92. This indicates that while the app generally continues to function even when there are errors, the higher standard deviation implies more variability in user experiences. Some users might have experienced difficulties with the app's fault tolerance, indicating a potential area for improvement.

In terms of Recoverability (*Kayang ibalik ng TalaBaybayin* Mobile Learning Application *ang data at maayos muli ang kalagayan nito matapos ang problema*), the app received a mean of 3.63 and a standard deviation of 0.74. This reflects that users generally feel confident in the app's ability to recover from issues and restore data. The relatively low standard deviation indicates consistent user experience regarding the app's recoverability.

The Mean for Reliability is 3.59, categorized as "Highly Acceptable," with a mean standard deviation of 0.81. The app is regarded as highly reliable, especially concerning faultlessness, availability, and recoverability. The moderate variation in standard deviations, particularly regarding fault tolerance, indicates potential areas for enhancing the app's reliability to provide a more consistent user experience

4.6 SECURITY

The TalaBaybayin mobile app integrates security features to safeguard user information and maintain confidentiality. Authentication through Firebase ensures secure login and password management, preventing unauthorized access. According to Rajappa et al. (2020), Shukla et al. (2021), and Saraf (2022), Firebase authentication and real-time databases provide reliable security measures that ensure data protection and user privacy in mobile applications.

Table 13 shows the mean distribution of respondents' responses in terms of security.

Table 13

Mean Distribution of Respondents' Responses in terms of Security

Security	Mean	Standard Deviation	Interpretation
1. Confidentiality	3.71	0.67	Highly Acceptable
2. Integrity	3.61	0.79	Highly Acceptable
3. Non-repudiation	3.60	0.72	Highly Acceptable
4. Accountability	3.57	0.86	Highly Acceptable
5. Authenticity	3.71	0.66	Highly Acceptable
6. Resistance	3.51	0.89	Highly Acceptable
Mean	3.62	3.62	Highly Acceptable

The table illustrates the respondents' assessment of the Security of the TalaBaybayin Mobile Learning Application, evaluated across six important dimensions: Confidentiality, Integrity, Non-repudiation, Accountability, Authenticity, and Resistance.

For Confidentiality, the application obtained a mean score of 3.71 and a standard deviation of 0.67. This suggests that users generally trust the app's capability to safeguard their personal information and maintain data privacy. The relatively low standard deviation indicates that most respondents share similar positive perceptions, showing consistent confidence in the app's data protection measures.

Regarding Integrity (Tinitiyak ng TalaBaybayin Mobile Learning Application na protektado laban sa pagbabago o pagbura ng walang pahintulot), the app achieved a mean of 3.61 and a standard deviation of 0.79. This implies that users believe the system effectively prevents unauthorized modifications or deletions of data. The moderate variation in responses indicates that while many users view the app as reliable in ensuring data integrity, others have slightly different levels of satisfaction.

For Non-repudiation (*Kayang patunayan ng TalaBaybayin* Mobile Learning Application *ang isang aksyon o pangyayari upang hindi na ito maitanggi*), the app scored a mean of 3.60 and a standard deviation of 0.72. This proves that users generally feel confident that the app can prove actions or events to prevent denial. The relatively low standard deviation indicates that most respondents have similar views on this feature.

In terms of Accountability (*Kayang itala ng TalaBaybayin* Mobile Learning Application *ang aksyon ng bawat gumagamit*), the app received a mean of 3.57 and a standard deviation of 0.86. This indicates that while users believe the app can track and record user actions, the higher standard deviation suggests more variability in user experiences, implying some users may have concerns or differing views regarding the app's accountability features.

For Authenticity (*Kayang patunayan ng TalaBaybayin* Mobile Learning Application *ang pagkakakilanlan ng isang tao o resource*), the app earned mean of 3.71 and a standard deviation of 0.66. This shows that users generally feel confident in the app's ability to verify identities and resources. The low standard deviation indicates a consistent positive experience across most users.

Regarding Resistance (*Kayang magpatuloy ang TalaBaybayin* Mobile Learning Application *kahit may masamang atake*), the app received a mean of 3.51 and a standard deviation of 0.89. This explains that users feel the app can continue functioning even in the event of a malicious attack. However, the higher standard deviation points to greater variation in user experiences, indicating that some users may have encountered issues or are less confident in the app's resistance to attacks.

The Mean for the table of Security is 3.62, which falls under the "Highly Acceptable" category with a mean standard deviation of 0.76. Overall, the app is perceived as highly secure, particularly in terms of confidentiality, authenticity, and integrity. However, the moderate variation in standard deviations, especially in areas like resistance and accountability, signifies that there are opportunities to enhance the app's security features to provide a more consistent and reliable experience for all users. Continuous updates and regular system audits are recommended to maintain strong protection against potential threats and ensure long-term data safety. Security remains a top priority. Users can trust their information is protected. Implementing advanced encryption methods and secure authentication protocols can further strengthen the overall security framework of the application.

4.7 MAINTAINABILITY

The TalaBaybayin mobile app is structured for maintainability to support efficient updates, feature improvements, and long-term functionality. Using a modular design built on React Native, developers can modify specific components or correct errors without affecting the rest of the system. This ensures the app remains stable and adaptable to future enhancements. Liu et al. (2024) and Ramachandrappa (2024) highlighted that React Native's modular structure and reusable components simplify maintenance and allow developers to implement system updates more efficiently. This approach reduces downtime and prevents major disruptions for users during updates. As a result, the app can continuously evolve while sustaining consistent performance and user satisfaction. The design also supports faster troubleshooting. It minimizes the risk of system-wide failures. It allows updates to be deployed more seamlessly. It ensures the platform remains reliable over time. It strengthens the overall sustainability of the application.

Table 14 shows the mean distribution of respondents' responses in terms of maintainability.

Table 14

Mean Distribution of Respondents' Responses in terms of Maintainability

Maintainability	Weighted Mean	Standard Deviation	Interpretation
1. Modularity	3.65	0.67	Highly Acceptable
2. Reusability	3.67	0.75	Highly Acceptable
3. Analysability	3.70	0.66	Highly Acceptable
4. Modifiability	3.59	0.78	Highly Acceptable
5. Testability	3.70	0.64	Highly Acceptable
Mean	3.66	0.70	Highly Acceptable

The Maintainability of the TalaBaybayin Mobile Learning Application is evaluated across five dimensions: Modularity, Reusability, Analysability, Modifiability, and Testability.

For Modularity (*Ang pagbabago sa isang bahagi ay kayang malimitahan ang epekto sa ibang bahagi*), the app received a mean of 3.65 and a standard deviation of 0.67. This indicates that, in general, the app is designed in a way that changes to one part do not significantly affect other areas. The relatively low standard deviation indicates that most users agree on the app's modular structure, with minimal variation in their responses.

Regarding Reusability (*Maaaring magamit ang* TalaBaybayin Mobile Learning Application *sa iba pang proyekto o bilang bahagi ng iba pang produkto*), the app earned a mean of 3.67 and a standard deviation of 0.75. This indicates that the app is perceived as useful for integration into other projects or as part of other products. The moderate standard deviation proves that while many users are satisfied with its reusability, there are some varying opinions on its flexibility for use in other contexts.

In terms of Analysability (*Maaaring masuri nang epektibo at mahusay ang TalaBaybayin* Mobile Learning Application *upang matukoy ang epekto ng inaasahang pagbabago sa isa o higit pang bahagi nito*), the app scored a mean of 3.70 and a standard deviation of 0.66. This indicates that users generally feel the app is suitable for analysis and identifying the effects of changes. The low standard deviation proves that most respondents share a similar positive outlook regarding its analysability.

When evaluating Modifiability (*Madaling baguhin ang TalaBaybayin App nang hindi nagkakaroon ng dagdag na mali o pagbaba ng kalidad*), the app received a mean of 3.59 and a standard deviation of 0.78. This reflects the general perception that the app is easy to modify without compromising its quality. However, the slightly higher standard deviation implies that some users experienced varying levels of ease when modifying the app, indicating potential areas for improvement.

For Testability (*Ang TalaBaybayin App ay may kakayahang magbigay-daan sa isang obhetibo at praktikal na pagsubok na maaaring idisenyo at isagawa upang matukoy kung natutugunan ang isang kinakailangan*), the app earned a mean of 3.70 and a standard deviation of 0.64. This signifies that users believe the app supports objective and practical testing to ensure that it meets its requirements. The low standard deviation signifies that responses were generally consistent, with little variation in perceptions of its testability.

The mean for Maintainability is 3.66, both of which fall under the "Highly Acceptable" category with a standard deviation of 0.70. Overall, the TalaBaybayin Mobile Learning Application is seen as highly maintainable, excelling in aspects such as modularity, reusability, and testability. The moderate variation in the standard deviations points that, while the app is largely well-regarded, there are areas where refinements could be made to further enhance its maintainability and adaptability across different user needs. his suggests that refining certain components could further streamline future updates. Continued improvement in this area will help ensure that the app remains scalable and adaptable to evolving user needs.

4.8 FLEXIBILITY

Flexibility is an important quality of software that allows it to adapt to various environments, accommodate changes in capacity, and be easily installed or replaced without disrupting the user experience. According to Galvez et al. (2023), applications that demonstrate high flexibility are more likely to maintain usability across different devices and user contexts. This ensures that users can access the app smoothly, whether on mobile phones or tablets, without performance issues. In the case of TalaBaybayin, flexibility supports continuous learning by allowing users to study anytime, anywhere. Moreover, this adaptability ensures that the app remains functional and relevant even as technology or user needs evolve. Table 15 presents the respondents' perceptions of the TalaBaybayin Mobile Learning Application in terms of its adaptability, scalability, installability, and replaceability.

Table 15

Mean Distribution of Respondents' Responses in terms of Flexibility

Flexibility	Mean	Standard Deviation	Interpretation
1. Adaptability	3.67	0.63	Highly Acceptable
2. Scalability	3.61	0.77	Highly Acceptable
3. Installability	3.61	0.82	Highly Acceptable
4. Replaceability	3.57	0.79	Highly Acceptable
Mean	3.62	0.75	Highly Acceptable

The flexibility of the TalaBaybayin Mobile Learning Application is evaluated across four important aspects: Adaptability, Scalability, Installability, and Replaceability.

For Adaptability *(Kayang iangkop ng TalaBaybayin App sa iba't ibang* hardware, software, o environment*)*, the app received a mean of 3.67 and a standard deviation of 0.63, indicating that users generally find the app highly adaptable to various hardware, software, and environments. The relatively low standard deviation reflects a strong consensus among users regarding its adaptability, with most sharing similar positive views. Regarding Scalability *(Kayang dagdagan o bawasan ng TalaBaybayin App ang kapasidad nito ayon sa pangangailangan)*, the app scored a mean of 3.61 and a standard deviation of 0.77, showing that users perceive it as capable of adjusting its capacity based on demand. The moderate standard deviation indicates some variability in opinions, implying that while most users find the scalability acceptable, a few may have differing experiences or expectations.

In terms of Installability *(Madaling mai-install at* ma-uninstall *ang TalaBaybayin App sa isang* environment*),* the app earned a mean of 3.61 and a standard deviation of 0.82, reflects that users generally find it easy to install and uninstall in different environments. However, the slightly higher standard deviation points to more varied experiences, indicating potential areas for improvement to streamline the installation and uninstallation process for all users. For Replaceability *(Kayang palitan ng TalaBaybayin App ang ibang produkto para sa parehong gamit/paggamit)*, the app received a mean of 3.57 and a standard deviation of 0.79, reflecting that users generally agree the app can replace other products for the same function. However, the moderate variability in standard deviation signifies some differing opinions on how seamlessly this replacement can happen.

The overall mean score for the flexibility evaluation is 3.62, with a standard deviation of 0.75, both falling under the “Highly Acceptable” category. This reflects that most users find the TalaBaybayin app adaptable and easy to use across different devices and settings. They appreciate its smooth installation process, stable performance, and ability to function well even when updates or changes occur. However, the moderate variation in responses indicates that some users may have encountered minor inconsistencies, such as differences in speed or layout depending on their device.

4.9 SAFETY

Safety in mobile learning applications ensures that users can interact with the system without encountering risks that may compromise their learning experience or the app’s functionality. According to Nolan and McDermott (2024), incorporating safety measures like hazard warnings, fail-safe modes, and operational constraints improves both reliability and user trust. Table 16 presents the respondents' perceptions of the TalaBaybayin Mobile Learning Application in terms of safety aspects such as operational constraints, risk identification, fail-safe mechanisms, hazard warnings, and safe integration.

Table 16

Mean Distribution of Respondents' Responses in terms of Safety

Safety	Mean	Standard Deviation	Interpretation
1. Operational Constraint.	3.68	0.76	Highly Acceptable
2. Risk Identification.	3.58	0.72	Highly Acceptable
3. Fail-Safe.	3.66	0.72	Highly Acceptable
4. Hazard Warning.	3.69	0.64	Highly Acceptable
5. Safe Integration.	3.71	0.62	Highly Acceptable
Mean	3.66	0.69	Highly Acceptable

Table 16 states that the Safety of the TalaBaybayin Mobile Learning Application is evaluated across five critical dimensions: Operational Constraint, Risk Identification, Fail -Safe, Hazard Warning, and Safe Integration.

For Operational Constraint (*Kayang panatilihin ng TalaBaybayin* Mobile Learning Application *ang ligtas na operasyon kahit may makatagpo na problema*), the app received a mean of 3.68 and a standard deviation of 0.76. This indicates that users generally feel the app can maintain safe operations even when issues arise. The moderate standard deviation reflects that while most users have a positive view of the app's operational reliability, there is some variability in their experiences.

For Risk Identification (Nakikita ng TalaBaybayin Mobile Learning Application ang mga pangyayaring maaaring magdulot ng panganib), the application obtained a mean score of 3.58 and a standard deviation of 0.72. This suggests that most respondents believe the app is effective in recognizing potential risks that may arise during use. The moderate standard deviation indicates some variation in user perceptions, though the majority of responses reflect a positive view of this feature.

In terms of Fail-Safe (Kayang mag-shift ng TalaBaybayin Mobile Learning Application sa ligtas na mode kung sakaling pumalya ito o bumalik sa ligtas na kondisyon), the app received a mean of 3.66 and a standard deviation of 0.72. This implies that users generally agree the app can transition to a secure mode or recover safely in the event of system failure. The moderate variation in responses shows minor differences in user experiences, yet the overall feedback remains favorable.

When it comes to Hazard Warning (Nagbibigay ang TalaBaybayin Mobile Learning Application ng babala kapag may hindi katanggap-tanggap na panganib), the application achieved a mean score of 3.69 and a standard deviation of 0.64. This demonstrates that most respondents agree the app effectively provides timely warnings when unacceptable risks are detected. The relatively low standard deviation suggests that users hold similar opinions regarding this safety feature.

Lastly, for Safe Integration (Kayang manatiling ligtas ang TalaBaybayin Mobile Learning Application kahit maisama sa iba pang bahagi o produkto), the app earned a mean of 3.71 and a standard deviation of 0.62. This indicates that users perceive the app as capable of maintaining security even when integrated with other components or systems. The low standard deviation reflects a strong consensus among respondents about the app’s stable and secure integration performance.

The mean for the Safety table is 3.66, corresponding to the “Highly Acceptable” category, with a standard deviation of 0.69. In summary, the app is perceived positively regarding Safety, particularly in its ability to maintain safe operations, identify risks, provide hazard warnings, and integrate securely with other systems. The moderate variability in standard deviations implies that there are some differences in user experiences, providing opportunities to improve the app's safety features and enhance its consistency across all user profiles.

The TalaBaybayin Mobile Learning App received an average rating of 3.64 based on ISO/IEC 25010:2023, showing that users generally find it highly acceptable. This suggests the app meets expectations in key areas like functionality, performance, compatibility, and security. A standard deviation of 0.74 shows some variation in user experiences, pointing to areas for improvement. Overall, feedback is positive, with room to fine-tune the app for even greater user satisfaction

CHAPTER V

SUMMARY AND RECOMMENDATIONS

This chapter provides an overview of the study, the conclusions drawn from the data analyzed and interpreted in the previous chapter, and the recommendations formulated based on the results.

Summary of Findings

The study centers on improving the learning and preservation of Baybayin, an ancient Filipino script, by creating the TalaBaybayin mobile application. This study applies a mixed-method approach, integrating survey-based feedback and structured interviews as primary data-gathering techniques, while adhering to an agile software development framework, focusing on Grade 7 students as the main user group. The application incorporates interactive lessons, practice exercises, quizzes, and gamified elements like leaderboards and badges to enhance user engagement and optimize learning results. This system delivers immediate feedback and monitors progress, presenting a contemporary, digital method for mastering Baybayin. The results indicate that the application successfully integrates educational technology with the preservation of culture, aiding in the resurgence of Baybayin within modern educational frameworks.

1. To determine the current state of studying Baybayin and its preservation among Filipino learners, particularly in terms of accessibility, interest, and the effectiveness of available materials through surveys and interviews.

The study revealed that Grade 7 students face significant challenges in retaining Baybayin due to limited access to engaging and interactive learning materials. The overall mean score for the preliminary survey on the difficulty of learning Baybayin among grade 7 students at Caniogan National High School is 2.47 with a standard deviation of 1.04 (see chapter 4 for the result), indicating that the students indeed have difficulty in retaining the Baybayin script outside the classroom setting. Traditional methods often fail to capture the students' interest, leading to difficulties in mastering and retaining the script. The lack of modern, accessible resources further hampers effective learning and retention of Baybayin. This underscores the need for innovative solutions, such as mobile applications, to create an interactive, engaging, and accessible platform that enhances the retention and preservation of Baybayin among young learners.

2. To design and develop a comprehensive platform for learning Baybayin, incorporating features that enhance learner engagement, facilitate progress tracking, and improve knowledge retention.

2.1. Login/Sign-up. A login and sign-up were implemented to ensure personalized access to the application. This allows users to create their individual accounts, enabling learners to conveniently monitor their progress and accomplishments over time. The login process also ensures that learners' data is safely stored and accessible only to verified users. This feature fosters a personalized learning experience, allowing users to revisit lessons and track improvements.

2.2. Lessons. The application provides a lesson plan that progresses from fundamental to more complex Baybayin learning. This systematic approach enables learners to advance at their own speed, beginning with the fundamental characters and progressing to more detailed elements of the script. The lessons aim to be engaging and educational, improving both comprehension and memory of Baybayin characters.

2.3. Leaderboard. A leaderboard feature was implemented to foster a competitive atmosphere within the application. Students can observe their ranking relative to other users depending on their advancement and achievement in diverse activities. This competitive aspect incentivizes users to persist in practice and enhance their skills, thereby enriching the learning experience.

2.4. Time-Based Scoring. A time-based scoring system was implemented to motivate learners to answer questions with precision. This feature awards points based on how quickly tasks are completed, aiming to create a sense of urgency and encourage learners to go beyond their usual performance. Time-based scoring introduces an exhilarating dimension to the learning process, motivating users to enhance both their speed and precision.

2.5. Quizzes. The app includes quizzes after each lesson to evaluate learners' understanding of the material. These quizzes not only provide immediate feedback but also include a passing rate. This ensures that learners must demonstrate a certain level of proficiency before advancing to the next lesson. The real-time feedback helps learners evaluate their comprehension, recognize areas needing enhancement, and reinforce the lesson content. By including a passing rate, the quizzes help maintain the integrity of the learning experience, guaranteeing that users gain the necessary understanding before advancing.

2.6. Progress tracking. The application includes a progress tracking dashboard, allowing learners to assess their development over time. Through monitoring completed lessons, quiz scores, and milestones, learners can perceive their progress and experience a sense of achievement. This feature promotes sustained engagement and aids learners in maintaining motivation throughout their educational journey.

2.7. Gamification features. Gamification elements were integrated into the application to augment user engagement. Elements like as points, levels, and badges augment the learning experience by offering incentives for progression. These gamified components transform the educational process into a game-like experience, maintaining learners' engagement and motivation over time.

2.8. Mini-games. The application features mini-games intended to enhance the enjoyment and interactivity of learning Baybayin. These games provide learners with the chance to practice Baybayin characters in a relaxed and entertaining setting. The program promotes repetitive practice through the integration of mini-games, thereby enhancing the memorization of Baybayin characters.

2.9. Multimedia sounds. Multimedia sounds were embedded in the app to improve the learning experience by providing auditory cues. This feature helps learners correctly pronounce Baybayin characters and phrases, enhancing their understanding of both the written and spoken aspects of the script. The use of sounds also increases the overall engagement of the learning process, making it more immersive and comprehensive.

2.10. Common Phrases. A section for common Baybayin phrases was added to the app to teach learners practical applications of the script. This feature helps users familiarize themselves with phrases that they can use in everyday conversations, making their learning experience more relevant and applicable. By learning commonly used phrases, learners can immediately start using Baybayin in real-life situations, deepening their connection with the language and script.

2.11. Completion Certificate. The inclusion of a Completion Certificate serves as a significant incentive tool within the mobile learning application. Upon completion of all lectures and quizzes, users will be awarded a certificate that validates their efforts and achievements, so augmenting their sense of success. This accreditation may serve as a tangible recognition of the learner's dedication and progress.

3. To integrate the following significant features into the design and development of the TalaBaybayin Mobile Learning Application

To deliver a well-rounded and interactive educational experience, the TalaBaybayin Mobile Learning Application was designed with several specialized features. These features were carefully integrated to enhance the learning, practice, and preservation of Baybayin, ensuring that users can interact with the script through multiple modes, including typing, handwriting, translation, and auditory reinforcement. Each feature is designed to be intuitive, interactive, and supportive of learners' progress, while also promoting cultural appreciation and retention of Baybayin knowledge. The significant features are described as follows:

3.1. BaybayinKeyboard. The app included a Baybayin Keyboard that allows learners to type Baybayin characters directly. This feature is designed to be intuitive, ensuring even beginners can input text accurately. The keyboard supports all Baybayin characters and provides visual guides to help learners select the correct symbols. All user inputs are stored in the database, enabling progress tracking and performance feedback. By combining digital practice with structured guidance, this feature helps learners become familiar with Baybayin in a modern, interactive format.

3.2. Handwriting Feature. The Handwriting Feature allows learners to practice writing Baybayin characters manually on the device screen. The system guides users with correct stroke order and character formation, reinforcing memorization and muscle memory. Each attempt is recorded and stored, enabling learners to review their performance and track improvements. Visual cues and scoring provide immediate feedback, helping learners refine their skills. This combination of traditional handwriting practice and digital interactivity strengthens both learning outcomes and engagement.

3.3. Tagalog to Baybayin Translation. The Baybayin translation feature was developed to provide a clear and accessible translation tool that converts common Tagalog words and phrases into Baybayin script. Learners can immediately see the connection between modern Tagalog and traditional script, which supports contextual learning and comprehension. By demonstrating practical applications of Baybayin in everyday language, this feature strengthens both reading and writing skills while promoting cultural appreciation, allowing learners to connect meaningfully with their Filipino heritage.

3.4. Text-to-Speech. The app included a text-to-speech feature that reads Baybayin characters, words, and phrases aloud, allowing learners to hear Tagalog pronunciations of the characters while following along visually. This reinforces memorization, improves reading fluency, and helps learners confidently articulate the script. In combining auditory and visual learning, the feature makes studying Baybayin more interactive and engaging, ensuring that learners not only understand the characters but also connect with Filipino culture in a meaningful and immersive way.

4. To evaluate the acceptability of the system using ISO/IEC 25010 product quality evaluation

The TalaBaybayin Mobile Learning Application was evaluated following the ISO/IEC 25010:2023 standard for software quality through User Acceptance Testing (UAT), utilizing a 4-point Likert Scale (Appendix I). A total of 100 respondents, including Grade 7 students, teachers, and IT professionals, shared their evaluations regarding the application's usability, functionality, and overall performance. The assessment concentrated on nine key software quality characteristics:

1. Functional Suitability. The app delivers Baybayin lessons, quizzes, and interactive exercises that effectively support learners in completing tasks, understanding the script, and practicing the correct usage of characters, ensuring that all educational objectives of the learning process are fully addressed.

2. Performance Efficiency. The system operates smoothly without significant delays or technical errors, providing a fast and responsive experience during lessons, exercises, and interactive activities, which allows learners to focus on content comprehension without interruptions or frustrations.

3. Compatibility. TalaBaybayin performs consistently across multiple device types and platforms, such as smartphones and tablets with different screen sizes and operating systems, ensuring that learners can access the application seamlessly on their preferred platform without encountering technical issues.

4. Interaction Capability. Users experience highly intuitive navigation, clear interface layouts, and immediate feedback through interactive elements such as quizzes, gamification features, leaderboards, and badges, all of which motivate continuous practice and make learning Baybayin engaging and enjoyable.

5. Reliability. The app consistently performs as intended under normal usage conditions, providing accurate results, stable operation during all activities, and dependable recording of progress and scores, which builds user confidence in the application as a trustworthy learning tool.

6. Security. Learners' personal data, practice records, and account information are safeguarded through proper security measures, ensuring privacy and protection against unauthorized access, while providing a safe learning environment, particularly important for younger users.

7. Maintainability. The system is designed to allow updates, modification, and enhancements to be implemented seamlessly and efficiently without affecting existing functionality or causing data loss, allowing future improvements and ongoing development guided by user input and evolving learning needs.

8. Flexibility. TalaBaybayin can adapt to diverse learning needs, user preferences, and various device environments, providing a customizable and inclusive experience that accommodates learners with different levels of knowledge and varying interaction styles.

9. Safety. measures within the app ensure a secure and positive learning environment, preventing inappropriate content or interactions, promoting responsible use, and encouraging users to practice and explore Baybayin confidently and safely.

The evaluation confirms that TalaBaybayin successfully fosters learners' interest, engagement, and proficiency in Baybayin while promoting cultural preservation. By making the script accessible, interactive, and appealing, the app encourages learners to explore their cultural heritage in a structured, gamified, and educationally meaningful way.

Users recognized the system as an effective tool that enhances learning outcomes and supports long-term cultural awareness, ensuring that the study's objectives of education, engagement, and preservation are fully achieved.

Conclusions

Based on the aforementioned data, a conclusion was reached on the study's assumptions. The TalaBaybayin Mobile Learning Application successfully achieved its intended objectives of enhancing the study and preservation of Baybayin among Filipino learners. The app effectively addressed the challenges identified in the study, particularly regarding the accessibility, engagement, and retention of Baybayin. It met the technical requirements and performed with integrity, as confirmed by the feedback from both learners and IT professionals who tested the app. Through its interactive features, gamification elements, and multimedia tools, the application significantly improved the learning experience, making Baybayin more accessible and enjoyable for students.

The app also provided an efficient, modern alternative to traditional methods of teaching Baybayin. As a result of integrating features such as the Baybayin keyboard, handwriting feature, Text-to-Baybayin translation, and text-to-speech, the application created an immersive learning environment that allowed students to practice Baybayin in diverse ways. This not only enhanced their understanding and retention of the script but also encouraged a deeper cultural appreciation.

Moreover, the progress tracking and quizzes with passing rates ensured that learners could monitor their improvement and gain mastery over the content. The gamification aspects, such as leaderboards and time-based scoring, further motivated students to continue learning and improving their skills. The Completion Certificate, received after completing all lessons and passing the quizzes, reinforces learner motivation by providing tangible recognition for their achievements, encouraging continued engagement and commitment to learning.

The success of the TalaBaybayin Mobile Learning Application is evident in its alignment with the study's objectives and the positive evaluation from its users. The app has effectively contributed to the preservation and promotion of Baybayin by providing a modern, accessible, and engaging platform for learners. It stands as a valuable tool for future generations to connect with their cultural heritage in an innovative and meaningful way.

Recommendations

After reviewing the study's findings and conclusions, the following recommendations were created to support future researchers in Information Technology who wish to conduct a similar study:

1. **Enable Offline Accessibility.** To promote inclusivity and ensure broader access, it is recommended that future versions of TalaBaybayin support offline functionality. This enhancement would allow users, particularly those in areas with unstable or limited internet connectivity, to continue engaging with the app's educational content without interruption. Offering offline access to lessons and practice activities would provide users with more flexibility and ensure uninterrupted learning, thus supporting consistent language acquisition.

2. **Integrate Social Media Sharing Features.** To foster community engagement and increase cultural awareness, incorporating social media sharing options into TalaBaybayin would be highly beneficial. Allowing users to share their progress, Baybayin

writings, and achievements on platforms such as Facebook, Instagram, and Twitter would not only allow them to showcase their learning journey but also inspire others to explore and adopt the Baybayin script. This feature would encourage peer-to-peer interaction and create a broader interest in preserving and using indigenous Filipino scripts.

3. Expand to Include Other Filipino Scripts and Language Support. Once TalaBaybayin has established a stable user base and proven effective in promoting the Baybayin script, it would be advantageous to expand the app's capabilities to include other indigenous Filipino writing systems, such as Hanunuo, Buhid, or Tagbanwa. Furthermore, integrating localization options, including translation features for regional Filipino dialects and English, would make the app accessible to a more diverse audience across the Philippines. Such expansion would help in the preservation of linguistic diversity and promote a greater appreciation for the country's indigenous writing systems.

4. Expand to Include Local Leaderboards and Community Rankings. Once TalaBaybayin has successfully implemented and stabilized its global leaderboard system, it would be beneficial to enhance the feature by incorporating local or community-based leaderboards. This expansion would allow users to view rankings within their respective regions, schools, or learning groups, fostering a sense of friendly competition and community engagement. By supporting localized leaderboards, TalaBaybayin can encourage users to connect and collaborate with fellow learners in their area, promoting active participation and sustained motivation. Moreover, such a feature would provide valuable insights into regional learning progress and engagement levels, contributing to the app's goal of preserving and revitalizing Baybayin literacy across diverse Filipino communities.

5. Integrate Speech-to-Text Functionality. Once TalaBaybayin established a stable system for text-based learning and interaction, it would be beneficial to expand its features by incorporating speech-to-text technology. This functionality would enable users to input spoken Filipino words or phrases and have them automatically converted into Baybayin text. Such a feature would enhance accessibility, particularly for users with limited typing skills or visual impairments and would make learning Baybayin more interactive and convenient. Additionally, integrating speech-to-text capabilities could encourage oral language practice and promote a deeper connection between spoken Filipino and its written Baybayin equivalent, further enriching the learning experience.

6. Implement Randomized Quiz Questions. To enhance user engagement and ensure a more dynamic learning experience, it is recommended that TalaBaybayin implement a randomized quiz question feature. This functionality would allow the app to shuffle or vary the set of questions each time a user takes a quiz, preventing memorization through repetition and encouraging genuine mastery of the Baybayin script. By generating unique quiz combinations for every attempt, users are continuously challenged to recall and apply their knowledge. Furthermore, this improvement would make the assessment process more interactive and adaptive, thereby increasing learning retention and maintaining long-term user interest.

7. Develop an Admin Side for Management and Monitoring. To ensure efficient management of the TalaBaybayin application, an admin side should be developed to oversee user activities, content updates, and system performance. This feature would let administrators manage learning materials, monitor quiz results, and analyze user engagement to enhance the app’s overall functionality as an educational platform for Baybayin learning.

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Appendix A

Communication Letters

REPUBLIC OF THE PHILIPPINES
BULACAN STATE UNIVERSITY
College of Information and Communications Technology

Level III Mastery in Quality Management 2022 | Quality Management System ISO 9001:2015 Certified | Times Higher Education Impact Rankings 2023 | Certified Dark Green School

CERTIFICATION

Date: **February 25, 2025**

This is to certify that the Capstone Project entitled "**TalaBaybayin: A Mobile Learning App for Guiding Filipino Learners in Mastering and Preserving Baybayin**" proposed by the following students was approved by the Capstone Coordinator and Capstone Adviser:

1. Francia, Christian	**4. Miramonte, Angelo**
2. Garcia, Kylee	**5. Teodoro, John Patrick**
3. Lapuz, Alyssa	**6.**

Gabriel M. Galang
Capstone Project Coordinator

Aaron Paul M. Dela Rosa 2/27/25
Capstone Adviser

Email Address: officeofthedean.cict@bulsu.edu.ph | **Trunkline: 9197800 Loc 1101 then 1102 Office of the Dean**
Bulacan State University Main Campus, Capitol Compound, McArthur Highway, Guinhawa, City of Malolos Bulacan

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The capstone adviser should determine the complexity level of the specific problem being addressed and the proposed solution, considering the duration of the project, the composition of the team, and the resources available.

The Adviser's commitments are:

1. Share his/her specialization in the capstone project.
2. guide the group in the research and development process by providing timely advice;
3. attend the group's project presentations and provide feedback that will ensure success;
4. review the project paper – its contents, grammar, and completeness, and prepare comments before its submission to Capstone Coordinator;
5. allocate consultation hours to the group; and
6. promote the value of hard work, communication, and integrity throughout the project's development by encouraging the group to work on their research; and
7. sign necessary documents needed by the group such as the Endorsement Form for the Proposal and Final defense, and Approval Sheet which indicates the successful completion of the final project.

☑ I confirm my agreement with the outlined terms above. I commit to adhering to these conditions and fulfilling my responsibilities as detailed.

Aaron Paul M. Dela Rosa 2/27/25
Capstone Adviser

REPUBLIC OF THE PHILIPPINES
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Brian Manansala,

Teacher II

Dear Mr. Brian Manansala,

We, the members of the Capstone Project Team for *Talabaybayin: A Mobile Learning Application for Guiding Filipino Learners in Mastering and Preserving Baybayin*, from the College of Information and Communications Technology – Bulacan State University, are reaching out to formally request your participation in an interview for our capstone project.

Our project aims to develop a mobile learning application that features interactive writing practice tools, assessment quizzes, and gamified elements designed to help high school students learn, master, and preserve the Baybayin script.

In line with this, we would greatly appreciate your time and insights as a Teacher II and a passionate advocate of Baybayin. Your expertise and advocacy will be invaluable in gathering essential data and requirements for the development of our system. Please be assured that all information shared during the interview will be treated with strict confidentiality and will solely contribute to the successful implementation of our project.

We look forward to your positive response and support. Please let us know a convenient time for the interview at your earliest convenience.

Thank you for your time and consideration.

Warm regards,

BSIT 3H-G2 Capstone Group (Talabaybayin: A mobile Learning Application for Guiding Filipino Learners in Mastering and Preserving Baybayin.)

Gabriel M. Galang, MSIT
Capstone Coordinator

Mr. Brian Manansala

Republic of the Philippines
City of Malolos, Bulacan
BULACAN STATE UNIVERSITY
College of Information and Communications Technology

ADVISER ENDORSEMENT FOR CAPSTONE FINAL DEFENSE

I am pleased to endorse the capstone paper titled **"TalaBaybayin: A Mobile Learning App For Guiding Filipino Learners In Mastering and Preserving Baybayin"** of the following students:

1. Francia, Christian B.	**4. Miramonte, Angelo**
2. Garcia, Kylee Remiel	**5. Teodoro, John Patrick**
3. Lapuz, Alyssa Mae	

After a comprehensive review of its Chapter I – V, I am confident in stating that the thesis is well-prepared and is now deemed ready for the upcoming final defense.

I recommend that the final be presented for the final defense stage, and I anticipate a compelling discussion that will further validate the robustness and significance of their final system.

0 9 OCT 2025

Aaron Paul M. Dela Rosa, MSIT
Capstone Adviser

Appendix B

Relevant Source Code

LessonScreen.tsx

```
import React, { useState, useEffect }
from 'react';
import {
  View,
  Text,
  ScrollView,
  TouchableOpacity,
  SafeAreaView,
  Dimensions,
  Alert,
} from 'react-native';
import { Ionicons } from '@expo/vector-
icons';
import { Lesson, BaybayinCharacter }
from '../../types/lesson';
import { useProgress } from
'../../contexts/ProgressContext';
import { Badge } from
'../../types/progress';
import BadgeEarnedModal from
'../common/BadgeEarnedModal';
import Aralin9CeremonyModal from
'../common/Aralin9CeremonyModal';
import MaxLevelCeremonyModal from
'../common/MaxLevelCeremonyModal';


import '../../global.css';

interface LessonScreenProps {
  lesson: Lesson;
  onBack: () => void;
  onComplete: () => void;
}

const { width } =
Dimensions.get('window');

export default function LessonScreen({
lesson, onBack, onComplete }:
LessonScreenProps) {
  const [currentSectionIndex,
setCurrentSectionIndex] = useState(0);
  const [startTime, setStartTime] =
useState<Date>(new Date());
  const [isCompleting, setIsCompleting]
= useState(false);
  const [showBadgeModal,
setShowBadgeModal] = useState(false);
  const [earnedBadge, setEarnedBadge]
= useState<Badge | null>(null);
  const [completionScore,
setCompletionScore] =
useState<number>(0);
  const [completionTimeSpent,
setCompletionTimeSpent] =
useState<number>(0);

  // Aralin 9 ceremony state
  const [showCeremonyModal,
setShowCeremonyModal] =
useState(false);
  const [ceremonySpecialBadge,
setCeremonySpecialBadge] =
useState<Badge | null>(null);

  // Max level ceremony state
  const [showMaxLevelModal,
setShowMaxLevelModal] =
useState(false);
  const [maxLevelBadge,
setMaxLevelBadge] = useState<Badge |
null>(null);

  const { completeLesson,
isLessonCompleted, awardBadge,
addExperience } = useProgress();

  const totalSections =
lesson.content.sections.length +
(lesson.content.summary ? 1 : 0); //
sections + summary
  const isLastSection =
currentSectionIndex >=
lesson.content.sections.length;

  useEffect(() => {
    setStartTime(new Date());
  }, []);
```

```
  // Reset modal state when lesson
changes
  useEffect(() => {
    setShowBadgeModal(false);
    setEarnedBadge(null);
    setCurrentSectionIndex(0);
    setCompletionScore(0);
    setCompletionTimeSpent(0);
  }, [lesson.id]);

  const handleComplete = async () => {
    // Prevent multiple completions
    if (isCompleting) {
      console.log('Lesson completion
already in progress, ignoring duplicate
call');
      return;
    }

    setIsCompleting(true);

    try {
      const endTime = new Date();
      const timeSpent =
Math.floor((endTime.getTime() -
startTime.getTime()) / 1000); // in
seconds
      const score =
calculateScore(timeSpent);

      // Calculate experience points that
will be earned (matching
ProgressContext logic)
      const wasAlreadyCompleted =
isLessonCompleted(lesson.id);
      const isFirstCompletion =
!wasAlreadyCompleted;
      // Give 85 points for lessons 1-8,
keep original calculation for lesson 9
      const experiencePoints =
isFirstCompletion ?
        (lesson.id === 'lesson_9' ? 15 +
Math.floor(score / 10) : 85) : 0;

      // Store completion data for modal
display - use experience points instead of
score

setCompletionScore(experiencePoints);
// Show experience points instead of
lesson score

setCompletionTimeSpent(timeSpent);

      console.log('About to complete
lesson:', {
        lessonId: lesson.id,
        lessonTitle: lesson.title,
        score,
        timeSpent,
        totalSections: totalSections,
        wasAlreadyCompleted
      });

      const completionResult = await
completeLesson(
        lesson.id,
        score,
        timeSpent,
        lesson.badge &&
!wasAlreadyCompleted ? lesson.badge :
undefined
      );
      console.log('LessonScreen: Lesson
completion successful',
completionResult);

      console.log('LessonScreen: All
completion tasks finished');

      // Check for max level achievement
first (highest priority)
      if
(completionResult?.maxLevelBadge &&
!wasAlreadyCompleted) {
        console.log('🏆 MAX LEVEL
ACHIEVED! Setting up ceremony...');

setMaxLevelBadge(completionResult.m
axLevelBadge);
```

```
        setShowMaxLevelModal(true);
        return; // Show max level ceremony
first, skip other modals
      }

      // For first completion: show badge
and summary modal
      if (!wasAlreadyCompleted) {
        // Calculate experience points for
display - 85 points for lessons 1-8,
original for lesson 9
        const experiencePoints = lesson.id
=== 'lesson_9' ? 15 + Math.floor(score /
10) : 85;

        // Store completion data for modal
display - use experience points

setCompletionScore(experiencePoints);

setCompletionTimeSpent(timeSpent);

        // Check for Aralin 9 ceremony
        if
(completionResult?.isAralin9Completio
n && lesson.id === 'lesson_9') {
          console.log('🎉 Aralin 9
completion ceremony!');

          // Set up ceremony modal data

setCeremonySpecialBadge(completionR
esult.specialBadge || null);
          setShowCeremonyModal(true);
        } else {
          // Regular badge flow for other
lessons
          let badgeToShow = null;

          // Priority 1: Special badge from
completing all lessons
          if
(completionResult?.specialBadge) {
            badgeToShow =
completionResult.specialBadge;
            console.log('🎉 Special badge
received:', badgeToShow);
          }
          // Priority 2: Regular lesson badge
(only for first completion)
          else if (lesson.badge) {
            badgeToShow = {
              id: String(lesson.badge.id || ''),
              name:
String(lesson.badge.name || ''),
              description:
String(lesson.badge.description || ''),
              icon: String(lesson.badge.icon ||
'🏆'),
              color: String(lesson.badge.color
|| '#10B981'),
              category: lesson.badge.category
|| 'lesson' as const,
              requirement:
String(lesson.badge.requirement || ''),
              earnedAt: new Date()
            };
          }

          setEarnedBadge(badgeToShow);
          setShowBadgeModal(true);
        };
        // Don't call onComplete() here -
wait for user to click "Magpatuloy"
      } else {
        // Retaking lesson: just complete
silently and go back
        onComplete();
      }

    } catch (error) {
      console.error('Error completing
lesson:', error);
      Alert.alert('Error', 'Hindi ma-save
ang progreso. Subukan ulit.');
    } finally {
      setIsCompleting(false);
    }
  };
```

```
  const handleBadgeModalClose = () =>
{
    setShowBadgeModal(false);
    setEarnedBadge(null);
    setCompletionScore(0);
    setCompletionTimeSpent(0);

    // Now call onComplete when user
actually clicks "Magpatuloy"
    onComplete();
  };

  const handleCeremonyModalClose = ()
=> {
    setShowCeremonyModal(false);
    setCeremonySpecialBadge(null);
    setCompletionScore(0);
    setCompletionTimeSpent(0);

    // Complete the lesson after ceremony
    onComplete();
  };

  const handleMaxLevelModalClose = ()
=> {
    setShowMaxLevelModal(false);
    setMaxLevelBadge(null);

    // After max level ceremony, continue
with normal flow
    onComplete();
  };



  const calculateScore =
(timeSpentSeconds: number): number
=> {
    // Base score
    let score = 80;

    // Time bonus (faster completion =
higher score)
    const timeBonus = Math.max(0, 20 -
Math.floor(timeSpentSeconds / 30));
    score += timeBonus;

    // Section completion bonus
    const completionBonus =
((currentSectionIndex + 1) /
totalSections) * 20;
    score += completionBonus;

    return Math.min(100, Math.max(50,
score));
  };

  const handleNext = () => {
    if (isLastSection) {
      // All sections completed
      handleComplete();
    } else {

setCurrentSectionIndex(currentSectionIn
dex + 1);
    }
  };

  const handlePrevious = () => {
    if (currentSectionIndex > 0) {

setCurrentSectionIndex(currentSectionIn
dex - 1);
    }
  };

  const getCurrentContent = () => {
    if (currentSectionIndex <
lesson.content.sections.length) {
      return
lesson.content.sections[currentSectionIn
dex];
    } else {
      // Return summary as final section
      return {
        title: 'Buod',
        content: lesson.content.summary ||
'Natapos na ang aralin.',
        examples: []
      };
    }
  };
```

```
const renderProgressBar = () => {
  return (
    <View className="flex-row justify-
center py-4 bg-white">
      {Array.from({ length: totalSections
}, (_, index) => (
        <View
          key={index}
          className={`w-6 h-6 rounded-
full mx-1 justify-center items-center ${
            index === currentSectionIndex
              ? 'bg-red-500'
              : index < currentSectionIndex
                ? 'bg-yellow-500'
                : 'bg-gray-300'
          }`}
        >
          {index < currentSectionIndex
&& (
            <Ionicons name="checkmark"
size={12} color="white" />
          )}
        </View>
      ))}
    </View>
  );
};

const currentContent =
getCurrentContent();

return (
  <SafeAreaView className="flex-1
bg-gray-50">
    {/* Header */}
    <View className="bg-blue-600
flex-row items-center px-4 py-3">
      <TouchableOpacity
onPress={onBack} className="mr-3">
        <Ionicons name="arrow-back"
size={24} color="white" />
      </TouchableOpacity>
      <View className="flex-1">
        <Text className="text-lg font-
bold text-white">{lesson.title}</Text>
        <Text className="text-sm text-
blue-100">
          Pahina {currentSectionIndex +
1} ng {totalSections}
        </Text>
      </View>
    </View>

    {/* Progress Dots */}
    {renderProgressBar()}

    {/* Content Display */}
    <ScrollView className="flex-1"
showsVerticalScrollIndicator={false}>
      <View className="m-4 bg-white
rounded-2xl p-6 shadow-sm">
        {/* Section Title */}
        <View className="mb-6">
          <Text className="text-2xl font-
bold text-primary text-center mb-2">
            {currentContent.title}
          </Text>
        </View>

        {/* Section Content */}
        <View className="mb-6">
          <Text className="text-base
text-gray-800 leading-6 text-justify">
            {currentContent.content}
          </Text>
        </View>

        {/* Characters Display (if
available) */}
        {currentContent.characters &&
currentContent.characters.length > 0 &&
(
          <View className="mb-6">
            <Text className="text-xl font-
bold text-gray-800 mb-4 text-
center">Mga Karakter</Text>
            <View className="flex-row
flex-wrap justify-center">

{currentContent.characters.map((charact
er, index) => (
```

```
                <View key={index}
className="bg-gray-50 rounded-xl p-4
m-2 items-center min-w-24">
                  <Text className="text-4xl
text-primary font-bold mb-
2">{character.character}</Text>
                  <Text className="text-sm
text-gray-600 font-
semibold">{character.romanized}</Text
>
                  <Text className="text-xs
text-gray-
500">{character.pronunciation}</Text>
                </View>
              ))}
            </View>
          </View>
        )}

        {/* Examples (if available) */}
        {currentContent.examples &&
currentContent.examples.length > 0 &&
(
          <View className="mb-6">
            <Text className="text-xl font-
bold text-gray-800 mb-4 text-
center">Mga Halimbawa</Text>

{currentContent.examples.map((exampl
e, index) => (
              <View key={index}
className="bg-gray-50 rounded-xl p-4
mb-3">
                <View className="flex-row
justify-between items-center">
                  <View className="flex-
1">
                    <Text className="text-2xl
text-primary font-bold mb-
1">{example.baybayin}</Text>
                    <Text className="text-lg
text-gray-800 font-
semibold">{example.romanized}</Text
>
                    <Text className="text-sm
text-gray-
600">{example.english}</Text>
                    {example.pronunciation
&& (
                      <Text className="text-
xs text-gray-500
italic">/{example.pronunciation}/</Text
>
                    )}
                  </View>
                </View>
              </View>
            ))}
          </View>
        )}

        {/* Cultural Note (for summary
section) */}
        {isLastSection &&
lesson.content.culturalNote && (
          <View className="bg-blue-50
rounded-xl p-4 mb-4">
            <Text className="text-lg font-
bold text-blue-800 mb-2">Kultural na
Tala</Text>
            <Text className="text-base
text-blue-700 leading-6">
              {lesson.content.culturalNote}
            </Text>
          </View>
        )}
      </View>
    </ScrollView>

    {/* Navigation */}
    <View className="flex-row p-4 bg-
white border-t border-gray-200">
      <TouchableOpacity
        className={`flex-1 flex-row
items-center justify-center py-3 px-4
rounded-lg mx-1 ${
          currentSectionIndex === 0 ? 'bg-
gray-100' : 'bg-yellow-500'
        }`}
        onPress={handlePrevious}
```

```
        disabled={currentSectionIndex
=== 0}
      >
        <Ionicons
          name="chevron-back"
          size={20}
          color={currentSectionIndex ===
0 ? '#D1D5DB' : '#FFFFFF'}
        />
        <Text className={`text-lg font-
semibold ml-1 ${
          currentSectionIndex === 0 ?
'text-gray-400' : 'text-white'
        }`}>
          Nauna
        </Text>
      </TouchableOpacity>

      <TouchableOpacity
        className={`flex-1 flex-row
items-center justify-center py-3 px-4
rounded-lg mx-1 ${
          isCompleting ? 'bg-gray-400' :
'bg-red-600'
        }`}
        onPress={handleNext}
        disabled={isCompleting}
      >
        <Text className="text-lg font-
semibold text-white mr-1">
          {isLastSection
            ? (isCompleting ? 'Tinutupos...'
: 'Tapusin ang Aralin')
            : 'Susunod'
          }
        </Text>
        <Ionicons name="chevron-
forward" size={20} color="white" />
      </TouchableOpacity>
    </View>

    {/* Badge Earned Modal - Shows
badge and summary for first completions
*/}
    <BadgeEarnedModal
      visible={showBadgeModal}
      badge={earnedBadge}
      score={completionScore || 0}
      timeSpent={completionTimeSpent
|| 0}

experienceGained={completionScore ||
0}
      completionType="lesson"
      wasRetake={false}

onClose={handleBadgeModalClose}
    />

    {/* Aralin 9 Ceremony Modal -
Special celebration for completing all
lessons */}
    <Aralin9CeremonyModal
      visible={showCeremonyModal}

specialBadge={ceremonySpecialBadge}
      score={completionScore || 0}
      timeSpent={completionTimeSpent
|| 0}

onClose={handleCeremonyModalClose
}
    />

    {/* Max Level Ceremony Modal -
Special celebration for reaching level 10
*/}
    <MaxLevelCeremonyModal
      visible={showMaxLevelModal}
      badge={maxLevelBadge}

onClose={handleMaxLevelModalClose}
    />

  </SafeAreaView>
 );
}
```

QuizScreen.tsx

```
import React, { useState, useEffect }
from 'react';
import {
```

```
  View,
  Text,
  TouchableOpacity,
  SafeAreaView,
  Alert,
  ScrollView,
  TextInput,
  Modal,
} from 'react-native';
import { Ionicons } from '@expo/vector-
icons';
import { Quiz, QuizQuestion,
QuestionType } from '../../types/quiz';
import { useProgress } from
'../../contexts/ProgressContext';
import { Badge } from
'../../types/progress';
import BadgeEarnedModal from
'../common/BadgeEarnedModal';
import Aralin9CeremonyModal from
'../common/Aralin9CeremonyModal';
import MaxLevelCeremonyModal from
'../common/MaxLevelCeremonyModal';
import LoadingIndicator from
'../common/LoadingIndicator';
import '../../global.css';

interface QuizScreenProps {
  quiz: Quiz;
  onBack: () => void;
  onComplete: (score: number,
totalPoints: number, timeSpent: number)
=> void;
}

export default function QuizScreen({
quiz, onBack, onComplete }:
QuizScreenProps) {
  // All hooks must be called before any
early returns
  const [currentQuestionIndex,
setCurrentQuestionIndex] = useState(0);
  const [answers, setAnswers] =
useState<{ [questionId: string]: string
}>({});
  const [skippedQuestions,
setSkippedQuestions] =
useState<Set<number>>(new Set());
  const [showingUnansweredOnly,
setShowingUnansweredOnly] =
useState(false);
  const [unansweredQuestionIndexes,
setUnansweredQuestionIndexes] =
useState<number[]>([]);
  const [timeLeft, setTimeLeft] =
useState(quiz?.timeLimit || 0);
  const [startTime] = useState(new
Date());
  const [questionStartTime,
setQuestionStartTime] = useState(new
Date());
  const [questionTimes,
setQuestionTimes] = useState<{
[questionId: string]: number }>({});
  const [showPointsGained,
setShowPointsGained] = useState<{
points: number; timeBonus: number;
isCorrect: boolean } | null>(null);
  const [totalPointsEarned,
setTotalPointsEarned] = useState(0);
  const [answeredQuestions,
setAnsweredQuestions] =
useState<Set<string>>(new Set());
  const [showBadgeModal,
setShowBadgeModal] = useState(false);
  const [earnedBadge, setEarnedBadge]
= useState<Badge | null>(null);
  const [completionScore,
setCompletionScore] =
useState<number>(0);
  const [completionTimeSpent,
setCompletionTimeSpent] =
useState<number>(0);
  const [completionExperience,
setCompletionExperience] =
useState<number>(0);

  // Enhanced time-based scoring UI
states
```

```
  const [questionTimer,
setQuestionTimer] =
useState<number>(0);
  const [currentTimeBonus,
setCurrentTimeBonus] =
useState<number>(0);
  const [totalTimeBonus,
setTotalTimeBonus] =
useState<number>(0);
  const [completionAccuracy,
setCompletionAccuracy] =
useState<number>(0);
  const [passed, setPassed] =
useState<boolean>(true);
  const [wasAlreadyCompleted,
setWasAlreadyCompleted] =
useState<boolean>(false);
  const [timeUpAlertShown,
setTimeUpAlertShown] =
useState<boolean>(false);
  const [showTimeUpModal,
setShowTimeUpModal] =
useState<boolean>(false);

  // Aralin 9 quiz ceremony state
  const [showQuizCeremonyModal,
setShowQuizCeremonyModal] =
useState(false);
  const [ceremonySpecialBadge,
setCeremonySpecialBadge] =
useState<Badge | null>(null);

  // Max level ceremony state
  const [showMaxLevelModal,
setShowMaxLevelModal] =
useState(false);
  const [maxLevelBadge,
setMaxLevelBadge] = useState<Badge |
null>(null);

  const { addExperience,
isQuizCompleted, awardBadge,
getQuizAttempts, completeQuiz,
getQuestionPoints } = useProgress();
```

```
  // Define handleTimeUp before
useEffect hooks to avoid dependency
issues
  const handleTimeUp = () => {
    if (timeUpAlertShown) return; //
Prevent multiple alerts

    setTimeUpAlertShown(true);
    setShowTimeUpModal(true);
  };

  // ALL useEffect hooks must be called
before any early returns
  // Timer effect
  useEffect(() => {
    if (!quiz?.timeLimit) return;

    const timer = setInterval(() => {
      setTimeLeft(prev => {
        if (prev <= 1 &&
!timeUpAlertShown) {
          handleTimeUp();
          return 0;
        }
        return prev - 1;
      });
    }, 1000);

    return () => clearInterval(timer);
  }, [quiz?.timeLimit,
timeUpAlertShown, handleTimeUp]);

  // Reset question timer when question
changes
  useEffect(() => {
    setQuestionStartTime(new Date());
    setQuestionTimer(0);
    setCurrentTimeBonus(0);
  }, [currentQuestionIndex]);

  // Question timer effect - tracks time
and calculates real-time bonus
  useEffect(() => {
    // Get current question safely for this
effect
    let effectCurrentQuestion = null;
```

```
    if (quiz?.questions) {
      if (showingUnansweredOnly) {
        if
(unansweredQuestionIndexes.length > 0
&&
            currentQuestionIndex >= 0 &&
            currentQuestionIndex <
unansweredQuestionIndexes.length) {
          const actualQuestionIndex =
unansweredQuestionIndexes[currentQue
stionIndex];
          if (actualQuestionIndex >= 0 &&
actualQuestionIndex <
quiz.questions.length) {
            effectCurrentQuestion =
quiz.questions[actualQuestionIndex];
          }
        }
      } else {
        if (currentQuestionIndex >= 0 &&
currentQuestionIndex <
quiz.questions.length) {
          effectCurrentQuestion =
quiz.questions[currentQuestionIndex];
        }
      }
    }

    if (!effectCurrentQuestion ||
answeredQuestions.has(effectCurrentQu
estion.id)) return;

    const timer = setInterval(() => {
      const now = new Date();
      const elapsedTime = (now.getTime()
- questionStartTime.getTime()) / 1000;
      setQuestionTimer(elapsedTime);

      // Calculate real-time time bonus
      const maxTimeForBonus = 10; //
seconds
      const timeBonus = Math.max(0,
Math.floor((maxTimeForBonus -
elapsedTime) *
(effectCurrentQuestion.points * 0.5 /
maxTimeForBonus)));
      setCurrentTimeBonus(timeBonus);
    }, 100); // Update every 100ms for
smooth UI

    return () => clearInterval(timer);
  }, [currentQuestionIndex,
questionStartTime, answeredQuestions,
quiz?.questions,
showingUnansweredOnly,
unansweredQuestionIndexes]);

  // Add null checks for quiz and
questions AFTER all hooks
  if (!quiz || !quiz.questions ||
quiz.questions.length === 0) {
    return <LoadingIndicator
text="Loading quiz..." />;
  }

  // Get current question with safety
checks
  let currentQuestion = null;
  if (showingUnansweredOnly) {
    // Safety checks for unanswered mode
    console.log('Unanswered mode -
currentQuestionIndex:',
currentQuestionIndex,
'unansweredQuestionIndexes:',
unansweredQuestionIndexes);
    if (unansweredQuestionIndexes.length
> 0 &&
        currentQuestionIndex >= 0 &&
        currentQuestionIndex <
unansweredQuestionIndexes.length) {
      const actualQuestionIndex =
unansweredQuestionIndexes[currentQue
stionIndex];
      console.log('Trying to get question at
actualQuestionIndex:',
actualQuestionIndex);
      if (actualQuestionIndex >= 0 &&
actualQuestionIndex <
quiz.questions.length) {
        currentQuestion =
quiz.questions[actualQuestionIndex];
```

```
      console.log('Successfully got
current question:', currentQuestion?.id);
     } else {
      console.warn('Invalid
actualQuestionIndex:',
actualQuestionIndex,
'quiz.questions.length:',
quiz.questions.length);
     }
    } else {
     console.warn('Invalid unanswered
mode state:', {
      unansweredLength:
unansweredQuestionIndexes.length,
      currentQuestionIndex,
      isInRange: currentQuestionIndex
>= 0 && currentQuestionIndex <
unansweredQuestionIndexes.length
     });
    }
  } else {
    // Normal mode
    if (currentQuestionIndex >= 0 &&
currentQuestionIndex <
quiz.questions.length) {
     currentQuestion =
quiz.questions[currentQuestionIndex];
    }
  }

 // If we still don't have a valid question,
show loading
 if (!currentQuestion) {
   console.warn('No current question
found - this might indicate an issue with
question navigation');

   // Try to recover by resetting to a valid
state
   if (showingUnansweredOnly &&
unansweredQuestionIndexes.length ===
0) {
     // No more unanswered questions,
switch back to normal mode
     console.log('Recovering: switching
back to normal mode');
     setShowingUnansweredOnly(false);

setCurrentQuestionIndex(quiz.questions.
length - 1);
     return <LoadingIndicator
text="Naproceed sa huling tanong..." />;
   } else if (showingUnansweredOnly
&& currentQuestionIndex >=
unansweredQuestionIndexes.length) {
     // Current index is out of bounds,
reset to first unanswered question
     console.log('Recovering: resetting to
first unanswered question');
     setCurrentQuestionIndex(0);
     return <LoadingIndicator
text="Bumabalik sa unang hindi
nasagot..." />;
   }

   return <LoadingIndicator
text="Naglo-load ang tanong..." />;
 }

 const selectedAnswer =
answers[currentQuestion?.id || ''];
 const isRetake =
isQuizCompleted(quiz.id);
 const currentAttempts =
getQuizAttempts(quiz.id);
 const maxAttempts = 3;
 const isLastAttempt = currentAttempts
>= maxAttempts - 1;

 // Helper functions for skip
functionality
 const getUnansweredQuestions = () =>
{
   return quiz.questions
     .map((_, index) => index)
     .filter(index => {
      const questionId =
quiz.questions[index].id;
      return !answers[questionId] ||
answers[questionId].trim() === '';
     });
 };
```

```
  const isCurrentlyOnLastQuestion = ()
=> {
    if (showingUnansweredOnly) {
      return currentQuestionIndex >=
unansweredQuestionIndexes.length - 1;
    }
    return currentQuestionIndex ===
quiz.questions.length - 1;
  };

  const hasUnansweredQuestions = () =>
{
    return
getUnansweredQuestions().length > 0;
  };

  const getCurrentQuestionNumber = ()
=> {
    if (showingUnansweredOnly) {
      return
unansweredQuestionIndexes[currentQue
stionIndex] + 1;
    }
    return currentQuestionIndex + 1;
  };

  const getTotalQuestions = () => {
    if (showingUnansweredOnly) {
      return
unansweredQuestionIndexes.length;
    }
    return quiz.questions.length;
  };

  const isLastQuestion =
isCurrentlyOnLastQuestion();

  // Calculate progress percentage for
debugging
  const progressPercentage =
((currentQuestionIndex + 1) /
getTotalQuestions()) * 100;

  // Debug log
  console.log('Quiz Progress:', {
    currentQuestionIndex,
    totalQuestions: quiz.questions.length,
    progressPercentage:
progressPercentage.toFixed(1) + '%'
  });

  // Show retake warning on mount
  // Removed attempt warning alerts for
cleaner UX

  const calculateQuestionScore =
(answerTime: number, basePoints:
number): { points: number; timeBonus:
number } => {
    // Time bonus calculation (max 10
seconds for full bonus)
    const maxTimeForBonus = 10; //
seconds
    const timeBonus = Math.max(0,
Math.floor((maxTimeForBonus -
answerTime) * (basePoints * 0.5 /
maxTimeForBonus)));
    const totalPoints = basePoints +
timeBonus;

    return {
      points: Math.max(basePoints,
totalPoints),
      timeBonus: timeBonus
    };
  };

  const showPointsAnimation = (points:
number, timeBonus: number, isCorrect:
boolean) => {
    setShowPointsGained({ points,
timeBonus, isCorrect });
    setTimeout(() =>
setShowPointsGained(null), 2000);
  };

  const handleAnswerSelect = (answer:
string) => {
    if (!currentQuestion) return;
```

```
    // This function is now ONLY for
multiple choice questions
    // Fill-in-blank questions use
handleFillInBlankTextChange
    if (currentQuestion.type ===
QuestionType.FILL_IN_BLANK) {
      console.warn('handleAnswerSelect
called for fill-in-blank question - this
should not happen');
      return; // Exit immediately - fill-in-
blank should use
handleFillInBlankTextChange
    }

    // For multiple choice questions,
prevent re-answering
    if
(answeredQuestions.has(currentQuestion
.id)) {
      return;
    }

    // Calculate time taken for this
question
    const answerTime = (new
Date().getTime() -
questionStartTime.getTime()) / 1000;

    // Store the answer
    setAnswers(prev => ({
      ...prev,
      [currentQuestion.id]: answer
    }));

    // If we're in unanswered mode and
this question now has an answer,
    // we need to update the unanswered
question indexes
    if (showingUnansweredOnly &&
answer && answer.trim() !== '') {
      // Update the unanswered questions
list to remove this question
      if (currentQuestionIndex <
unansweredQuestionIndexes.length) {
        const currentActualIndex =
unansweredQuestionIndexes[currentQue
stionIndex];
        const updatedUnanswered =
unansweredQuestionIndexes.filter(index
=> index !== currentActualIndex);

setUnansweredQuestionIndexes(updated
Unanswered);

        // If this was the last unanswered
question in the current list,
        // we might need to adjust the
current index
        if (currentQuestionIndex >=
updatedUnanswered.length &&
updatedUnanswered.length > 0) {

setCurrentQuestionIndex(updatedUnans
wered.length - 1);
        } else if
(updatedUnanswered.length === 0) {
          // No more unanswered questions,
switch back to normal mode

setShowingUnansweredOnly(false);

setCurrentQuestionIndex(quiz.questions.
length - 1);
        }
      }
    }

    // Store the time taken
    setQuestionTimes(prev => ({
      ...prev,
      [currentQuestion.id]: answerTime
    }));

    // Mark question as answered
    setAnsweredQuestions(prev => new
Set(prev).add(currentQuestion.id));

    // Check if answer is correct
    const isCorrect = answer ===
currentQuestion.correctAnswer;
```

```
  if (isCorrect) {
    // Correct answer: add points with
time bonus
    const scoreData =
calculateQuestionScore(answerTime,
currentQuestion.points);
    setTotalPointsEarned(prev => prev +
scoreData.points);
    setTotalTimeBonus(prev => prev +
scoreData.timeBonus);

showPointsAnimation(scoreData.points,
scoreData.timeBonus, true);
  } else {
    // Wrong answer: just show visual
feedback, no point deduction
    showPointsAnimation(0, 0, false);
  }

  // Auto-advance to next question or
handle end of quiz after a short delay
  setTimeout(() => {
    // IMPORTANT: If we're in
unanswered mode, don't auto-advance to
prevent conflicts
    if (showingUnansweredOnly) {
      return; // User must manually
navigate in unanswered mode
    }

    if (currentQuestionIndex ===
quiz.questions.length - 1) {
      // Last question - check for
unanswered questions before submitting
      if (hasUnansweredQuestions()) {
        // Switch to showing only
unanswered questions
        const unanswered =
getUnansweredQuestions();
        if (unanswered.length > 0) {

setUnansweredQuestionIndexes(unansw
ered);

setShowingUnansweredOnly(true);
          setCurrentQuestionIndex(0); //
Start from first unanswered question
        } else {
          handleSubmitQuiz();
        }
      } else {
        // All questions answered, now
submit
        handleSubmitQuiz();
      }
    } else {
      // Move to next question

setCurrentQuestionIndex(currentQuestio
nIndex + 1);
      setQuestionStartTime(new Date());
    }
  }, 2000); // 2 second delay to show the
feedback
 };

 // Separate handler for fill-in-blank text
input (no auto-advance)
 const handleFillInBlankTextChange =
(text: string) => {
  if (!currentQuestion) return;

  console.log('Fill-in-blank text
change:', text.length, 'characters');

  // Just store the text without any
validation, auto-advance, or index
manipulation
  setAnswers(prev => ({
   ...prev,
   [currentQuestion.id]: text
  }));

  // Note: We don't update unanswered
question indexes here to prevent auto-
navigation
  // The indexes will be updated when
the user manually clicks "Next" and the
answer is submitted
 };
```

```
  const submitFillInBlankAnswer =
(answer: string) => {
    if (!currentQuestion ||
answeredQuestions.has(currentQuestion.
id)) return;

    // Calculate time taken for this
question
    const answerTime = (new
Date().getTime() -
questionStartTime.getTime()) / 1000;

    // Store the time taken
    setQuestionTimes(prev => ({
      ...prev,
      [currentQuestion.id]: answerTime
    }));

    // Mark question as answered
    setAnsweredQuestions(prev => new
Set(prev).add(currentQuestion.id));

    // If we're in unanswered mode,
update the unanswered questions list
    if (showingUnansweredOnly) {
      const currentActualIndex =
unansweredQuestionIndexes[currentQue
stionIndex];
      const updatedUnanswered =
unansweredQuestionIndexes.filter(index
=> index !== currentActualIndex);

setUnansweredQuestionIndexes(updated
Unanswered);

      // Adjust currentQuestionIndex if
needed to prevent loading issues
      if (updatedUnanswered.length ===
0) {
        // No more unanswered questions,
but don't auto-navigate - let handleNext
handle it
        console.log('All unanswered
questions completed');
      } else if (currentQuestionIndex >=
updatedUnanswered.length) {
        // Current index is now out of
bounds, move to the last available
question

setCurrentQuestionIndex(updatedUnans
wered.length - 1);
      }
      // Note: If currentQuestionIndex <
updatedUnanswered.length, we stay at
the same position
      // which will now show the next
unanswered question
    }

    // Check if answer is correct (trim for
comparison)
    const isCorrect =
answer.trim().toLowerCase() ===
currentQuestion.correctAnswer.toLower
Case();

    if (isCorrect) {
      // Correct answer: add points with
time bonus
      const scoreData =
calculateQuestionScore(answerTime,
currentQuestion.points);
      setTotalPointsEarned(prev => prev +
scoreData.points);
      setTotalTimeBonus(prev => prev +
scoreData.timeBonus);

showPointsAnimation(scoreData.points,
scoreData.timeBonus, true);
    } else {
      // Wrong answer: just show visual
feedback, no point deduction
      showPointsAnimation(0, 0, false);
    }

    // Auto-advance to next question or
handle end of quiz after a short delay
    setTimeout(() => {
      // IMPORTANT: If we're in
unanswered mode, don't auto-advance to
prevent conflicts
```

```
    if (showingUnansweredOnly) {
      return; // User must manually
navigate in unanswered mode
    }

    if (currentQuestionIndex ===
quiz.questions.length - 1) {
      // Last question - check for
unanswered questions before submitting
      if (hasUnansweredQuestions()) {
        // Switch to showing only
unanswered questions
        const unanswered =
getUnansweredQuestions();
        if (unanswered.length > 0) {

setUnansweredQuestionIndexes(unansw
ered);

setShowingUnansweredOnly(true);
          setCurrentQuestionIndex(0); //
Start from first unanswered question
        } else {
          handleSubmitQuiz();
        }
      } else {
        // All questions answered, now
submit
        handleSubmitQuiz();
      }
    } else {
      // Move to next question

setCurrentQuestionIndex(currentQuestio
nIndex + 1);
      setQuestionStartTime(new Date());
    }
  }, 2000); // 2 second delay to show the
feedback
 };

 const handleNext = () => {
  // For fill-in-blank questions, submit
the answer when moving to next (only if
answered)
  if (currentQuestion?.type ===
QuestionType.FILL_IN_BLANK &&
selectedAnswer &&
!answeredQuestions.has(currentQuestion
.id)) {

submitFillInBlankAnswer(selectedAnsw
er);
  }

  if (isLastQuestion) {
   // Check if there are unanswered
questions
    if (!showingUnansweredOnly &&
hasUnansweredQuestions()) {
      // Switch to showing only
unanswered questions
      const unanswered =
getUnansweredQuestions();
      if (unanswered.length > 0) {

setUnansweredQuestionIndexes(unansw
ered);

setShowingUnansweredOnly(true);
        setCurrentQuestionIndex(0); //
Start from first unanswered question

        Alert.alert(
          'May Hindi Pa Nasagot',
          `May ${unanswered.length}
pang tanong na hindi pa nasagot.
Sagutan mo muna ang lahat bago mo
ma-submit ang quiz.`,
          [{ text: 'OK' }]
        );
        return; // Don't submit yet, go to
unanswered questions first
      } else {
        // No unanswered questions found,
submit the quiz
        handleSubmitQuiz();
      }
    } else if (showingUnansweredOnly
&& hasUnansweredQuestions()) {
```

```
        // Still in unanswered mode and
there are still unanswered questions
        Alert.alert(
          'May Hindi Pa Nasagot',
          'Sagutan mo muna ang lahat ng
tanong bago mo ma-submit ang quiz.',
          [{ text: 'OK' }]
        );
        return; // Don't submit yet
      } else {
        // All questions are now answered,
submit the quiz
        handleSubmitQuiz();
      }
    } else {
      // Move to next question
      console.log('Moving to next
question. Current mode:',
showingUnansweredOnly ? 'unanswered'
: 'normal');
      console.log('Current index:',
currentQuestionIndex, 'Total
questions/unanswered:',
showingUnansweredOnly ?
unansweredQuestionIndexes.length :
quiz.questions.length);

      if (showingUnansweredOnly) {
        // In unanswered mode, make sure
we don't go out of bounds
        if (currentQuestionIndex + 1 <
unansweredQuestionIndexes.length) {

setCurrentQuestionIndex(currentQuestio
nIndex + 1);
        } else {
          console.warn('Attempted to
navigate beyond unanswered questions');
        }
      } else {
        // Normal mode

setCurrentQuestionIndex(currentQuestio
nIndex + 1);
      }
    }
  };

  const handlePrevious = () => {
    if (currentQuestionIndex > 0) {

setCurrentQuestionIndex(currentQuestio
nIndex - 1);
    }
  };

  const handleSubmitQuiz = async () =>
{
    console.log('QuizScreen:
handleSubmitQuiz called');

    // Safety check: Don't submit if there
are still unanswered questions
    if (hasUnansweredQuestions()) {
      console.log('QuizScreen: Cannot
submit - still has unanswered questions');
      Alert.alert(
        'May Hindi Pa Nasagot',
        'Hindi mo pa masasagutan ang lahat
ng tanong. Sagutan mo muna ang lahat
bago mag-submit.',
        [{ text: 'OK' }]
      );
      return;
    }

    const endTime = new Date();
    const timeSpent =
Math.floor((endTime.getTime() -
startTime.getTime()) / 1000);

    // Check if this is a retake (no points
for retakes)
    const isRetake =
isQuizCompleted(quiz.id);

    // Calculate detailed score with time
bonuses and track which questions were
answered correctly
    let correctAnswers = 0;
    let totalPossiblePoints = 0;
    let totalTimeBonus = 0;
```

```
    let newPointsThisAttempt = 0;

    // Prepare question data for the
completeQuiz function with robust
checks
    let questionData: { id: string;
correctAnswer: string; points: number
}[] = [];

    if (quiz.questions &&
Array.isArray(quiz.questions) &&
quiz.questions.length > 0) {
      questionData =
quiz.questions.map(question => ({
        id: question.id,
        correctAnswer:
question.correctAnswer,
        points: question.points
      }));
    }

    if (!quiz.questions ||
quiz.questions.length === 0 ||
questionData.length === 0) {
      console.error('Quiz questions are
undefined, empty, or questionData failed
to prepare');
      Alert.alert('Error', 'Hindi ma-load
ang mga tanong ng quiz');
      return;
    }

    quiz.questions.forEach(question => {
      totalPossiblePoints +=
question.points;

      if (answers[question.id] ===
question.correctAnswer) {
        correctAnswers++;
        const answerTime =
questionTimes[question.id] || 10; //
Default to 10s if no time recorded
        const scoreData =
calculateQuestionScore(answerTime,
question.points);
        totalTimeBonus +=
scoreData.timeBonus;

        // Check if we haven't earned points
for this question before
        const alreadyEarnedForQuestion =
getQuestionPoints(question.id);
        if (alreadyEarnedForQuestion ===
0) {
          newPointsThisAttempt +=
scoreData.points;
        }
      }
    });

    const accuracy = (correctAnswers /
quiz.questions.length) * 100;
    const quizPassed = accuracy >=
quiz.passingScore; // Use quiz's passing
score
    setPassed(quizPassed);

    console.log('=== ARALIN 1 QUIZ
ACCURACY DEBUG ===');
    console.log('Quiz ID:', quiz.id);
    console.log('Quiz Title:', quiz.title);
    console.log('Total Questions:',
quiz.questions.length);
    console.log('Question IDs:',
quiz.questions.map(q => q.id));
    console.log('Answers Object:',
answers);
    console.log('Correct Answers Count:',
correctAnswers);
    console.log('Questions with
Answers:', Object.keys(answers).length);

    // Detailed answer checking
    quiz.questions.forEach((question,
index) => {
      const userAnswer =
answers[question.id];
      const isCorrect = userAnswer ===
question.correctAnswer;
      console.log(`Q${index + 1}
(${question.id}): "${userAnswer}" vs
```

```
"${question.correctAnswer}" =
${isCorrect ? 'CORRECT' :
'WRONG'}`);
   });

   console.log('Raw Accuracy
Calculation:',
`${correctAnswers}/${quiz.questions.le
ngth} * 100 = ${accuracy}`);
   console.log('Displayed Accuracy:',
`${accuracy.toFixed(1)}%`);

console.log('===================
====================');

   console.log('Quiz completion check:',
{
     correctAnswers,
     totalQuestions:
quiz.questions.length,
     accuracy: accuracy.toFixed(2),
     passingScore: quiz.passingScore,
     quizPassed,
     quizId: quiz.id,
     accuracyCalculation:
`${correctAnswers}/${quiz.questions.le
ngth} * 100 =
${accuracy.toFixed(2)}%`,
     passCondition:
`${accuracy.toFixed(2)} >=
${quiz.passingScore} = ${quizPassed}`
   });

   // Check if quiz was already
completed BEFORE this attempt
(copying lesson pattern)
   const quizAlreadyCompleted =
isQuizCompleted(quiz.id);

setWasAlreadyCompleted(quizAlreadyC
ompleted);

   // Complete the quiz and get actual
new points earned (this will save the
progress)
   let quizResult = null;

   // CRITICAL: Only complete quiz
and award points/badges if user
PASSED
   if (!isRetake && quizPassed &&
!quizAlreadyCompleted) { // Only
complete if passed and not already
completed
     console.log('✅ QUIZ PASSED -
Awarding points and badges');
     try {
       // Final safety check before calling
completeQuiz
       if (!questionData ||
questionData.length === 0) {
         console.error('questionData is
invalid before calling completeQuiz:',
questionData);
         Alert.alert('Error', 'Hindi ma-save
ang quiz results dahil sa problema sa
mga tanong.');
         return;
       }

       quizResult = await
completeQuiz(quiz.id, accuracy,
newPointsThisAttempt, timeSpent,
answers, questionData);

       // Note: completeQuiz already
handles experience points internally, no
need to call addExperience again
     } catch (error) {
       console.error('Error completing
quiz:', error);
       Alert.alert('Error', 'Hindi ma-save
ang quiz results. Subukan ulit.');
       return;
     }

     // Store completion data for modal
display (copying lesson pattern)

setCompletionScore(newPointsThisAtte
mpt); // Use time-based score instead of
quizResult.pointsGained
```

```
setCompletionTimeSpent(timeSpent);
      setCompletionAccuracy(accuracy);

setCompletionExperience(quizResult?.experienceGained || 0);
      setPassed(quizPassed); // Ensure passed state is set correctly

    console.log('About to complete quiz:', {
      quizId: quiz.id,
      quizTitle: quiz.title,
      timeBasedScore: newPointsThisAttempt,
      accuracy,
      quizResult,
      timeSpent,
      wasAlreadyCompleted: quizAlreadyCompleted,
      correctAnswers,
      totalPossiblePoints,
      totalTimeBonus,
      passed: quizPassed
    });      // Quiz completion already happened above at line 277
      console.log('QuizScreen: Quiz completion successful');

      // Check for max level achievement after quiz completion
      if (quizResult?.maxLevelBadge && !quizAlreadyCompleted) {
        console.log('🏆 MAX LEVEL ACHIEVED IN QUIZ! Setting up ceremony...');

setMaxLevelBadge(quizResult.maxLevelBadge);
        setShowMaxLevelModal(true);
        return; // Show max level ceremony first, skip other modals
      }

      // For first completion: show badge and summary modal (copying lesson pattern)
      if (!quizAlreadyCompleted) {
        // Check for Aralin 9 quiz ceremony
        if (quizResult?.isAralin9QuizCompletion && quiz.id === 'aralin9_quiz') {
          console.log('🎉 Aralin 9 quiz completion ceremony!');

          // Set up ceremony modal data

setCeremonySpecialBadge(quizResult.specialBadge || null);

setShowQuizCeremonyModal(true);
        } else {
          // Regular badge flow for other quizzes
          // Check if badge should be awarded (quiz has badge, passed, and within attempt limit)
          // CRITICAL: Double-check that user actually passed before awarding badge
          if (quiz.badge && currentAttempts < maxAttempts && quizPassed && accuracy >= quiz.passingScore) {
            console.log('🏆 Awarding badge - user passed with accuracy:', accuracy.toFixed(2) + '%');
            const newBadge = {
              id: String(quiz.badge.id || ''),
              name: String(quiz.badge.name || ''),
              description: String(quiz.badge.description || ''),
              icon: String(quiz.badge.icon || '🏆'),
              color: String(quiz.badge.color || '#10B981'),
              category: quiz.badge.category || 'quiz' as const,
```

```
          requirement:
String(quiz.badge.requirement || ''),
          earnedAt: new Date()
        };
        setEarnedBadge(newBadge);

        // Also award the badge in the
progress context
        await awardBadge(
          quiz.badge.id,
          quiz.badge.name,
          quiz.badge.description,
          quiz.badge.icon,
          quiz.badge.color,
          quiz.badge.category,
          quiz.badge.requirement
        );
      } else {
        console.log('🚫 No badge
awarded - conditions not met:', {
          hasBadge: !!quiz.badge,
          withinAttemptLimit:
currentAttempts < maxAttempts,
          quizPassed,
          actualAccuracy:
accuracy.toFixed(2) + '%',
          requiredAccuracy:
quiz.passingScore + '%'
        });
        setEarnedBadge(null);
      }

      setShowBadgeModal(true);
      // Don't call onComplete() here -
wait for user to click "Magpatuloy"
    }
  } else {
    // Retaking quiz: just complete
silently and go back (copying lesson
pattern)
    console.log('QuizScreen: Calling
onComplete for retake', { accuracy,
quizResult, timeSpent });
    onComplete(accuracy,
quizResult?.pointsGained || 0,
timeSpent);
      }
    } else if (quizPassed &&
quizAlreadyCompleted) {
      // Quiz passed but already completed
before - no experience/badges
      setCompletionScore(0); // No points
for retaking

setCompletionTimeSpent(timeSpent);
      setCompletionAccuracy(accuracy);
      setCompletionExperience(0); // No
experience for retaking
      setEarnedBadge(null); // No badge
for retaking
      setPassed(true); // Important: Set
passed to true since quiz was passed

      console.log('Quiz passed but already
completed:', { accuracy,
wasAlreadyCompleted:
quizAlreadyCompleted, passed: true });
      setShowBadgeModal(true); // Still
show modal but with retake message
    } else {
      console.log('❌ QUIZ FAILED - No
points or badges awarded');
      // Failed the quiz (didn't pass) - set
completion data but no
experience/badges
      setCompletionScore(0); // No points
for failing

setCompletionTimeSpent(timeSpent);
      setCompletionAccuracy(accuracy);
      setCompletionExperience(0); // No
experience for failing
      setEarnedBadge(null); // No badge
for failing
      setPassed(false); // Important: Set
passed to false since quiz failed

      // CRITICAL: Ensure no progress is
saved for failed attempts
      console.log('Quiz failed:', {
        accuracy: accuracy.toFixed(2),
        passed: false,
```

```
      requiredAccuracy:
quiz.passingScore,
      wasCompleteQuizCalled: false,
      wasBadgeAwarded: false,
      experienceGained: 0
    });
    setShowBadgeModal(true); // Still
show modal to display failure and allow
retake
  }
 };

 const showResults = (score: number,
pointsEarned: number, timeSpent:
number) => {
  // TODO: Implement results screen
  console.log('QuizScreen: showResults
calling onComplete', { score,
pointsEarned, timeSpent });
  onComplete(score, pointsEarned,
timeSpent);
 };

 const handleBadgeModalClose = () =>
{
  setShowBadgeModal(false);
  setEarnedBadge(null);
  setCompletionScore(0);
  setCompletionTimeSpent(0);
  setCompletionAccuracy(0);

  // Now call onComplete when user
actually clicks "Magpatuloy" (copying
lesson pattern)
  console.log('QuizScreen:
handleBadgeModalClose calling
onComplete');
  onComplete(0, 0, 0); // Values don't
matter since completion already
happened
 };

 const
handleQuizCeremonyModalClose = ()
=> {
  setShowQuizCeremonyModal(false);
  setCeremonySpecialBadge(null);
  setCompletionScore(0);
  setCompletionTimeSpent(0);
  setCompletionAccuracy(0);

  // Complete the quiz after ceremony
  console.log('QuizScreen:
handleQuizCeremonyModalClose
calling onComplete');
  onComplete(0, 0, 0); // Values don't
matter since completion already
happened
 };

 const handleMaxLevelModalClose = ()
=> {
  setShowMaxLevelModal(false);
  setMaxLevelBadge(null);

  // After max level ceremony, continue
with normal flow
  onComplete(0, 0, 0); // Values don't
matter since completion already
happened
 };

 const handleRetakeQuiz = () => {
  // Reset quiz state for retake
  setCurrentQuestionIndex(0);
  setAnswers({});
  setSkippedQuestions(new Set());
  setShowingUnansweredOnly(false);
  setUnansweredQuestionIndexes([]);
  setTimeLeft(quiz.timeLimit || 0);
  setQuestionStartTime(new Date());
  setQuestionTimes({});
  setTotalPointsEarned(0);
  setAnsweredQuestions(new Set());
  setTotalTimeBonus(0);
  setShowBadgeModal(false);
  setEarnedBadge(null);
  setCompletionScore(0);
  setCompletionTimeSpent(0);
  setCompletionAccuracy(0);
  setCompletionExperience(0);
  setPassed(true);
```

```
    console.log('Quiz retake initiated');
  };

  const formatTime = (seconds: number)
=> {
    const mins = Math.floor(seconds /
60);
    const secs = seconds % 60;
    return
`${mins}:${secs.toString().padStart(2,
'0')}`;
  };

  const renderQuestion = () => {
    if (!currentQuestion) {
      return (
        <View className="bg-white
rounded-2xl p-6 shadow-sm">
          <Text className="text-lg font-
semibold text-gray-600 text-
center">Naglo-load ng tanong...</Text>
        </View>
      );
    }

    switch (currentQuestion.type) {
      case
QuestionType.CHARACTER_RECOG
NITION:
        return (
          <View className="bg-white
rounded-2xl p-6 shadow-sm">
            <Text className="text-lg font-
semibold text-gray-800 text-center mb-8
leading-
7">{currentQuestion.question}</Text>
            <View className="items-center
mb-8 p-8 bg-gray-50 rounded-xl">
              <Text className="text-6xl
text-primary font-bold">

{currentQuestion.baybayinCharacter}
              </Text>
            </View>
            {renderOptions()}
          </View>
        );

      case
QuestionType.ROMANIZED_TO_BAY
BAYIN:
        return (
          <View className="bg-white
rounded-2xl p-6 shadow-sm">
            <Text className="text-lg font-
semibold text-gray-800 text-center mb-8
leading-
7">{currentQuestion.question}</Text>
            <View className="items-center
mb-8 p-8 bg-gray-50 rounded-xl">
              <Text className="text-5xl
text-primary font-bold">

{currentQuestion.romanizedCharacter}
              </Text>
            </View>
            {renderOptions()}
          </View>
        );

      case
QuestionType.FILL_IN_BLANK:
        return (
          <View className="bg-white
rounded-2xl p-6 shadow-sm">
            <Text className="text-lg font-
semibold text-gray-800 text-center mb-8
leading-
7">{currentQuestion.question}</Text>
            <View className="mb-6">
              <Text className="text-sm
font-medium text-gray-600 mb-2">Isulat
ang inyong sagot:</Text>
              <TextInput
                className="bg-gray-50
border-2 border-gray-300 rounded-xl px-
4 py-3 text-lg font-medium text-gray-
800 text-center"
                style={{
```

```
              borderColor: selectedAnswer
&& selectedAnswer.length > 0 ?
'#6B7280' : '#D1D5DB',
              backgroundColor: '#F9FAFB'
            }}
            placeholder="Mag-type dito..."

placeholderTextColor="#9CA3AF"
            value={selectedAnswer || ''}

onChangeText={handleFillInBlankText
Change}
            autoCapitalize="none"
            autoCorrect={false}
            textAlign="center"
            multiline={false}
            returnKeyType="done"
          />
          <Text className="text-xs text-
gray-500 text-center mt-2">
            💡 I-type ang tamang sagot at
pindutin ang "Next" para sa susunod na
tanong
          </Text>
        </View>
      </View>
    );

    default:
      return (
        <View className="bg-white
rounded-2xl p-6 shadow-sm">
          <Text className="text-lg font-
semibold text-gray-800 text-center mb-8
leading-
7">{currentQuestion.question}</Text>
          {renderOptions()}
        </View>
      );
    }
  };

  const renderOptions = () => {
    if (!currentQuestion ||
!currentQuestion.options) return null;

    const isQuestionAnswered =
answeredQuestions.has(currentQuestion.
id);

    return (
      <View className="gap-3">

{currentQuestion.options.map((option,
index) => {
          const isSelected = selectedAnswer
=== option;
          const isCorrect = option ===
currentQuestion.correctAnswer;
          const isWrong =
isQuestionAnswered && isSelected &&
!isCorrect;
          const shouldShowCorrect =
isQuestionAnswered && isCorrect;

          return (
            <TouchableOpacity
              key={index}
              className={`bg-gray-100 p-4
rounded-xl border-2 flex-row items-
center justify-center ${
                isSelected ? 'bg-yellow-50
border-yellow-500' : 'border-transparent'
              } ${isWrong ? 'bg-red-50
border-red-500' : ''} ${
                shouldShowCorrect ? 'bg-
yellow-50 border-yellow-500' : ''
              }`}
              onPress={() =>
handleAnswerSelect(option)}

disabled={isQuestionAnswered}
            >
              <Text className={`text-base
text-gray-800 text-center font-medium
${
                isSelected ? 'text-green-600
font-bold' : ''
              } ${isWrong ? 'text-red-600
font-bold' : ''} ${
                shouldShowCorrect ? 'text-
green-600 font-bold' : ''
```

```
          }`}>
            {option}
          </Text>
          {shouldShowCorrect && (
            <Ionicons name="checkmark-
circle" size={20} color="#10B981"
className="absolute right-4" />
          )}
          {isWrong && (
            <Ionicons name="close-circle"
size={20} color="#EF4444"
className="absolute right-4" />
          )}
        </TouchableOpacity>
      );
    })}
  </View>
  );
 };

 return (
  <SafeAreaView className="flex-1
bg-gray-50">
    {/* Header */}
    <View className="bg-blue-600
flex-row items-center px-4 py-3">
      <TouchableOpacity
onPress={onBack} className="mr-3">
        <Ionicons name="arrow-back"
size={24} color="white" />
      </TouchableOpacity>
      <View className="flex-1">
        <Text className="text-lg font-
bold text-white">
          {quiz.title} {isRetake &&
'(Retake)'}
        </Text>
        <Text className="text-sm text-
blue-100">
          {showingUnansweredOnly
            ? `Hindi Pa Nasagot
${currentQuestionIndex + 1} ng
${unansweredQuestionIndexes.length}`
            : `Tanong
${getCurrentQuestionNumber()} ng
${quiz.questions.length}`
```

```
          }
        </Text>
        {showingUnansweredOnly && (
          <Text className="text-xs text-
yellow-300">
            🔄 Ulit sa mga hindi pa
nasagot
          </Text>
        )}
      </View>
      <View className="items-end">
        <Text className="text-white
text-base font-bold mb-1">
          {totalPointsEarned} pts
        </Text>
        {totalTimeBonus > 0 && (
          <Text className="text-yellow-
300 text-xs font-semibold mb-1">
            ⚡ Bonus: +{totalTimeBonus}
          </Text>
        )}
        {quiz.timeLimit && (
          <View className="flex-row
items-center">
            <Ionicons name="time-outline"
size={16} color="white" />
            <Text className="text-white
text-base font-bold ml-
1">{formatTime(timeLeft)}</Text>
          </View>
        )}
      </View>
    </View>

    {/* Progress Bar */}
    <View className="bg-secondary-50
px-4 py-2 border-b border-secondary-
200">
      <View className="h-2 bg-
secondary-200 rounded-full overflow-
hidden">
        <View
          className="h-full bg-secondary
rounded-full"
          style={{ width:
`${progressPercentage}%` }}
```

```
          />
        </View>
        <Text className="text-xs text-
secondary-600 text-center mt-1">
          {showingUnansweredOnly
            ? `Hindi Pa Nasagot
${currentQuestionIndex + 1} sa
${unansweredQuestionIndexes.length}`
            : `Tanong
${getCurrentQuestionNumber()} sa
${quiz.questions.length}`
          }
        </Text>
      </View>

      {/* Question Content */}
      <ScrollView className="flex-1 p-4"
showsVerticalScrollIndicator={false}>
        {/* Real-time Time Bonus Indicator
*/}
        {currentQuestion &&
!answeredQuestions.has(currentQuestion
.id) && (
          <View className="bg-orange-
500 rounded-xl p-4 mb-4 shadow-lg"
style={{
            shadowColor: '#F97316',
            shadowOffset: { width: 0, height:
2 },
            shadowOpacity: 0.3,
            shadowRadius: 4,
            elevation: 6,
          }}>
            <View className="flex-row
items-center justify-between">
              <View className="flex-row
items-center">
                <View className="bg-
yellow-400 rounded-full p-1 mr-3">
                  <Ionicons name="flash"
size={18} color="white" />
                </View>
                <Text className="text-white
font-bold text-base">Dagdag puntos sa
bilis</Text>
              </View>
              <View className="flex-row
items-center bg-yellow-400 rounded-lg
px-3 py-1">
                <Text className="text-white
text-lg font-
bold">+{currentTimeBonus}</Text>
                <Text className="text-
orange-800 text-sm font-semibold ml-
1">pts</Text>
              </View>
            </View>
            <View className="mt-3">
              <Text className="text-orange-
100 text-sm font-medium">
                ⏱️
{questionTimer.toFixed(1)}s • Sagot
nang mabilis para sa bonus!
              </Text>
              <View className="mt-2 h-2
bg-orange-300 rounded-full overflow-
hidden">
                <View
                  className="h-full bg-
yellow-300 rounded-full"
                  style={{
                    width: `${Math.max(0, (10 -
questionTimer) / 10 * 100)}%`
                  }}
                />
              </View>
              <Text className="text-orange-
200 text-xs mt-1 text-center">
                {currentTimeBonus > 0 ? '🔥
Mabilis ka!' : questionTimer > 10 ? '⏰
Walang bonus' : '⚡ Bilisan para sa
bonus!'}
              </Text>
            </View>
          </View>
        )}

        {renderQuestion()}

        {/* Points Animation Overlay */}
        {showPointsGained && (
```

```
        <View className="absolute
inset-0 justify-center items-center bg-
black/40 z-50">
          <View className={`px-8 py-6
rounded-2xl shadow-xl mx-4 ${
            showPointsGained.isCorrect ?
'bg-green-500' : 'bg-red-500'
          }`} style={{
            shadowColor:
showPointsGained.isCorrect ? '#10B981'
: '#EF4444',
            shadowOffset: { width: 0,
height: 4 },
            shadowOpacity: 0.3,
            shadowRadius: 8,
            elevation: 10,
          }}>
            {showPointsGained.isCorrect ?
(
              <>
                <Text className="text-white
text-2xl font-bold text-center mb-2">
                  🎉 Tama! 🎉
                </Text>
                <Text className="text-white
text-xl font-bold text-center mb-2">

+{showPointsGained.points} points!
                </Text>

{showPointsGained.timeBonus > 0 ? (
                  <View className="bg-
yellow-400 rounded-lg px-4 py-2 mt-2">
                    <Text className="text-
orange-900 text-base font-bold text-
center">
                      ⚡ Dagdag puntos sa
bilis: +{showPointsGained.timeBonus}
                    </Text>
                    <Text className="text-
orange-800 text-sm text-center">
                      Mabilis ka! 🔥
                    </Text>
                  </View>
                ) : (
                  <Text className="text-
green-200 text-sm text-center mt-1">
                    Base score:
{showPointsGained.points} pts
                  </Text>
                )}
              </>
            ) : (
              <>
                <Text className="text-white
text-xl font-semibold text-center mb-1">
                  ❌ Mali
                </Text>
                <Text className="text-red-
200 text-sm text-center">
                  Subukan ulit sa susunod!
                </Text>
              </>
            )}
          </View>
        </View>
      )}
    </ScrollView>

    {/* Navigation */}
    <View className="flex-row p-4 bg-
white border-t border-gray-200">
      <TouchableOpacity
        className={`flex-1 flex-row
items-center justify-center py-3 px-4
rounded-lg mx-1 ${
          currentQuestionIndex === 0 ?
'bg-gray-300 opacity-50' : 'bg-yellow-
500'
        }`}
        onPress={handlePrevious}
        disabled={currentQuestionIndex
=== 0}
      >
        <Ionicons name="chevron-back"
size={20} color={currentQuestionIndex
=== 0 ? '#9CA3AF' : '#FFFFFF'} />
        <Text className={`text-base
font-medium ml-1
${currentQuestionIndex === 0 ? 'text-
```

```
gray-500' : 'text-
white'}`}>Nauna</Text>
        </TouchableOpacity>

        <TouchableOpacity
          className={`flex-1 flex-row
items-center justify-center py-3 px-4
rounded-lg mx-1 ${
            (showingUnansweredOnly &&
!selectedAnswer) ? 'bg-gray-300
opacity-50' : 'bg-blue-600'
          }`}
          onPress={handleNext}

disabled={showingUnansweredOnly
&& !selectedAnswer}
        >
          <Text className={`text-base
font-medium mr-1 ${
            (showingUnansweredOnly &&
!selectedAnswer) ? 'text-gray-500' :
'text-white'
          }`}>
            {isLastQuestion
              ? (showingUnansweredOnly
&& hasUnansweredQuestions() ? 'Hindi
Pwede Ipasa' : 'Ipasa')
              : (selectedAnswer ? 'Susunod' :
(showingUnansweredOnly ? 'Sagutan
Muna' : 'Laktawan'))
            }
          </Text>
          <Ionicons name="chevron-
forward" size={20} color={
            (showingUnansweredOnly &&
!selectedAnswer) ? '#9CA3AF' : 'white'
          } />
        </TouchableOpacity>
      </View>

      {/* Badge Earned Modal */}
      <BadgeEarnedModal
        visible={showBadgeModal}
        badge={earnedBadge}
        score={completionScore || 0}
        timeSpent={completionTimeSpent
|| 0}
        timeBonus={totalTimeBonus}
        accuracy={completionAccuracy}

experienceGained={completionExperien
ce}
        passed={passed}

wasAlreadyCompleted={wasAlreadyCo
mpleted}
        passingScore={quiz.passingScore}
        completionType="quiz"
        wasRetake={false}

onClose={handleBadgeModalClose}
        onRetake={handleRetakeQuiz}
      />

      {/* Aralin 9 Quiz Ceremony Modal -
Special celebration for completing all
quizzes */}
      <Aralin9CeremonyModal

visible={showQuizCeremonyModal}

specialBadge={ceremonySpecialBadge}
        score={completionScore || 0}
        timeSpent={completionTimeSpent
|| 0}
        type="quiz"

onClose={handleQuizCeremonyModalC
lose}
      />

      {/* Max Level Ceremony Modal -
Special celebration for reaching level 10
*/}
      <MaxLevelCeremonyModal
        visible={showMaxLevelModal}
        badge={maxLevelBadge}

onClose={handleMaxLevelModalClose}
      />
```

```
      {/* Custom Time Up Modal - No
blinking! */}
      <Modal
        visible={showTimeUpModal}
        transparent={true}
        animationType="fade"
        onRequestClose={() => {}} //
Prevent back button dismiss
      >
        <View className="flex-1 justify-
center items-center bg-black/60 px-6">
          <View className="bg-white
rounded-2xl p-8 w-full max-w-sm
shadow-2xl">
            {/* Header */}
            <View className="items-center
mb-6">
              <View className="bg-red-100
rounded-full p-4 mb-4">
                <Ionicons name="time"
size={40} color="#DC2626" />
              </View>
              <Text className="text-2xl
font-bold text-black text-center">
                ⏰ Tapos na ang Oras!
              </Text>
            </View>

            {/* Content */}
            <View className="mb-8">
              <Text className="text-base
text-black text-center leading-6">
                Naubos na ang
{Math.floor((quiz.timeLimit || 0) / 60)}
minutong oras para sa quiz na ito.
              </Text>

              <View className="bg-blue-50
rounded-lg p-4 mt-4 border border-blue-
200">
                <Text className="text-blue-
800 font-semibold text-center mb-1">
                  📊 Mga Nasagot
                </Text>
                <Text className="text-blue-
700 text-lg font-bold text-center">

{Object.keys(answers).length} sa
{quiz.questions.length} na tanong
                </Text>
              </View>

              <Text className="text-base
text-black text-center mt-4">
                Ano ang gusto ninyong gawin?
              </Text>
            </View>

            {/* Buttons */}
            <View className="gap-3">

              <TouchableOpacity
                className="bg-gray-200
rounded-xl py-4 px-6"
                onPress={() => {

setShowTimeUpModal(false);
                  onBack();
                }}
              >
                <Text className="text-black
text-lg font-bold text-center">
                  Bumalik sa Quiz List
                </Text>
              </TouchableOpacity>
            </View>
          </View>
        </View>
      </Modal>
    </SafeAreaView>
  );
}
```

Appendix C

Research Instrument

CAP 401 – Capstone Pr oject and Research 2
RESEARCH INSTRUMENT (ISO/IEC 25010:2023)

Please fill in the necessary information needed for your demographic profile. Additionally, using a scale of 4 (Strongly Agree) to 1 (Strongly Disagree), please rate the developed application with the following criteria based on the ISO/IEC 25010:2023 by marking (✓) the box.

Thank you very much.

DEMOGRAPHIC INFORMATION	
Full name *(optional)*	
Role	

A. FUNCTIONAL SUITABILITY	**4**	**3**	**2**	**1**
Functional Completeness. Naibibigay ng TalaBaybayin App ang lahat ng kinakailangang function para magawa ang mga nakatakdang gawain at layunin ng mga gumagamit.	☐	☐	☐	☐
Functional Correctness. Nagbibigay ang TalaBaybayin App ng tamang resulta kapag ginamit ng mga nakatakdang gumagamit.	☐	☐	☐	☐
Functional Appropriateness. Naibibigay ng TalaBaybayin App ang mga function na nakakatulong para matapos ang mga gawain at layunin.	☐	☐	☐	☐

B. PERFORMANCE EFFICIENCY	**4**	**3**	**2**	**1**
Time Behavior. Ginagawa ng TalaBaybayin App ang tungkulin nito sa tamang oras at bilis ayon sa kailangan.	☐	☐	☐	☐
Resource Utilization. Hindi gumagamit ang TalaBaybayin App ng sobra sa itinakdang dami ng resources kapag gumagana.	☐	☐	☐	☐
Capacity. Natutugunan ng TalaBaybayin App ang mga pangangailangan kaugnay ng pinakamataas na hangganan ng sistem.	☐	☐	☐	☐

C. COMPATIBILITY	**4**	**3**	**2**	**1**
Co-existence. Kayang gumana ng TalaBaybayin App ng maayos habang kasama o ka-share ng ibang produkto nang hindi nakakasira dito.	☐	☐	☐	☐
Interoperability. Kayang makipagpalitan ng impormasyon ng TalaBaybayin App sa ibang produkto at magamit ito nang sabay.	☐	☐	☐	☐

D. INTERACTION CAPABILITY	**4**	**3**	**2**	**1**
Appropriateness Recognizability. Nakikita ng mga gumagamit na ang TalaBaybayin App ay bagay sa kanilang pangangailangan.	☐	☐	☐	☐
Learnability. Madaling matutunan ng mga gumagamit kung paano gamitin ang mga function ng TalaBaybayin App sa takdang oras.	☐	☐	☐	☐

Operability. May mga function at katangian ang TalaBaybayin App na madaling kontrolin at patakbuhin.	□	□	□	□
User Error Protection. Kayang pigilan ng TalaBaybayin App ang maling pagpapatakbo.	□	□	□	□
User Engagement. Kaaya-aya at nakakaengganyong gamitin ang TalaBaybayin App para ipagpatuloy ang pakikipag-ugnayan.	□	□	□	□
Inclusivity. Maaaring gamitin ang TalaBaybayin App ng iba't ibang uri ng tao.	□	□	□	□
User Assistance. Magagamit ng mga tao na may iba't ibang kakayahan at katangian ang TalaBaybayin App para maabot ang kanilang layunin.	□	□	□	□
Self-Descriptiveness. Kayang magpakita ng tamang impormasyon ang TalaBaybayin App upang madaling maintindihan ng gumagamit kung paano ito gagamitin.	□	□	□	□

E. RELIABILITY	**4**	**3**	**2**	**1**
Faultlessness. Gumagana ng tama ang TalaBaybayin App sa normal na paggamit.	□	□	□	□
Availability. Naa-access at nagagamit ang TalaBaybayin App kapag kinakailangan.	□	□	□	□
Fault Tolerance. Gumagana pa rin ang TalaBaybayin App kahit may error sa hardware o software.	□	□	□	□
Recoverability. Kayang ibalik ng TalaBaybayin App ang data at maayos muli ang kalagayan nito matapos ang problema.	□	□	□	□

F. SECURITY	**4**	**3**	**2**	**1**
Confidentiality. Tinitiyak ng TalaBaybayin App na tanging awtorisadong tao lang ang may access sa data.	□	□	□	□
Integrity. Tinitiyak ng TalaBaybayin App na protektado laban sa pagbabago o pagbura ng walang pahintulot.	□	□	□	□
Non-repudiation. Kayang patunayan ng TalaBaybayin App ang isang aksyon o pangyayari upang hindi na ito maitanggi.	□	□	□	□
Accountability. Kayang itala ng TalaBaybayin App ang aksyon ng bawat gumagamit.	□	□	□	□
Authenticity. Kayang patunayan ng TalaBaybayin App ang pagkakakilanlan ng isang tao o resource.	□	□	□	□
Resistance. Kayang magpatuloy ang TalaBaybayin App kahit may masamang atake.	□	□	□	□

G. MAINTAINABILITY	**4**	**3**	**2**	**1**
Modularity. Ang pagbabago sa isang bahagi ay kayang malimitahan ang pekto sa ibang bahagi.	□	□	□	□
Reusability. Maaaring magamit ang TalaBaybayin App sa iba pang proyekto o bilang bahagi ng iba pang produkto.	□	□	□	□
Analysability. Maaaring masuri nang epektibo at mahusay ang TalaBaybayin App upang matukoy ang epekto ng inaasahang pagbabago	□	□	□	□

sa isa o higit pang bahagi nito, upang matukoy ang mga kakulangan o sanhi ng pagkabigo, o upang tukuyin ang mga bahagi na kailangang baguhin.				
Modifiability. Madaling baguhin ang TalaBaybayin App nang hindi nagkakaroon ng dagdag na mali o pagbaba ng kalidad.	☐	☐	☐	☐
Testability. Ang TalaBaybayin App ay may kakayahang magbigay-daan sa isang obhetibo (objective) at praktikal na pagsubok na maaaring idisenyo at isagawa upang matukoy kung natutugunan ang isang kinakailangan.	☐	☐	☐	☐

H. FLEXIBILITY	**4**	**3**	**2**	**1**
Adaptability. Kayang iangkop ng TalaBaybayin App sa iba't ibang hardware, software, o environment.	☐	☐	☐	☐
Scalability. Kayang dagdagan o bawasan ng TalaBaybayin App ang kapasidad nito ayon sa pangangailangan.	☐	☐	☐	☐
Installability. Madaling mai-install at ma-uninstall ang TalaBaybayin App sa isang environment.	☐	☐	☐	☐
Replaceability. Kayang palitan ng TalaBaybayin App ang ibang produkto para sa parehong gamit/paggamit.	☐	☐	☐	☐

I. SAFETY	**4**	**3**	**2**	**1**
Operational Constraint. Kayang panatilihin ng TalaBaybayin App ang ligtas na operasyon kahit may makatagpo na problema.	☐	☐	☐	☐
Risk Identification. Nakikita ng TalaBaybayin App ang mga pangyayaring maaaring magdulot ng panganib.	☐	☐	☐	☐
Fail-Safe. Kayang mag-shift ng TalaBaybayin App sa ligtas na mode kung sakaling pumalya ito o bumalik sa ligtas na kondisyon.	☐	☐	☐	☐
Hazard Warning. Nagbibigay ang TalaBaybayin App ng babala kapag may hindi katanggap-tanggap na panganib.	☐	☐	☐	☐
Safe Integration. Kayang manatiling ligtas ng TalaBaybayin App kahit maisama sa iba pang bahagi o produkto.	☐	☐	☐	☐

Comments/Suggestions/Recommendations:

____

____

____

____

THANK YOU!

Evaluated by:

____________________________/_____________

Signature/Date Signed

TalaBaybayin: A Mobile Learning Application for Guiding Filipino Learners in Mastering and Preserving Baybayin

We are 4th-year students from section 4H-G2 of the College of Information and Communications Technology at Bulacan State University, conducting a capstone research project titled TalaBaybayin.

TalaBaybayin is a mobile learning application designed to help Filipino learners master and preserve Baybayin, an ancient script that is an essential part of our cultural heritage. Your responses will help us understand user familiarity, preferences, and learning needs, allowing us to create a more engaging and effective Baybayin learning experience.

All information you provide will remain confidential and will be used solely for academic research purposes. We truly appreciate your time and valuable feedback!

Email: pipoycapstone0322@gmail.com

Privacy Statement

We are committed to comply with the Philippine Data Privacy Act of 2012 or RA 10173 in handling your data with confidentiality and anonymity. Data will only be presented as a summary or will be reported only as a collective combined total. By accepting the private policy statement of the university, you entrust your data with us and we will do our best to keep your information confidential.

☐ I agree

General Survey Questions

Name (Surname, First name, Last name M.I)

Mayroon ka bang ginagamit na mobile device?

☐ Meron

☐ Wala

Anong uri ng mobile phone ang kasalukuyan mong ginagamit?

☐ Android

☐ iPhone (iOS)

☐ None

☐ Other: ___________

Anong grade level ka ngayong taon?

☐ Grade 7

☐ Grade 8

☐ Grade 9

☐ Grade 10

May kaalaman o pamilyaridad ka ba sa Baybayin?

☐ Meron

☐ Wala

May araling Baybayin ba sa inyong asignaturang Filipino?

☐ Meron

☐ Wala

Kapag “Meron” ang iyong sagot, saan mo ito unang nalaman o narinig?

☐ Paaralan

☐ Pamilya o Komunidad

☐ Online/Social Media

☐ Aklat o Babasahin

☐ Other: ___________

May karanasan ka na ba sa paggamit ng Baybayin sa pagbasa o pagsusulat?

☐ Meron

☐ Wala

Alin sa mga sumusunod ang sa tingin mo mahalaga sa isang app para matuto ng Baybayin? (Piliin ang lahat ng naaangkop)

☐ Interactive Lessons

☐ Practice Exercises for Handwriting

☐ Quizzes and Assessments

☐ Progress Tracking

☐ Gamification (Points, Badges)

☐ Autosave ng Ginagawang Baybayin

☐ Text-to-Baybayin Translation

☐ Leaderboard

☐ Mini Games

☐ Text-to-Speech

May interes ka ba na sumubok matuto ng Baybayin sa isang mobile app?

☐ Meron

☐ Wala

Gamified Section

Nagkaroon ka na ba ng karanasan sa paggamit ng learning app gaya ng Duolingo, Quizizz, o Quizlet?

☐ Meron

☐ Wala

Saan ka mas nasisiyahan sa pag-aaral: kapag may laro o kapag walang laro?

☐ Meron

☐ Wala

May nabibigay ba sa sayo ng motibasyon ang mga learning app?

☐ Meron

☐ Wala

Maaari bang maituro ang Baybayin sa pamamagitan ng isang mobile app?

☐ Oo

☐ Hindi

Ano ang mas makakatulong para mapabilis ang pagkatuto mo ng Baybayin?

☐ May Mobile App

☐ Wala

Interesado akong matuto ng Baybayin gamit ang isang mobile app.

☐ Oo

☐ Hindi

Mayroon ka bang libreng oras para matutunan o magamit ang TalaBaybayin app?

☐ Meron

☐ Wala

Mayroon ka bang interes sa paggamit ng TalaBaybayin app para matapos ang mga aralin at makakuha ng badge?

☐ Meron

☐ Wala

Ire-rekomenda mo ba ito sa iba?

☐ Oo

☐ Hindi

Pagbasa ng Baybayin

Scale: 1 = Hindi Mahirap | 2 = Medyo Mahirap | 3 = Katamtamang Hirap | 4 = Sobrang Hirap

1. Bumasa (makilala) ng isang titik sa Baybayin → ☐1 ☐2 ☐3 ☐4

2. Bumasa ng Baybayin na may kudlit → ☐1 ☐2 ☐3 ☐4

3. Bumasa ng isang salita sa Baybayin → ☐1 ☐2 ☐3 ☐4

4. Bumasa ng isang pangungusap sa Baybayin → ☐1 ☐2 ☐3 ☐4

5. Makilala ang magkakahawig na titik → ☐1 ☐2 ☐3 ☐4

6. Bumasa ng mahahabang salita → ☐1 ☐2 ☐3 ☐4

7. Bumasa ng Baybayin nang walang gabay → ☐1 ☐2 ☐3 ☐4

8. Pagbasa ng Baybayin nang mabilis at tama → ☐1 ☐2 ☐3 ☐4

9. Pag-unawa sa isang salita → ☐1 ☐2 ☐3 ☐4

10. Pag-unawa sa isang pangungusap → ☐1 ☐2 ☐3 ☐4

Pagsulat ng Baybayin

Scale: 1 = Hindi Mahirap | 2 = Medyo Mahirap | 3 = Katamtamang Hirap | 4 = Sobrang Hirap

1. Magsulat ng isang titik sa Baybayin → ☐1 ☐2 ☐3 ☐4

2. Magsulat ng isang titik na may kudlit → ☐1 ☐2 ☐3 ☐4

3. Magsulat ng isang salita sa Baybayin → ☐1 ☐2 ☐3 ☐4

4. Magsulat ng isang pangungusap sa Baybayin → ☐1 ☐2 ☐3 ☐4

5. Isulat ang iyong pangalan sa Baybayin → ☐1 ☐2 ☐3 ☐4

6. Madali kong naisusulat ang aking pangalan → ☐1 ☐2 ☐3 ☐4

7. Madali kong natutukoy ang tamang mga simbolo → ☐1 ☐2 ☐3 ☐4

8. Madali kong natutukoy ang tamang mga simbolo (duplicate Q) →☐1 ☐2 ☐3 ☐4

9. Natutukoy ko agad ang tamang simbolo kapag sumusulat → ☐1 ☐2 ☐3 ☐4

10. Naisusulat ko nang tama ang kombinasyon ng patinig at katinig → ☐1 ☐2 ☐3 ☐4

Appendix D

Plagiarism Certificate

PLAGIARISM CERTIFICATION

This is to certify that the research paper detailed below has been evaluated by online anti-plagiarism software. The contents and material were found outstanding and below the permissible limit and indicated a no percentage score in AI detection of content copied.

Title of the document	: Talabaybayin: A Mobile Learning Application for Guiding Filipino Learners in Mastering and Preserving Baybayin
Document type	: Capstone
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Appendix E

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Appendix F

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AUTHOR/S: Christian B. Francia, Kylee Remiel L. Garcia, Alyssa Mae V. Lapuz, Angelo C. Miramonte, John Patrick D. Teodoro
ADVISER: Aaron Paul M. Dela Rosa, MSIT
DATE:
SUBMITTED:
COLLEGE/PROGRAM: College of Information and Communications Technology/ Bachelor of Science in Information Technology

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Appendix G

IMRaD Format

TALABAYBAYIN: A MOBILE LEARNING APPLICATION FOR GUIDING FILIPINO LEARNERS IN MASTERING AND PRESERVING BAYBAYIN

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Abstract

This study, TalaBaybayin: A Mobile Learning Application for Guiding Filipino Learners In Mastering And Preserving Baybayin focuses on a mobile learning app that helps Filipino learners master and preserve the ancient pre-colonial writing system known as Baybayin. It promotes keeping this script alive and getting good at it through an interactive setup with gamified elements on a phone app. The app targets mainly Grade 7 students since they cover Baybayin in the Matatag curriculum from the Department of Education, which stresses learning and grasping the old Filipino script. Built with React Native Skia and following the Agile Software Development Life Cycle, the app mixes in multimedia lessons, handwriting recognition, quizzes, and fun features like ranks, badges, and leaderboards to boost involvement and make it easier to use.

The system's effectiveness was evaluated using the ISO/IEC 25010:2023 software quality standard through a mixed-method approach involving 100 respondents composed of students, teachers, IT professionals, and industry experts. The TalaBaybayin application received an average rating of 3.64 with a standard deviation of 0.74, indicating that users generally find the system highly acceptable in terms of functionality, performance, compatibility, and security. These findings demonstrate that TalaBaybayin provides an effective, interactive, and culturally relevant learning experience that supports both educational and cultural preservation goals.

Keywords: *TalaBaybayin, Baybayin, mobile learning, ISO/IEC 25010, cultural preservation.*

INTRODUCTION

Technological advancements have transformed industries through the rise of mobile applications. This shift from personal computers to smartphones has changed how people use technology, driving growth in communication, education, and business (Kemp, 2025; Thomas & Devi, 2021).

In education, mobile learning provides flexibility, accessibility, and personalized learning experiences (Lazaro & Duart, 2023). It enhances student engagement and performance through interactive environments tailored to different learning needs (Kaur et al., 2022). Gamification further increases motivation by adding features such as points, ranks, and badges (Abdeen & Albiladi, 2021; Oliveira et al., 2024).

Mobile technology also plays a key role in preserving indigenous writing systems (Motorola, 2022). For instance, Motorola's Cherokee language interface and Thailand's digital preservation of the Tai Tham script show how digital platforms can protect traditional languages and cultural heritage (Dao, 2023; Kittipalo et al., 2022).In the Philippines, the Baybayin script faces challenges due to the dominance of the Latin alphabet, risking its decline (Galvez et al., 2023).

While legislative efforts have been made to protect Baybayin, there is still a lack of interactive digital tools to aid its learning (Lacorte et al., 2024). Mobile learning apps could revitalize Baybayin by making it more accessible and engaging for grade 7 students and independent learners (Camba, 2021).

To address this gap, the researchers developed TalaBaybayin: A Mobile Learning Application for Guiding Filipino Learners in Mastering and Preserving Baybayin, which features a Baybayin keyboard, interactive writing practice tools, quizzes, and gamified elements to support learning and mastery of the script.

Through achievement ranks and interactive challenges, the mobile application encourages active participation of the students and helps preserve Baybayin in a modern and engaging way. The app incorporates multimedia elements like audio pronunciations to aid auditory learners in mastering Baybayin sounds. The features like this encourage the users to interactively learn the scripts in a fun way.

Review of Related Literature

Baybayin literacy, as examined in this study, refers to a learner's capacity to identify, read, and write Baybayin symbols (Prado, 2024). Research shows that while Baybayin is part of Filipino heritage, exposure to the script was interrupted during colonization, leading to the dominance of the Latin alphabet in both education and media (Galvez et al., 2023). Although Baybayin resurfaced in recent years as a form of cultural identity, classroom instruction has not kept pace with students' need for interactive and engaging learning materials. Printed modules and brief discussions often serve only as introductions, leaving learners without opportunities for practice and guided application (Dela Cruz et al., 2020; Escote, 2023).

Several studies show that students learn symbolic writing systems more effectively when teaching methods involve repetition, feedback, and opportunities for hands-on practice (Prado, 2024; Solares, 2024). However, many Baybayin-related materials available online function only as translators or reference charts rather than instructional tools (Bernardo & Estuar, 2025; Perin et al., 2024). These resources do not track progress, assess learning, or reinforce writing skills, which are necessary for mastery.

Mobile-assisted language learning (MALL) has been found to address these instructional gaps. Lazaro and Duart (2023) highlight that learning through smartphones allows students to learn at their preferred pace and setting. Apps that incorporate visual, auditory, and interactive elements help sustain attention and allow students to review lessons independently. Kaur et al. (2022) further explain that mobile learning environments cater to diverse learners and can increase academic performance.

Motivation also plays a crucial role in sustaining learning. Gamification implemented through badges, ranks, points, and leaderboards has been proven to increase motivation and persistence when learning new skills (Oliveira et al., 2024; Abdeen & Albiladi, 2021). Platforms like Quizlet and Duolingo illustrate how learners remain engaged when progress is visible and rewarded (Pitriani et al., 2024; Steiner, 2024). This supports the integration of gamification into language learning applications.

Technology has also become a medium for cultural preservation. International examples, such as Motorola's Cherokee language integration and Thailand's digital preservation of the Tai Tham script, show that mobile technology can revive endangered writing systems (Dao, 2023; Kittipalo et al., 2022). These initiatives demonstrate that technology can bridge cultural traditions and modern learning environments.

Locally, the MATATAG Curriculum strengthens the need for Baybayin integration in education. The Department of Education (DepEd, 2024) emphasizes foundational learning, cultural relevance, and contextualized activities. Baybayin is introduced in Grade 7 Filipino under multimodal literacy, which highlights the exploration of indigenous writing systems.

Despite these opportunities, existing Baybayin resources are limited to reference tools and manual worksheets. None combines lesson delivery, handwriting practice, gamification, and learning progress tracking in one application. This gap becomes the foundation for the development of TalaBaybayin, which integrates mobile learning, practice features, feedback, and gamification aligned with curriculum goals.

Research Objectives

The study aims to create a mobile learning application that supports Grade 7 learners in developing literacy and appreciation of the Baybayin writing system. The application, named TalaBaybayin, utilizes interactive lessons, handwriting practice, and gamified learning elements to encourage continuous engagement and mastery. This research was guided by both a general objective and several specific objectives.

General Objective

To design, develop, and evaluate TalaBaybayin, a mobile learning application that provides an interactive platform for Grade 7 learners to practice and master Baybayin while contributing to cultural preservation.

Specific Objectives

1. Assess the current condition of Baybayin learning among Filipino learners, particularly in terms of access to learning materials, student interest, and challenges encountered in traditional instruction.
2. Design and develop a mobile learning application that incorporates features for learning Baybayin.
3. Integrate specialized tools that reinforce Baybayin literacy and writing accuracy.
4. Evaluate the application’s acceptability and software quality using the ISO/IEC 25010:2023 model.

METHODS

This section details the methodological framework adopted for developing and assessing the TalaBaybayin mobile learning application. It encompasses the research design, participant details, and instrumentation, along with explanations of data collection procedures and analytical approaches.

Research Design

To address the challenges Filipino 7th-grade high school students faced in learning and preserving the Baybayin script, the study used a mixed-methods methodology, namely an exploratory sequential model. In a single investigation, mixed methods research, which first appeared in the late 1970s, combines qualitative and quantitative data collection methodologies (Lall, 2021). Creswell (2014) characterized mixed methods as an investigative strategy that blends or associates qualitative and quantitative elements. Furthermore, Battista and Torre (2023) noted that this approach unites qualitative and quantitative strategies to attain a profound comprehension of intricate phenomena. Researchers can strengthen the credibility of findings, cross-validate data, and gain deeper insights into investigative concerns by utilizing the advantages of both paradigms. This framework encouraged the integration of various viewpoints, making it easier to collect complex qualitative data in addition to exacting quantitative assessments, strengthening the study's overall robustness.

This design suited the study by permitting iterative app refinement based on early qualitative feedback, followed by quantitative confirmation of efficacy and acceptance. The exploratory sequential approach was ideal for the complex facets of Baybayin learning, merging user-focused qualitative data with empirical quantitative metrics to produce a culturally appropriate and technically viable educational solution.

Respondents of the Study

The target population consisted of Filipino seventh-grade high school learners, educators, Baybayin advocates, and IT specialists proficient in software creation and assurance. Purposive sampling selected 70 seventh-grade students from Caniogan National High School who had interacted with the instructional module directly. Despite the sample originating from one institution, comparable learning environments and module use across four schools suggest these participants represent the wider seventh-grade demographic. Diversity was ensured by including students from various class sections.Expert sampling, a purposeful variation, was used for participants other than primary users. This approach ensures the participation of experts in system design, facilitating in-depth app analysis. As a result, the College of Information and Communication Technology (CICT) at Bulacan State University (BulSU) hired ten IT professors with specializations in programming, web design, and database management. They are in a good position to assess app acceptability and technical standards because of their software experience.

Additionally, 10 industry IT specialists, including developers, engineers, and quality testers, were involved. These professionals reviewed the app against coding norms and best practices, offering critiques to elevate quality. Quality testers examined functionality and usability, pinpointing errors to verify adherence to performance benchmarks. Their combined input refined the app to meet industry and user standards.To validate the app's quality, feedback was solicited from 10 teachers and 70 seventh-grade students at Caniogan High School. Selections reflected varied teaching and learning backgrounds with the script, ensuring insights from students, instructors, and enthusiasts with hands-on experience.

The total cohort numbered 100, distributed as 70 students, 10 teachers, 10 BulSU IT experts, and 10 industry specialists. Criteria included relevant Baybayin or software experience; exclusions applied to those lacking it for pertinent input. Ethical protocols involved securing informed consent, maintaining response anonymity and privacy, and following data protection rules. The research complied with institutional ethics board standards, with voluntary participation after clear explanations of goals and processes.

Instruments of the Study

The study employed multiple instruments to collect data on the usability, functionality, and effectiveness of the TalaBaybayin mobile learning application. A four-point Likert scale survey was used to assess both the level of difficulty students experienced in learning Baybayin and the acceptability of the application, with 1 indicating Not Difficult or Not Acceptable and 4 indicating Highly Difficult or Highly Acceptable (Mendenhall, Beaver, & Beaver, 2012; Triola, 2018). The evaluation of the TalaBaybayin application's acceptability and software quality was guided by the ISO/IEC 25010:2023 Software Quality and Acceptance Model. This model provides a comprehensive framework for assessing key quality attributes such as functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability (ISO/IEC, 2023). Participants included seventy Grade 7 students, ten Filipino teachers, ten IT faculty members from Bulacan State University, and ten IT industry professionals. Expert evaluation tools ensured that IT professionals could assess backend efficiency, user interface functionality, and compliance with software development best practices. To validate the application's performance, Black Box Testing was used to examine functionality, interface behavior, and user experience without reference to the internal code structure, while User Acceptance Testing (UAT) and integration testing verified that the system satisfied functional and non-functional requirements (Ostrand, 2002). This approach ensured that all app features, such as the Baybayin keyboard and interactive quizzes, operated correctly under various user scenarios.

Data Gathering Procedures

Data were collected in multiple stages to evaluate the TalaBaybayin mobile learning application. First, classroom observations of Grade 7 students at Caniogan National High School were conducted to identify challenges in learning the Baybayin script. Semi-structured interviews with Filipino teachers provided insights into instructional needs, learning gaps, and recommendations for system features. These qualitative findings guided the preliminary design of the application. After development, students completed structured surveys using a four-point Likert scale to assess learning difficulty, while teachers evaluated instructional relevance and usability. Expert sampling was employed to select ten IT faculty members from Bulacan State University and ten IT industry professionals to assess the system's technical quality, usability, and adherence to software development standards. Participants interacted with all application features, including the Baybayin keyboard, handwriting module, translation tools, quizzes, mini-games, and gamified learning elements. Iterative Black Box Testing, User Acceptance Testing (UAT), and integration testing ensured proper functionality, smooth module interaction, and user satisfaction. Data collection occurred over several weeks, with student observations and teacher interviews conducted during school hours and expert assessments carried out in laboratory or office settings. Collected data were analyzed using descriptive statistics, including mean and standard deviation, to identify trends, strengths, weaknesses, and areas for improvement, informing iterative refinements to enhance learning outcomes and support Baybayin preservation.

Data Analysis

In this study, data analysis employed both qualitative and quantitative approaches, following an exploratory sequential mixed-methods design. Teacher interviews and classroom observations were first analyzed using thematic analysis to uncover challenges students faced in learning Baybayin and to guide the design of the TalaBaybayin application. Dahal et al. (2024) note that thematic analysis is effective for identifying patterns, key themes, and meaningful insights from textual data, offering a deeper understanding that informs practical interventions. For the quantitative portion, descriptive statistics were calculated to summarize survey responses from students, teachers, and IT professionals. Measures such as mean and standard deviation were used to assess system usability, learning difficulty, and overall acceptability (Mendenhall, Beaver, & Beaver, 2012; Triola, 2018). Microsoft Excel was used to perform all statistical computations. Results were interpreted by comparing mean scores to the four-point Likert scale, with higher values indicating greater difficulty or acceptability, and standard deviations reflecting variability among respondents. This analysis highlighted the application's strengths and areas for improvement, guiding iterative enhancements to ensure TalaBaybayin effectively supported student learning while meeting software quality standards.

Ethical considerations

The study followed strict ethical guidelines to protect all participants. Written consent forms were obtained from students and their parents or guardians prior to participation, while teachers and expert evaluators also provided informed consent. Participant confidentiality and anonymity were maintained by recording and storing all data securely, with access limited to the research team. Personal identifiers were excluded from the analysis and reporting of results. The research adhered to institutional ethical standards, and no funding sources or conflicts of interest influenced the study. All participants were informed that their involvement was voluntary and that they could withdraw at any time without any consequences. Additional care was taken to ensure that all procedures aligned with the principles of respect, fairness, and transparency throughout the research process. Participants were also given the opportunity to ask questions or seek clarifications before agreeing to take part in the study.

RESULTS AND DISCUSSION

Interpretation and Analysis of Results

The preliminary survey was conducted with 70 students from Caniogan High School to assess the level of difficulty they experience in learning the Baybayin script.

Using the scale, 1 represents Not Difficult and 4 represents Highly Difficult, allowing the researchers to quantify and interpret student struggles in a structured manner.

Table 1

Level of Difficulty in Learning Baybayin Among Grade 7 Students of Caniogan National High School

Items	**Mean**	**Standard Deviation**	**Descriptive Interpretation**
1. Bumasa (makilala) ng isang titik sa Baybayin.	2.34	1.19	Slightly Difficult
2.Bumasa ng Baybayin na may kudlit (hal. kudlit sa itaas o ibaba).	2.23	0.96	Slightly Difficult
3.Bumasa ng isang salita sa Baybayin.	2.49	1.11	Slightly Difficult
4.Bumasa ng isang pangungusap na nakasulat sa Baybayin.	2.57	0.93	Moderately Difficult
5. Makilala ang magkakahawig na titik sa Baybayin (hal. *ba–pa*, *ka–ga*).	2.49	1.11	Slightly Difficult
6.Bumasa ng mahahabang salita sa Baybayin.	2.71	1.07	Moderately Difficult
7. Bumasa ng Baybayin nang walang kopya o gabay	3.10	0.97	Moderately Difficult
8. Pagbasa ng Baybayin nang mabilis at tama.	2.61	1.05	Moderately Difficult
9. Pag-unawa sa isang salitang nakasulat sa Baybayin.	2.44	1.09	Slightly Difficult
10. Pag-unawa sa isang pangungusap na nakasulat sa Baybayin.	2.43	1.05	Slightly Difficult
11. Magsulat ng isang titik sa Baybayin.	2.10	1.21	Slightly Difficult
12. Magsulat ng isang titik sa Baybayin na may kudlit.	2.31	1.08	Slightly Difficult
13. Magsulat ng isang salita sa Baybayin.	2.27	1.05	Slightly Difficult
14. Magsulat ng isang pangungusap sa Baybayin.	2.49	0.95	Slightly Difficult
15. Isulat ang iyong pangalan sa Baybayin.	2.29	1.06	Slightly Difficult
16. Madali kong naisusulat ang aking pangalan gamit ang Baybayinsalita sa Baybayin.	2.23	0.97	Slightly Difficult
17. Madali kong natutukoy ang tamang mga simbolo ng Baybayin.	2.57	0.93	Moderately Difficult
18. Madali kong natutukoy ang tamang mga simbolo ng Baybayin.	2.46	1.0	Slightly Difficult
19. Natutukoy ko agad ang tamang simbolo kapag nagsusulat ng isang salita.	2.59	1.09	Moderately Difficult
20. Naisusulat ko nang tama ang kombinasyon ng mga patinig at katinig sa Baybayin.	2.64	1.02	Moderately Difficult
Mean	2.47	1.04	Moderately Difficult

Before evaluating the application's software quality, a preliminary survey was conducted among seventy Grade 7 students to identify their difficulties in learning Baybayin. The overall mean score for the preliminary survey on the difficulty of learning Baybayin among grade 7 students at Caniogan National High School is 2.47, which falls within the moderately difficult range, with a standard deviation of 1.04. The results show that most respondents find Baybayin reading and writing tasks moderately difficult, with variations based on familiarity and skill.

After addressing these issues through the design of interactive and gamified features, the TalaBaybayin application was evaluated using the ISO/IEC 25010:2023 model.

Table 2

Summary Overview of the Respondents' Ratings Based on ISO/IEC 25010 Quality Evaluation Criteria

Items	Mean	Standard Deviation	Descriptive Interpretation
Functional Suitability	3.62	0.74	Highly Acceptable
Performance Efficiency	3.53	0.81	Highly Acceptable
Compatibility	3.63	0.71	Highly Acceptable
Interaction Capability	3.85	0.72	Highly Acceptable
Reliability	3.59	0.81	Highly Acceptable
Security	3.62	0.77	Highly Acceptable
Maintainability	3.66	0.70	Highly Acceptable
Flexibility	3.62	0.75	Highly Acceptable
Safety	3.66	0.69	Highly Acceptable
Overall Mean	3.64	0.74	Highly Acceptable

The assessment of the TalaBaybayin mobile learning application yielded an average score of 3.64, accompanied by a standard deviation of 0.74. This outcome reveals that participants regarded the platform as exceptionally suitable across aspects like operational capabilities, ease of use, and overall efficiency. It appears the application aligned well with what users anticipated, delivering a seamless educational journey.

The findings further confirm the application's alignment with the ISO/IEC 25010:2023 software quality model, particularly in the areas of *Functional Suitability*, *Performance Efficiency*, and *Usability*. Consistent positive feedback from students and teachers reflects how the system's design successfully resolved the common challenges of conventional instruction, especially the lack of interactive practice and immediate feedback identified by Dela Cruz et al. (2020) and Escote (2023).

These results also parallel the conclusions of Kaur et al. (2022) and Lazaro and Duart (2023), who emphasized that mobile learning enhances accessibility, engagement, and individualized pacing. The integration of gamified elements such as points, ranks, and badges further reinforced learner motivation, supporting the observations of Oliveira et al. (2024) and Abdeen and Albiladi (2021) on the positive impact of gamification on sustained participation. Users reported high levels of enjoyment and engagement while using the app, aligning with the findings of Steiner (2024) and Pitriani et al. (2024), who associated gamification with persistence and long-term learning commitment.

From a cultural perspective, the positive evaluation of TalaBaybayin supports the assertions of Camba (2021) and Galvez et al. (2023) that digital platforms can serve as powerful tools for revitalizing Filipino heritage through technology-mediated learning. The inclusion of handwriting modules and translation tools enabled learners to connect traditional writing practices with modern educational technologies, mirroring the digital preservation initiatives described by Kittipalo et al. (2022) and Dao (2023) in their studies of indigenous scripts.

Limitations of the Study

TalaBaybayin offers strong tools for learning Baybayin, but it has several key limitations that affect its use. The app targets students, teachers, and Baybayin fans, which limits its appeal to others, like non-Filipino learners or those interested in different scripts. It focuses on basic to intermediate learners and does not include advanced features for deep linguistic study or research.

The app requires smartphones or tablets, so people without these devices cannot use it. This is a problem in rural areas with limited technology. Users with old devices or poor internet may struggle with videos, games, or forums. The app only works on Android and has no web version, excluding iOS users or those who prefer computers. Features like games and forums depend on user participation, which may not always happen. Tests give quick feedback but lack the personal touch of a teacher's review. This can miss individual needs, such as cultural details or different learning styles.

The app teaches reading and writing but lacks deeper context, like history or culture. It does not offer custom paths for different interests or skills. Overall, these issues suggest ways to improve, like adding web access or better offline options, to make the app more useful while keeping its educational value. Future updates could address these to reach more people.

CONCLUSION AND RECOMMENDATIONS

This study aimed to examine the current state of Baybayin learning, develop an enhanced mobile platform, integrate specialized instructional tools, and evaluate its software quality using the ISO/IEC 25010:2023 model. For Objective 1, to assess the current condition of Baybayin learning among Filipino learners in terms of access to materials, student interest, and challenges in traditional instruction, the findings significant deficiency in dynamic and current materials. Educators mentioned depending on manual conversions and outdated paper-based techniques, which hindered student involvement and complicated maintaining lesson momentum. A poll of 70 seventh-grade pupils yielded an average challenge rating of 2.47, suggesting Baybayin is somewhat tough, largely because of insufficient user-friendly and captivating educational resources. For Objective 2, to design and develop a comprehensive platform that enhances engagement, supports progress tracking, and improves knowledge retention, the TalaBaybayin application was created with essential features such as secure log-in/sign-up, interactive lessons, quizzes, leaderboards, time-based scoring, multimedia sounds, mini-games, progress tracking, common phrases, and completion certificates.

These components were adapted from the issues identified in the data collection process and were specifically designed to strengthen student motivation and learning continuity. Furthermore, to ensure a more holistic and engaging experience, features were carefully integrated to enhance the learning, practice, and preservation of Baybayin, ensuring that users can interact with the script through multiple modes, including typing, handwriting, translation, and auditory reinforcement. For Objective 3, to integrate specialized features that enhance the user experience and learning process, the app included a Baybayin keyboard, handwriting and tracing tools, a text-to-Baybayin converter, and text-to-speech support, all of which encouraged practical application and consistent practice. Lastly, for Objective 4, to evaluate the application’s acceptability and software quality based on ISO/IEC 25010:2023, the TalaBaybayin mobile learning app achieved an average rating of 3.64 with a standard deviation of 0.74, indicating high user acceptability across functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety. Although some variation in responses was noted, the overall findings confirm that the application effectively addresses the challenges identified at the start of the study. Users recognized the system not only as an effective learning tool but also as a platform that enhances learning outcomes and supports long-term cultural awareness, ensuring that the study's objectives of education, engagement, and preservation are fully achieved.

Drawing from the study's findings, a number of suggestions have been outlined to elevate the TalaBaybayin app's performance. For starters, adding offline capabilities is suggested to assist students in locations with unreliable or absent internet service, guaranteeing constant availability of tutorials and interactive tasks. This would also boost fairness and give users the freedom to progress through content on their own schedule. Additionally, it's proposed to widen the app's coverage by integrating more traditional Philippine writing systems like Hanunuo, Buhid, and Tagbanwa, coupled with translations into prominent Filipino languages and English. Lastly, creating an administrative panel for oversight and control could strengthen the app's utility, enabling supervisors to handle educational content, track user interactions, review assessment outcomes, and evaluate participation levels.

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Appendix H

User Guide

1. Introduction

TalaBaybayin is a mobile learning app designed to help users learn, write, and translate Baybayin the ancient Filipino script in an engaging and interactive way. The app promotes cultural preservation while supporting students, educators, and enthusiasts who wish to rediscover this unique writing system.

With gamified lessons, quizzes, practice modules, and progress tracking, TalaBaybayin encourages users to advance through levels, earn achievements, and even receive certificates for their accomplishments.

---

2. Getting Started

2.1 Launching the App

After installation, tap on the **TalaBaybayin** icon on your device to open the app.

2.2 Signing Up / Logging In

- New users can **sign up** using an email address or Google account.
- Existing users can **log in** with their registered credentials..

After logging in, the user is directed to the **Dashboard**.

---

3. Dashboard

The **Dashboard** serves as the main hub of the app

Displays the user's **current rank**, **total points**, and **daily streak**.

---

4. Common Phrases

This section provides everyday Filipino expressions translated into Baybayin. Users can:

- View Baybayin**, Tagalog, English script** side-by-side

This helps learners apply Baybayin in real-life communication contexts.

---

5. Lessons

The **Lessons** section introduces users to the fundamentals of Baybayin:

- **Basic Characters:** Vowels and Consonants
- **Kudlit Marks:** Modifiers that change vowel sounds
- **Combination Rules:** How symbols connect to form words
- **Word Construction:** Writing simple words and sentences

Unlocking new lessons depends on the user's **lesson progress**, motivating continued learning.

---

6. Quiz

After completing lessons, users can test their knowledge through quizzes. Features include:

- Multiple-choice and fill-in-the-blank questions
- True or False
- Score tracking and retries

Quizzes are essential for progressing to advanced levels and unlocking new content.

---

7. Practice Writing

The **Practice Writing** section allows users to trace and freely write Baybayin characters using their device's touchscreen.
Features include:

- **Save the progress** for each letter
- **Reset and clear** buttons for repeated practice

---

8. Translator with Baybayin Keyboard

The **Translator** tool enables users to convert Filipino or English words into Baybayin.
Key features:

- **Custom Baybayin keyboard** for direct input
- **Instant translation** between text and Baybayin
- Option to **copy** translated text
- Audio playback for pronunciation assistance

This feature helps users apply their learning in creative and practical ways.

---

9. Progress

The **Progress** screen allows users to track their overall learning journey within the app.

Displayed details include:

- Number of **lessons completed**
- **Completed Quiz** and
- **Rank** and **Badges** earned
- **Overall completion percentage**

Certificate Feature

Upon completing all lessons, quizzes, and mini-games, users will **earn a Digital Completion Certificate** as proof of mastery.

- The certificate includes the **learner's name**, **completion date**, and **achievement rank**.
- Displays all lessons and completed quiz and their accuracy.
- It can be **viewed, downloaded, or shared** directly from the Progress screen.

This encourages consistent learning and gives users a sense of accomplishment.

---

10. Leaderboard

The **Leaderboard** displays the top learners in the app, ranked by:

- Total points earned
- Rank

Users can see their current position and strive to improve their ranking by completing lessons and engaging with the app daily.

---

11. Mini Games

The **Mini Games** section allows users to reinforce their Baybayin knowledge through fun and engaging activities. These games are designed to make learning interactive and enjoyable while improving memory, recognition, and writing accuracy.

Available Games:

1. **Memory Match**
   - Objective: Find matching pairs of Baybayin and Tagalog words.
   - Difficulty: Easy
   - This game helps learners associate Baybayin symbols with their Tagalog equivalents.
2. **Choose the Baybayin**
   - Objective: Select the correct Baybayin character for each alphabet.
   - Difficulty: Easy
   - Strengthens symbol recognition and reinforces letter association.
3. **Word Scramble**
   - Objective: Arrange the letters to form the correct Baybayin word.
   - Difficulty: Medium
   - Enhances vocabulary and spelling through interactive challenges.
4. **Symbol Quiz**
   - Objective: Identify the correct Baybayin symbol in a fast-paced quiz.
   - Difficulty: Medium
   - Tests quick recall and visual recognition under time pressure.
5. **Pattern Recognition**
   - Objective: Follow the Baybayin stroke patterns correctly.
   - Difficulty: Hard
   - Improves handwriting accuracy and understanding of Baybayin writing order.

---

12. Profile Settings

The **Profile Settings** screen allows users to:

- View information of the user
- Access multimedia

---

13. Logout

To securely exit the app, users can tap the **Logout** button in the Profile Settings or Dashboard menu.
This action signs the user out of their account and returns them to the **Login screen**.

Appendix I

Pictures of Data Gathering


BULSU
CREM

Appendix J

Sample Input Output Reports

TalaBaybayin
SERTIPIKO NG PAGTATAPOS
TalaBaybayin Learning Program
Pinatutunayan nitong si
SUNSHINE, ARRAY S
ay nakumpleto ng matagumpay ang lahat ng aralin at pagsusulit sa TalaBaybayin at naging isang tunay na Baybayin Master
Lahat ng Aralin Nakumpleto
Lahat ng Pagsusulit Naipassa
Baybayin Master
Petsa ng pagkakamit: October 27, 2025
9/9
Natapos na Aralin
9/9
Natapos na Pagsusulit
2,407
Experience Points
Mga Score sa Pagsusulit
Aralin 1 80%
Aralin 2 90%
Aralin 3 80%
Aralin 4 87%
Aralin 5 87%
Aralin 6 80%
Aralin 7 80%
Aralin 8 90%
Aralin 9 90%
Ginawa ng
TalaBaybayin
Baybayin Learning Application
2025-10-27

Appendix K

Evaluation Tool or Test Document

ISO 25010:2023 RESPONSES

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yin App ng kabala-kapag ng Tala	Baybayin App kahit	Column 43	Column 44
5	5	None	
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5	5	Magaling!	
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4	4	This app is amazing because it help Filipino students to improve baybayin	
4	3	Okay lang po yung app	
5	5	Magaling	
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4	4	None	
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4	3	asal bata ay mapagamit sa pag-sulat o paggamit ng baybayin, halimbawa po, sa APP may isang laro na kung saan gagawa ng i	
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4	4	Good	
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3	3	Nothing	
4	4	Good Application	
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3	3	ter if I have access to the app itself since the questions mostly rely on the usage and adaptability of the app. Overall nice UI for t	
3	3	The app is a creative and educational way to reconnect with Filipino culture. The project has great potential.	
4	3		
3	3	Nothing. I'd say its a very good application that can help people learn Baybayin. I'll probably use it once you publish it!	
3	4		
4	4		
4	4	N/a	
3	4	sers can explore the app you plan to use without having to log in. This is because some users prefer to explore apps immediatel	
4	4		
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4	4	Pagkatuto sa pamana ng Talabaybayin	

Appendix L

Capstone Application Form

Republic of the Philippines
City of Malolos, Bulacan
BULACAN STATE UNIVERSITY
College of Information and Communications Technology

BAGONG PILIPINAS

Management System ISO 9001:2015 | Level III Recognition for Mastery in Quality Management 2022 | THE Impact Rankings 2023 +1001 | Dark Green School Certified | World University Rankings Ranked 684 | WURI The WORLD UNIVERSITY RANKINGS for INNOVATION

APPLICATION FOR CAPSTONE PROJECT FINAL DEFENSE

Capstone/Project Design Details	
Complete Title	TalaBaybayin: A Mobile Learning App For Guiding Filipino Learners In Mastering and Preserving Baybayin
Academic Year/Semester	2025-2026
Capstone/Project Design Adviser	Aaron Paul M. Dela Rosa, MSIT
Chairman	Renato L. Adriano II, MSIT
Panel Members	Edwin S. Garcia, MSIT
	John Lemuel C. Salazar

Capstone/Project Design Group		
Student Number	LName, FName MI	Signature
Team Leader *2022100524*	Francia, Christian B.	
Members 2022105410	Garcia, Kylee Remiel L.	
2022100864	Lapuz, Alyssa Mae V.	
2020103001	Miramonte, Angelo C	
2022100746	Teodoro, John Patrick D.	

I certify that I have read the Capstone/Project Design manuscript presented by the student in connection with this application for final defense and classify the same as eligible for defense within the schedule/deadlines set by the Department Head.

Republic of the Philippines
City of Malolos, Bulacan
BULACAN STATE UNIVERSITY
College of Information and Communications Technology

Management System ISO 9001:2015 | Level III Recognition for Mastery in Quality Management 2022 | THE Impact Rankings 2023 +1001 | Dark Green School Certified | World University Rankings Ranked 684 | WURI The WORLD UNIVERSITY RANKINGS for INNOVATION

Capstone/Project Design Adviser		
Complete Name	Signature	Date
Aaron Paul M. Dela Rosa, MSIT		09 OCT 2025

Capstone Project Coordinator		
Complete Name	Signature	Date
Menirissa D De Belen, MSIT		10/09/25

Program Coordinator		
Complete Name	Signature	Date
Jayson A. Batoon, DIT		

Program Chair		
Complete Name	Signature	Date
Edwin S. Garcia, MSIT		10/10/2025

INSTRUCTIONS TO THE STUDENT

1. This form must be accomplished in quadruplicate (4 copies) when all necessary signatures have been completed. Application forms with incomplete information and signatures will not be accepted for processing.
2. Upon submission, the Accounting Office will make an assessment of the relevant fees for the application.
3. Pay the assessed amount at the Cashier's Office.
4. Submit three (3 copies) together with the official receipt to the following:
   - Office of the Dean
   - Program Chair
   - Capstone Coordinator
5. Coordinate the schedule and venue of the defense with the Program Chair.

collegeemail@bulsu.edu.ph| 044-919-7800
Main Campus, Bulacan State University
Guinhawa, City of Malolos, Bulacan

BU S
Bulacan State Univ

Republic of the Philippines
City of Malolos, Bulacan
BULACAN STATE UNIVERSITY
College of Information and Communications Technology

WURI

TERMS AND CONDITIONS	
1.	To be able to apply for Capstone/Project Design Final defense, the student must be enrolled in Capstone Project/Project Design course during the term.
2.	A verdict resulting to revision in Capstone/Project Design is classified as "Incomplete." To qualify for completion, revisions must be approved and reported by the Chair of the Defense Panel to the Capstone/Project Design Coordinator within the specified date agreed upon by the Panel Members. In case of failure to comply with the revisions, the "Incomplete" is automatically converted to "Failed," in which case the student has to reenroll the Capstone/Project Design.

STUDENT CONFORME

I have read and understood the "Instructions" and "Terms and Conditions" stipulated above and agree to the same.

Christian Bayan Francia
SIGNATURE OVER PRINTED NAME / DATE

VALIDATION OF PAYMENT

OR Number: 2608856
Date: 10/10/2025 Amount: Php 1,800

Appendix M

One Page Curriculum Vitae

Christian B. Francia

Pungo, Calumpit, Bulacan | pipoyfrancia0322@gmail.com@gmail.com | 0960-428-7867

EDUCATION

Bulacan State University - Main Campus, Malolos, Bulacan 2022-2025
Bachelor of Science in Information Technology

La Consolacion University Philippines, Malolos, Bulacan 2020-2022
Senior High School - Technical-Vocational-Livelihood (TVL) Computer Progamming

Caniogan High School, Calumpit, Bulacan 2016-2020
Junior High School

Pungo Elementary School, Calumpit, Bulacan

CERTIFICATION

- **Salesforce Supported Virtual Internship Program 2025 – Certificate of Completion** - (March – May 2025, Philippines)

- **iLEAP Course on Intellectual Property - Certificate of Completion -** Modules completed: Introduction to Intellectual Property, Copyright, Trademark, and Patent – (December 2, 2024)

- **SpeakIT Seminar (Bulacan State University) - Certificate of Appreciation** – (November 22, 2024)

SKILLS

- Web Development
- Front-end Developer
- Game Development

PROJECTS

TalaBaybayin
- Mobile learning application designed to help Filipino learners master and preserve **Baybayin**, the pre-colonial Filipino writing system.

Lost in the Amber Age - 2D Platformer
- Developed a game concept featuring a young protagonist transported to the Jurassic period, blending action-adventure gameplay with educational storytelling. Designed engaging platforming challenges, dynamic encounters with prehistoric creatures, and narrative elements centered on exploration.

Kylee Remiel L. Garcia

Frances, Calumpit, Bulacan | kyleeremielzz@gmail.com@gmail.com | 0953-473-1458

EDUCATION

Bulacan State University - Main Campus, Malolos, Bulacan 2022-2025
Bachelor of Science in Information Technology

Frances National High School, Calumpit, Bulacan 2020-2022
Senior High School -Science, Technology, Engineering, and Mathematics (STEM)
Frances National High School, Calumpit, Bulacan 2016-2020
Junior High School

Frances Elementary School, Calumpit, Bulacan

CERTIFICATION

- **Salesforce Supported Virtual Internship Program 2025 – Certificate of Completion** - (March – May 2025, Philippines)

- **iLEAP Course on Intellectual Property - Certificate of Completion -** Modules completed: Introduction to Intellectual Property, Copyright, Trademark, and Patent – (December 2, 2024)

- **Cisco Network Basics**

SKILLS

• Frontend: HTML, CSS, JavaScript
• Backend: PHP
• Languages: Java, C++
• Professional: Communication, leadership, collaboration, time management, critical thinking, creativity, documentation

PROJECTS

• **Hospital Management System Website** – Group project with full CRUD; contributed to development and documentation.
• **Personal Portfolio Website** – Built with HTML, CSS, JavaScript, and PHP to showcase skills and background.
• **Talabaybayin Mobile App** – Led documentation; applied NativeWind for efficient cross-platform styling in React Native.

Alyssa Mae V. Lapuz

Marilao, Bulacan | lapuzalyssamae02@gmail.com@gmail.com | 0994-224-0034

EDUCATION

Bulacan State University - Main Campus, Malolos, Bulacan 2022-2025
Bachelor of Science in Information Technology

Barcelona Academy, Marilao, Bulacan 2020-2022
Senior High School - STEM

Regina Coeli Educational Centre Foundation, Caloocan 2016-2020
Junior High School

CERTIFICATION

- **Salesforce Supported Virtual Internship Program 2025 – Certificate of Completion** - (March – May 2025, Philippines)

SKILLS

- Web Development
- Front-end Developer
- Game Development

PROJECTS

TalaBaybayin

- Mobile learning application designed to help Filipino learners master and preserve **Baybayin**, the pre-colonial Filipino writing system.

Lost in the Amber Age - 2D Platformer

- Developed a game concept featuring a young protagonist transported to the Jurassic period, blending action-adventure gameplay with educational storytelling. Designed engaging platforming challenges, dynamic encounters with prehistoric creatures, and narrative elements centered on exploration.

Angelo C. Miramonte

Bulakan, Bulacan | angelomiramonte32@gmail.com@gmail.com | 0993-581-3681

EDUCATION

Bulacan State University - Main Campus, Malolos, Bulacan
Bachelor of Science in Information Technology

San Francisco Xavier High Shool,
San Francisco, Bulakan
Senior High School – TVL
Junior High School

CERTIFICATION

- NCII – Animation
- SpeakIT: Crack the Code to Effective Communication
- Networking Devices and Initial Configuration – Cisco Networking Academy
- Computer Hardware Basics – Cisco Networking Academy (2023)
- Networking Basics – Cisco Networking Academy (2023)
- Networking Addressing and Basic Trobleshooting – Cisco Networking Academy (2024)

SKILLS

- Web Development
- UI/UX Design
- Networking
- Microsoft Office

ACADEMIC PROJECTS

TalaBaybayin

- A Mobile learning application designed to help Filipino learners to master Baybayin.

Lost in the Amber Age - 2D Platformer

- Educational game with Jurassic theme.

JOHN PATRICK TEODORO

Sta. Maria, Bulacan 3022 | 0949-639-3389 | Pteodoro021@gmail.com

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INTERNSHIP APPLICANT

Enthusiastic Information Technology student seeking a 500-hour internship starting December 2025, with a focus on UI/UX Design, Software Testing, and Graphic Design. Passionate about creating innovative, user-centered solutions and eager to apply classroom knowledge to real-world projects. Committed to learning new technical and professional skills while contributing to a collaborative team environment.

EDUCATION

Bachelor of Science in Information Technology (BSIT)
Bulacan State University – Malolos, Bulacan
2022 – 2026 (Expected Graduation)

ACADEMIC PROJECTS

- **TalaBaybayin Learning Application** — Interactive mobile learning app showcasing the ancient Filipino script. Built with React Native and Firebase; designed in Figma. Contributed to UI/UX design and documentation.
- **Hospital Management System Website** — Java and MySQL-based system streamlining hospital operations with full CRUD functionality. Contributed to documentation and system structuring.
- **Lost in the Amber Age** — Educational Unity-based dinosaur-themed game. Contributed main game assets and documentation.

SKILLS

- **UI/UX Design:** Wireframes, prototypes, and user-centered interface designs using Figma
- **Programming & Development:** Java, Python
- **Game Development:** Unity, asset creation, educational game design
- **Documentation & Reporting:** Strong technical documentation skills
- **Collaboration & Teamwork:** Effective in both creative and technical team roles